How Ricardian Are We?*

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Abstract

Not very. To answer this question, we conjecture that households might be non-Ricardian because they do not have rational expectations over their future tax burden. From this assumption, we derive a behavioral consumption function, where households act as if bonds are net wealth, and consume out of taxes and transfers. The coefficient on taxes is determined by the attenuation present in households' behavioral expectations. This consumption function also nests other causes of non-Ricardianism, including liquidity constraints and overlapping generations. To estimate the coefficient, we derive a Bayesian limited information method that uses a large number of macroeconomic shocks from the literature as instrumental variables. We find that households internalize only 9% of their future taxes. In a general equilibrium model, this low value implies that public borrowing explains the majority of the net foreign asset decline and current account deficits in the US over the last 50 years.

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"... the people who pay the taxes never so estimate them, and therefore do not manage their private affairs accordingly."

— David Ricardo (1820)

1 Introduction

Neoclassical theory predicts that *ceteris paribus* private consumption should be unaffected by government borrowing if we are *Ricardian* (Barro, 1974). But, how Ricardian are we? What share of future taxes do households save to pay for? The answer is critical for the design of major macroeconomic policies – e.g. fiscal stimulus, public finance, and crisis response – and yet we have little idea what the number is.

We address this crucial question by bringing modern methods to bear on a classic problem: estimating the consumption function. A large literature in the 1980s investigated the Ricardian question by using aggregate time series regressions to estimate how taxes affect consumption on average.[1] However, the effect of taxes on consumption is unlikely to be identified in these regressions (Feldstein, 1982; Cardia, 1997). This is because demand shocks (i.e. omitted variables in the consumption function) affect consumption directly, but can also affect taxes, income, and other determinants in general equilibrium. Mainstream work with this strategy largely ceased, with the Ricardian question unanswered.

Time series evidence is necessary. After the aggregate consumption function approach was abandoned, microeconomic studies flourished. Studies using interpersonal variation in taxes, transfers, and consumption are valuable for many reasons, but because of the *missing intercept problem*, they cannot (nor do they claim to) answer whether US households are Ricardian on average. Microdata are useful for estimating an individual's marginal propensity to consume (MPC), but not the aggregate *MPC out of debt-financed transfers* (MPC-DFT). To do so, macro evidence must be incorporated, either through full general equilibrium modeling (e.g. Angeletos et al. (2024)) or some semi-structural approach (e.g. Wolf (2023)). We adopt a minimal structure by assuming a single structural equation: a behavioral consumption function.

Our strategy is to estimate the consumption function using a battery of macroeconomic shocks as instrumental variables (IV), by drawing on careful identification in the recent applied macro literature. A valid IV approach will resolve the endogeneity problems that

[1] Well-known examples include Barro and Feldstein (1978), Tanner (1979), Seater and Mariano (1985), and especially Kormendi (1983), which spawned a decade of comments and replies in the American Economic Review (Barth et al., 1986; Modigliani and Sterling, 1986; Kormendi and Meguire, 1986; Feldstein and Elmendorf, 1990; Modigliani and Sterling, 1990; Kormendi and Meguire, 1990; Graham, 1995; Kormendi and Meguire, 1995).

prevented identification from traditional regressions using aggregate time series. This approach is inspired by Barnichon and Mesters (2020), who suggest using exogenous shocks to estimate structural macroeconomic equations. The consumption function is a valuable application, and we build on their method in order to do so. Specifically, we develop **B-HIVE**: the **B**ayesian **H**ybrid **IV E**stimator. **B-HIVE** is a Bayesian limited information framework for using many external instruments to estimate macro models. This framework is a valuable enhancement for three reasons. First, we have designed it to accommodate a large number of instruments. This is important because, in contrast to the original Barnichon-Mesters application, the consumption function features many endogenous regressors. Second, the set of well-identified potential instruments from the literature cover a wide range of time periods. Our Bayesian method easily accommodates periods with missing data. Third, we can leverage all the usual strengths of Bayesian estimation, among which the most relevant is the use of informed priors. Macroeconomic shocks are notoriously weak instruments (Barnichon and Mesters, 2020; Lewis and Mertens, 2025) so disciplining the parameter space with a Bayesian prior is valuable. For example, while there is uncertainty about the degree to which households internalize future taxes, the theories considered here place θ between zero and one.

To derive our consumption function, we assume that Ricardian equivalence fails because of a behavioral friction: agents do not forecast their future taxes with rational expectations. Instead, they discount future taxes with a behavioral factor, similar to a cognitive discounting bias (Gabaix, 2020). This behavioral approach is useful for letting the data identify the degree of Ricardianism without prejudice. Other mechanisms that break Ricardian equivalence – such as finite lifetimes (Blanchard, 1985) or liquidity constraints (Campbell and Mankiw, 1989) – give clear predictions about the coefficient on taxes in the consumption function. Microeconomic evidence implies strong priors about the sizes of these particular frictions, but not about the quantitative effects of non-rational expectations. However, we also show that models featuring an alternative mechanism for the Ricardian equivalence failure imply an isomorphic consumption function to ours. Thus we interpret our estimates as describing non-Ricardianism in general, independent of the fundamental cause.[2]

We find that US consumers are far from Ricardian. The relevant parameter is the share of future taxes that households internalize in their consumption function. When households are rational, this share is one, and Ricardian equivalence holds. In contrast, our results imply that households internalize only a fraction of future taxes. In our baseline estimate with aggregate consumption, the point estimate is 9.2%. Simultaneously, our Bayesian method

[2]With additional assumptions on parameter values, it is possible to attribute non-Ricardian behavior to different mechanisms. When we do so we find that a discounting wedge alone cannot explain both the estimated wealth MPC and θ. Eichenbaum et al. (2025) draw the same conclusion in an experiment where survey respondents may randomly receive an explanation of the tax implications from a hypothetical transfer.

estimates the marginal propensity to consume out of debt-financed transfers (MPC-DFT); for Ricardian households this would be zero, but we estimate roughly 0.519%. We also estimate an annualized MPC out of net worth of 2.32%. This wealth MPC is somewhat below microeconomic estimates for illiquid wealth, which tend to range from 3% - 7% (Mian et al., 2013; Di Maggio et al., 2020; Chodorow-Reich et al., 2021). Our MPC–DFT is smaller than estimates from recent one-off fiscal transfers (Parker et al., 2013, 2022; Baker et al., 2023), survey-based thought experiments (Colarieti et al., 2024; Eichenbaum et al., 2025), or randomized payments (Boehm et al., 2025).

What are the aggregate consequences of this non-Ricardian behavior? To find out, we embed our mechanism in a two-country general equilibrium model, and conclude that the dramatic rise in US government debt is responsible for the majority of the decline in the US net foreign asset (NFA) position and current account deficits over the last 50 years.[3] For this exercise, we take an otherwise standard neoclassical model, calibrated to reflect the US and rest of the world from 1976–2025, and give households the degree of non-Ricardianism that we estimate in the time series. Then, we feed in a series of fiscal shocks to recover the 50-year trend in debt, and quantify the implications for the US' external balances; the model explains 71.0% of the observed NFA decline. The open economy element is crucial: in a closed non-Ricardian economy, government borrowing substantially crowds out capital, but in an open economy, most of its borrowing is financed from abroad. This is consistent with the US experience, which has seen regular current account deficits and a 75.4 percentage point decline in NFA/GDP over the 50 years, but no clear reduction in capital investment. Finally, this exercise provides a useful validation of our main result; we concluded from short-run time series variation that households are far from Ricardian. But there is also information in the data's long-run trends, and we learn that rationalizing these trends requires severe non-Ricardianism.

The general equilibrium model also cleanly demonstrates the econometric challenges that our estimation strategy resolves. The model features a variety of structural shocks, which we use in a population regression exercise to illustrate the estimation biases of different approaches. When households experience demand shocks (stochastic residuals in the Euler equation) OLS estimates of the consumption function are biased. Estimates of consumption responses to structural tax shocks also do not recover the relevant coefficient. And when demand shocks are serially correlated, the classic strategy of using lagged aggregates as instruments fails as well. However, our strategy of using many structural shocks as IVs – even when measured with error – consistently estimates the consumption function.

[3] A long literature studies the "twin deficits" hypothesis that government borrowing causes current account deficits. Non-Ricardian households are often postulated to be a cause, whether due to finite horizons (Blanchard, 1985; Kumhof and Laxton, 2013) or incomplete markets (Aggarwal et al., 2023). Still, how large the quantitative effect is remains an open question.

Literature: This paper joins a reinvigorated literature studying the causes and consequences of non-Ricardian behavior. A common characteristic of recent and contemporaneous work on this topic is that the degree of non-Ricardianism is assumed ex ante: some microfoundation causes the failure of Ricardian equivalence, and microdata evidence is used to calibrate the relevant parameters. These are valuable exercises, but their conclusions are sensitive to the magnitude of the degree of non-Ricardianism. Therefore our time series evidence is valuable to discipline this literature, particularly because there are many different microfoundations that can all contribute to non-Ricardianism in the aggregate consumption function.[4]

One related paper deserves particular attention for using alternative methods to estimate non-Ricardianism as caused by behavioral expectations. In contemporaneous work, Eichenbaum et al. (2025) rely on survey evidence; they randomly assigned thought experiments to consumers and ask for the resulting consumption plans. They estimate substantially larger MPCs than our approach, closer to the micro estimates of MPC out of liquid wealth than total wealth. However, despite the dissimilar methods, both of our studies agree: consumers are far from Ricardian, and standard models require a behavioral friction to fully explain the wedge.

Other mechanisms used to study non-Ricardian behavior by relaxing full information rational expectations include finite planning horizons (Woodford, 2019; Woodford and Xie, 2022; Lustenhouwer and Mavromatis, 2023), level-k thinking (Bianchi-Vimercati et al., 2024), and adaptive learning (Evans et al., 2012; Eusepi and Preston, 2018; Branch and Gasteiger, 2023). Woodford (2013) reviews older results along these lines. Other work applying modern quantitative methods to tradition non-Ricardian mechanisms including finite lifetimes (Aguiar et al., 2023; Angeletos et al., 2024, 2026; Rachel and Ravn, 2025), constrained hand-to-mouth agents Galí et al. (2007); Nisticò (2016); Orchard et al. (2023); Mian et al. (2025), heterogeneous agents with borrowing constraints (Hagedorn et al., 2019; Auclert et al., 2024) and distortionary taxation (Bianchi and Melosi, 2022).

Our **B-HIVE** method has its roots in likelihood-based treatments of instrumental variable-based inference going back to Anderson and Rubin (1949, 1950), who introduced the limited information maximum likelihood (LIML) method. A Bayesian treatment of this approach was first discussed by Drèze (1976) and has since been extended in many directions (Kleibergen and Zivot, 2003; Chao and Phillips, 2002; Koop et al., 2012). Early work recognized endogeneity problems in the estimation of the consumption function and made

[4]In addition to the semi-structural approaches described thus far, another strategy is to estimate fully structural models with aggregate time series rather than calibrating to match micro evidence. For example, Brzoza-Brzezina et al. (2025) study a fully structural behavioral DSGE model where cognitive discounting (Gabaix, 2020) causes the failure of Ricardian equivalence; estimated with standard Bayesian methods on European data, they find that consumers are even less Ricardian than in our own results.

some attempts to resolve them using lagged aggregate variables as IVs. See for example Hayashi (1982) or Feldstein (1982). Seater and Mariano (1985) argued that lagged aggregates are not valid IVs; we concur and demonstrate as much in our population regression exercise. Using instrumental variables to estimate one equation of an equilibrium model, where lagged observables are the instruments, has been popular beyond the consumption function papers we have mentioned above, in particular in the New Keynesian literature – see, for example, Galí and Gertler (1999); Sbordone (2002); Mavroeidis et al. (2014). The approach of using lagged observables as instruments has also been criticized there (Nason and Smith, 2008; Mavroeidis, 2005). We instead use instruments for structural shocks, following the lead of Barnichon and Mesters (2020), who introduce this idea in the context of estimating a single structural equation. Del Negro et al. (2020) provide a related Bayesian implementation in their analysis of the Phillips curve. Their estimator follows a two-step approach: they first use an identified shock to trace out impulse responses, and then estimate the Phillips curve in the variation spanned by those responses. Crawley and Gamber (2026) follows a related strategy, using impulse responses to macro shocks for Bayesian estimation of parameterized Jacobian matrices that relate consumption to income and interest rates in a HANK model. **B-HIVE** instead estimates the structural equation, the reduced-form dynamics of the macro variables, and the shock measurement equations jointly, so uncertainty about the dynamics, the instruments, and the structural coefficients is reflected in the posterior. Several frequentist approaches instrument for multiple endogenous regressors in a structural macroeconomic equation: Caldara and Kamps (2017) do so to estimate a fiscal policy rule by IV, while Adams and Barrett (2025) estimate multiple Taylor-type monetary policy rules by IV. Lewis and Mertens (2022) extend the Barnichon and Mesters (2020) approach using forecast errors and additional instrument lags to add statistical power.

The remainder of the paper is organized as follows. Section 2 lays out the theoretical framework and derives the behavioral consumption function. Section 3 describes our Bayesian limited information estimator. Section 4 describes the data and estimation results. Section 5 explores the macroeconomic implications of our findings. Finally, Section 6 concludes.

2 The Consumption Function

This section derives the consumption function from the relevant equilibrium conditions. This is useful to do before proceeding to the full model in Section 5, because it clarifies how general our result is.

In Section 2.1 we derive a generic consumption function that is determined by only two

equations that are featured in many models: a household budget constraint and an Euler equation pricing non-contingent bonds.

Then in Section 2.2 we derive a specific *behavioral* form of the consumption function that we can plausibly estimate in the data. This form uses two additional ingredients: the government budget constraint and a behavioral relationship between agents' expectations and the rational expectation.

2.1 The Generic Consumption Function

Consider an economy which satisfies, among other things, the following two equilibrium conditions. First is the representative household's budget constraint:

$$B_{t-1} + R_t^K K_{t-1} + Y_t^N = C_t + T_t + Q_t B_t + K_t \tag{1}$$

where B_{t-1} is risk-free government debt acquired in the previous period, K_{t-1} is capital (and/or other financial assets) with R_t^K the net-of-depreciation return.[5] Y_t^N is non-financial income, C_t is consumption, T_t is taxes, and Q_t is the price of new government debt. T_t represents the total taxes paid by households, which appears in the budget constraint whether taxes are lump-sum or distortionary.

The second equation is the household's Euler equation for pricing the debt:

$$Q_t = \beta \tilde{\mathbb{E}}_t \left[\frac{u'(C_{t+1})}{u'(C_t)} \right] + Z_t^d \tag{2}$$

where $\tilde{\mathbb{E}}_t[\cdot]$ is a (possibly non-rational) expectation operator, $u'(C_t)$ denotes a household's marginal utility of consumption, and Z_t^d is an exogenous intertemporal wedge.[6]

The linearized forms (not *log*-linearized) of these two equations are

$$n_{t-1} + y_t = c_t + \tau_t + q_t \overline{B} + \beta n_t \tag{3}$$

$$q_t = \beta \tilde{\mathbb{E}}_t [\gamma \frac{1}{\overline{C}} (c_t - c_{t+1})] + z_t^d \tag{4}$$

where lower-case variables denote deviations from the steady state. $y_t = y_t^N + \overline{K} r_t^k$ represents *net* income, while financial wealth n_t is defined as

$$n_t \equiv b_t + \overline{R^k} k_t \tag{5}$$

with the assumption that $\overline{R^k} = 1/\beta$. Finally, $\gamma \equiv -\frac{u''(\overline{C})\overline{C}}{u'(\overline{C})}$ denotes the steady state

[5] Specifically, R_t^K is the gross financial return, so it includes the original capital value plus net capital income. For example, in the model in Section 5, which has rental rate R_t, the return is $R_t^K = R_t + 1 - \delta$.

[6] See Adams (2026) for a technical characterization of behavioral expectations operators.

coefficient of relative risk aversion.

The consumption function is most concise with some recursive notation. The linearized tax present value equation is

$$\tilde{v}_t^\tau = \tau_t + \beta \tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^\tau]$$

Similarly, define a present value equation for the remaining income component

$$\tilde{v}_t^y = y_t + \beta \tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^y] \tag{6}$$

for government spending

$$\tilde{v}_t^g = g_t + \beta \tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^g] \tag{7}$$

and the present value (or "consol value") of future one-period bonds by

$$\tilde{v}_t^q = q_t + \beta \tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^q] \tag{8}$$

We also use this approach to define an exogenous "demand" factor from the intertemporal wedges, which affects the consumption equation:

$$\zeta_t = -\frac{\overline{C}}{\gamma} z_t^d + \beta \tilde{\mathbb{E}}_t[\zeta_{t+1}] \tag{9}$$

The present value variables are written with tildes to denote that they depend on the behavioral expectation $\tilde{\mathbb{E}}$. If expectations are rational ($\mathbb{E}$) then we write them without the tilde, e.g. v_t^τ. As yet, we make no assumptions about the subjective expectations operator $\tilde{\mathbb{E}}_t$ other than linearity and that agents know current variables with certainty, e.g. $\tilde{\mathbb{E}}_t[n_t] = n_t$.

Proposition 1 *The linear consumption function is*

$$c_t = (1-\beta)\left(n_{t-1} + \tilde{v}_t^y - \tau_t - \beta \tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^\tau]\right) + \left(\frac{\overline{C}}{\gamma} - (1-\beta)\overline{B}\right)\tilde{v}_t^q + \zeta_t \tag{10}$$

Proof: Appendix A

Proposition 1 is derived only using the household budget constraint and Euler equation on risk-free bonds. It holds in the lion's share of representative agent models. The Euler equation is the only one that is directly affected by expectations; this is the channel through which distorted beliefs enter the consumption function.

However, the generic consumption function (10) is not well-suited to be estimated directly. $\tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^\tau]$ is not typically observed – if it was, we could directly estimate how $\tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^\tau]$ depends on the rational forecast without bothering with the consumption function at all.

So instead, we need an additional model equation to transform the consumption function into a form that can plausibly be estimated in the data.

2.2 A Behavioral Consumption Function

We now introduce a third model equation, the budget constraint for a government that issues non-contingent debt:[7]

$$B_{t-1} + G_t = T_t + Q_t B_t \tag{11}$$

where G_t is government expenditure. The linearized form is

$$b_{t-1} = \tau_t - g_t + \overline{B} q_t + \beta b_t \tag{12}$$

assuming that the steady state bond price is $\beta = \overline{Q}$.

We also now make an assumption about the relationship between the behavioral and rational expectation operators. Their forecasts for discounted future taxes is

$$\tilde{\mathbb{E}}_t[\tilde{v}^{\tau}_{t+1}] = \theta \mathbb{E}_t[v^{\tau}_{t+1}] \tag{13}$$

for some scalar θ. Several types of behavioral expectations lead to equation (13)'s wedge representation. This includes the popular cognitive discounting assumption (Gabaix, 2020), while Adams (2026) shows that, to first order, it can represent certain other biases and information frictions. Crucially, the θ parameter only disciplines the expectations of *taxes*. While households may forecast income, prices, or other variables non-rationally, we are agnostic about how those expectations are formed, and do not assume that they are determined by θ.

Proposition 2 *If the government budget constraint is given by equation* (12) *and expectations satisfy equation* (13)*, then the consumption function can be expressed as*

$$c_t = (1-\beta)\left(n_{t-1} - \theta b_{t-1} + \tilde{v}^y_t - (1-\theta)\tau_t - \theta v^g_t + \theta \overline{B} v^q_t\right) + \left(\frac{\overline{C}}{\gamma} - (1-\beta)\overline{B}\right)\tilde{v}^q_t + \zeta_t \tag{14}$$

Proof. Iterate the government budget constraint (12) and take rational expectations:

$$b_{t-1} = \tau_t - g_t + \overline{B} q_t + \mathbb{E}_t\left[\sum_{j=1}^{\infty} \beta^j (\tau_{t+j} - g_{t+j} + \overline{B} q_{t+j})\right]$$

[7] The non-contingent assumption is not important. We make this assumption for clarity, so that readers do not wonder whether default is important for our non-Ricardianism. In Appendix D.1, we show that if debt is risky, the behavioral consumption function is unchanged.

$$= \tau_t - g_t + \overline{B} q_t + \beta \mathbb{E}_t \left[v^{\tau}_{t+1} - v^{g}_{t+1} + \overline{B} v^{q}_{t+1} \right]$$
$$= \tau_t + \beta \mathbb{E}_t \left[v^{\tau}_{t+1} \right] - v^{g}_{t} + \overline{B} v^{q}_{t}$$

Combine with equation (13) to find

$$\beta \tilde{\mathbb{E}}_t [\tilde{v}^{\tau}_{t+1}] = \theta \left(b_{t-1} - \tau_t + v^{g}_{t} - \overline{B} v^{q}_{t} \right)$$

then use this result to replace $\beta \tilde{\mathbb{E}}_t [\tilde{v}^{\tau}_{t+1}]$ in the consumption function (10). ■

Why is the behavioral consumption function in equation (14) more useful for our estimation strategy than the generic consumption function in equation (10)? By incorporating the restriction implied by the government budget constraint, the behavioral attenuation θ shows up directly in the coefficient on taxes τ_t. Because θ does not just enter through expectations (as in equation (10)), this allows our consumption function to nest a variety of causes of non-Ricardianism. Moreover, this representation implies a straightforward testable hypothesis: if agents are Ricardian then $\theta = 1$, and the coefficient on taxes is zero.

Moreover, the variables in the behavioral consumption function (14) can plausibly be observed. Consumption c_t, financial net worth n_{t-1}, government debt b_{t-1}, and taxes τ_t are all directly measured. $\tilde{v}^{q}_{t}$ is the market price of a consol bond; it can be inferred with a model from the yield curve or simply approximated with long-term debt. v^{q}_{t} and v^{g}_{t} are rational expectations, which can be estimated.[8] The most problematic variable is the household's expected discounted value of future net income $\tilde{v}^{y}_{t}$. One way to account for this variable is to include survey data on household income forecasts, although we will consider several alternative approaches.

Proposition 2's behavioral consumption function also illustrates why the missing intercept problem prevents the use of cross-sectional data to estimate Ricardian equivalence. As mentioned in the introduction, a large literature uses household-level variation to estimate how government transfers affect consumption. Comparing two households with different transfer receipts – but the same future tax burden – estimates the MPC $1 - \beta$. This is useful information, but it is not informative about Ricardian equivalence. This is because the *MPC out of debt-financed transfers* (MPC-DFT) is $(1-\beta)(1-\theta)$. To ascertain if households are Ricardian, there must also be variation in their future tax burdens; this is why time series evidence will be essential for our application.

[8] The appearance of these rational expectations terms does not imply that any agents in the economy must have rational expectations. Rather, they are a result of the requirement that the government budget constraint must hold in all future periods; it would be equally valid to use future realizations, but this would prevent estimation because the longest horizons would exceed the sample window.

2.3 Alternative Justification for the Behavioral Consumption Function: Discounting Wedges

How important is the behavioral mechanism for our empirical work? Not particularly important: many alternative mechanisms for non-Ricardianism lead to similar conclusions regarding the consumption function. In this section, we show that a wedge between the household's and government's discount factors leads to the same reduced form as our behavioral consumption function. Then, in Appendix D, we show that this is true of other mechanisms, including hand-to-mouth households (Campbell and Mankiw, 1989).

A discounting wedge is a popular feature for modeling failures of Ricardian equivalence. For example, the wedge nests the inter-generational explanation for non-Ricardianism (Blanchard, 1985) whereby agents have some probability of death, after which they do not receive utility (or at least receive diminished utility from their dynasty). Alternatively, the wedge can capture occasionally binding liquidity constraints (Farhi and Werning, 2019); Angeletos et al. (2024) argue that this mechanism leads to similar conclusions as in a full heterogeneous agents model.[9] The wedge can also represent distortionary taxation.[10]

With a discounting wedge ω, the assumptions for the generic consumption function in Proposition 1 still hold, albeit with discount factor $\beta\omega$. But we relax the assumption that the steady state bond price β is equal to the household's discount factor, so Proposition 2 no longer follows and rational expectations households may not be Ricardian.

And yet, Proposition 3 states that the resulting consumption function is *isomorphic* to the behavioral consumption function from Proposition 2, under the assumption that the present value of taxes follows an AR(1) process as in Eichenbaum et al. (2025). The usual Ricardian coefficient $1-\theta$ – which captures the deviation from rational expectations in the rest of the paper – now measures how far the discounting wedge ω is from one. That is to say: households and governments discounting at different rates is equivalent to households misextrapolating future tax liabilities.

Proposition 3 *If the government discounts by β and the household discounts by $\beta\omega$, the*

[9] Berger et al. (2023) derive a time-varying wedge that fully capture the distortions to the representative Euler equation caused by imperfect risk sharing in heterogeneous agent models. They find that in ordinary times, these wedges do not contribute much to aggregate volatility. In Appendix D.6, we show how the time-varying wedge enters the behavioral consumption function. And in Section 4.3, we extend our baseline estimation procedure to account for the effects of imperfect risk sharing.

[10] It is well known that distortionary taxation can break Ricardian equivalence, but not all distortionary taxes do so. Proposition 2 uses only budget constraints and an Euler equation to derive the behavioral consumption function; therefore, to distort consumption responses, the tax must distort intertemporal decision making through the Euler equation. For example, capital taxation can do so. However, as Trostel (1993) shows, labor taxation does not necessarily: reducing consumption solely by reducing income is not non-Ricardian.

present value of taxes is AR(1) given by

$$v_t^\tau = \rho_\tau v_{t-1}^\tau + \varepsilon_t^\tau \tag{15}$$

and households have rational expectations, then the consumption function can be expressed as

$$c_t = (1-\beta\omega)\left(n_{t-1} - \theta b_{t-1} + \tilde{v}_t^y - (1-\theta)\tau_t - \theta v_t^g + \theta\overline{B}v_t^q\right) + \left(\frac{\overline{C}}{\gamma} - (1-\beta\omega)\overline{B}\right)\tilde{v}_t^q + \zeta_t \tag{16}$$

where

$$\theta = \omega\frac{1-\beta\rho_\tau}{1-\beta\omega\rho_\tau} \tag{17}$$

where for quantity x, v_t^x denotes the present value using the government's discount factor β, while $\tilde{v}_t^x$ denotes using the household's discount factor $\beta\omega$:

$$v_t^x = x_t + \beta\mathbb{E}_t[v_{t+1}^x] \qquad \tilde{v}_t^x = x_t + \beta\omega\mathbb{E}_t[\tilde{v}_{t+1}^x]$$

Proof: Appendix A

Given this equivalence, why do we bother with the behavioral expectations, when a household-government discounting wedge is already standard in our theories? Because a discounting wedge is too tightly disciplined by the microfoundations. For example, when the wedge is determined by the overlapping generations (OLG) structure so that ω represents the survival rate, ω is necessarily close to 1 so equation (17) implies that the tax coefficient θ cannot be much less than 1 either.[11] In contrast, we have very loose priors about θ as a cognitive discounting coefficient. Therefore, with behavioral expectations as the source of non-Ricardianism, we are free to let the time series speak for themselves by choosing a relatively uninformed prior.[12]

3 B-HIVE: the Bayesian Hybrid IV Estimator

In this section, we describe **B-HIVE**, a Bayesian limited information approach that allows us to estimate the parameters of our consumption function without having to take a stance on a full equilibrium model. To do this, we borrow insights from the literature on limited information maximum likelihood (Anderson and Rubin, 1949), and in particular Bayesian implementations of this idea (Drèze, 1976; Kleibergen and Zivot, 2003)[13] - we discuss below

[11] Angeletos et al. (2024) set the OLG survival rate to $\omega = 0.865$.

[12] Moreover, we show in Section 4.4 that if one is willing to impose additional structure, it is possible to jointly identify the discounting wedge and behavioral attenuation parameters.

[13] Such a Bayesian implementation has been used by, for example, Lubik and Schorfheide (2007).

how a Bayesian approach is natural in situations with possibly weak identification and small sample sizes. We cast our estimation in terms of a linear Gaussian state space model, which allows us to efficiently deal with different sample sizes for various variables as well as possibly mismeasured data. We next describe the state space model in detail before turning to specifics of the estimation.

3.1 The State Space Model

The main econometric problem is that the regressors in the consumption function are endogenous. The unobserved consumption-equation shock ζ_t can move consumption directly and can also move net worth, debt, taxes, income, government spending, interest rates, and other predictors through equilibrium relationships. As discussed in the introduction and illustrated in the population regression exercise in Section 5.3, OLS is biased because ζ_t affects both consumption and equilibrium predictors. **B-HIVE** addresses this problem by estimating the consumption equation jointly with a reduced-form law of motion for the predictors, while using external macroeconomic shocks to isolate variation in the predictors that is orthogonal to ζ_t.

The structural equation is the consumption function. We write the estimating equation as[14]

$$c_t = \phi_0 + \phi_n\, n_{t-1} + \phi_b\, b_{t-1} + \phi_\tau\, \tau_t + \phi_y y_t + \phi_q q_t + \sum_{j\in\{y,g,q\}} \phi_{v^j}\, v_t^j + \zeta_t \tag{18}$$

The simple behavioral model in Section 2 implies

$$\phi_n = 1-\beta, \qquad \phi_\tau = -(1-\beta)(1-\theta), \qquad \theta = 1 + \phi_\tau/\phi_n.$$

In the empirical benchmark, we estimate the coefficients in equation (18), subject to the prior restrictions $\phi_n \in (1-\beta, 1)$ and $\phi_\tau \in (-\phi_n, 0)$. These priors imply a uniform prior for θ on $(0,1)$ (Appendix I).

We interpret $\theta \in [0,1]$ as the degree to which households internalize future taxes. Under the behavioral assumptions in Section 2.2, θ has the structural interpretation of an attenuation in expected future taxes. But as we show in Section 2.3, other causes of non-Ricardianism give alternative structural interpretations for θ. Still, regardless of the specific mechanism, θ is a structural parameter that represents a wedge distorting how consumers respond to future taxes.

[14] y_t and q_t appear separately from v_t^y and v_t^q because Proposition 2 contains the perceived present values $\tilde{v}_t^y$ and $\tilde{v}_t^q$. Mirroring our treatment of taxes, their separate inclusion allows for scaling between perceived expectations of future income and bond prices and their rational counterparts. Current values are known and are not scaled. For example, if perceived discounted future income is a_y times its rational counterpart, then $\tilde{v}_t^y = y_t + a_y(v_t^y - y_t) = (1-a_y)y_t + a_y v_t^y$. This mapping implies the restriction $\phi_y + \phi_{v^y} = \phi_n$. We impose this equality only in Section 4.3's income-sum restriction robustness check.

Let $X_t \in \mathbb{R}^m$ collect the predictors used in equation (18) and its present-value terms. In the benchmark, $X_t = (n_t, b_t, \tau_t, y_t, g_t, q_t)'$ and $m = 6$; the data section and Table 1 describe their construction. Consumption is not included in X_t because it is governed by the structural equation.

We assume that X_t follows a VAR(p), with $p = 4$ in the application, driven by a joint vector of structural shocks:

$$X_t = \mu_X + A_1 X_{t-1} + \cdots + A_p X_{t-p} + G\varepsilon_t, \qquad \varepsilon_t = (\varepsilon_{c,t}, \varepsilon'_{X,t})' \sim N(0, I_{m+1}).$$

Let h_c denote the loading of the normalized consumption-equation innovation in the consumption equation. The matrix G is $m \times (m+1)$. The first shock, $\varepsilon_{c,t}$, is the innovation to the consumption residual, while the remaining m shocks are non-consumption structural shocks. The benchmark consumption residual follows

$$\zeta_t = \rho_\zeta \zeta_{t-1} + h_c \varepsilon_{c,t}.$$

We estimate ρ_ζ jointly with the other parameters. The first column of G need not be zero. Thus current and past consumption innovations can affect X_t, and the right-hand-side variables in equation (18) can be correlated with the consumption residual. This is the source of endogeneity in the state-space system.

The VAR also disciplines the forward-looking terms in the consumption function. Define the companion state

$$\overline{X}_t = (X'_t, X'_{t-1}, \ldots, X'_{t-p+1})'.$$

Then the VAR can be written as

$$\overline{X}_t = \overline{\mu} + F\overline{X}_{t-1} + \overline{G}\varepsilon_t,$$

where F is the transition matrix for the stacked state $\overline{X}_t$. Thus powers of F generate the forecasts used in the present-value terms. We use a common estimated discount factor β for the y, g, and q terms. Absorbing their constant components into ϕ_0, their state-dependent components are

$$v_t^j = s'_j (I_{mp} - \beta F)^{-1} \overline{X}_t, \qquad j \in \{y, g, q\}.$$

The vector s_j selects the relevant element of $\overline{X}_t$. We estimate β jointly with the other state-space parameters. Appendix H gives the matrix construction. Parameter draws that violate the stability restrictions have zero prior density.

The instruments are noisy measurements of the non-consumption shocks:

$$w_t = \mu_w + M_X X_{t-1} + M\varepsilon_t + \eta_t, \qquad \eta_t \sim N(0, \Sigma_\eta).$$

The vector w_t contains the external macro shock series. The term $M_X X_{t-1}$ allows predictable contamination by lagged observables. The matrix M maps structural shocks into the instruments; in our baseline, each row of M has one allowed nonzero loading. The exclusion restriction is that instruments do not load on $\varepsilon_{c,t}$, so the first column of M is restricted to zero. The relevance restriction is that the instruments load on non-consumption shocks that move the endogenous regressors through G.

This setup maps directly into standard IV intuition. Variation in X_t induced by $\varepsilon_{c,t}$ is endogenous because it is correlated with ζ_t. Variation in X_t induced by the non-consumption shocks measured by w_t is identifying variation. These shocks may affect consumption through the variables in X_t and the present-value terms, but they do not enter the consumption residual. This is why external macro shocks can identify the consumption equation even though the predictors in X_t are endogenous.

B-HIVE is limited-information in this sense: it does not estimate a full general equilibrium model. It estimates one structural equation together with a reduced-form law of motion for the variables needed to evaluate that equation and construct expectations. The resulting state-space representation lets us evaluate the likelihood with missing instruments, measurement error in the instruments, weak instruments, and latent present-value objects. The same representation also makes extensions modular. For example, stochastic volatility can be introduced by adding latent volatility states that scale the innovations in the VAR or the measurement equations; conditional on those states, the system remains a state-space model with the same identifying restrictions. We use this likelihood function to conduct Bayesian inference.

Why do we use a Bayesian approach? After all, the previous literature that has pioneered the use of structural shocks as instruments (Barnichon and Mesters, 2020) has used a frequentist approach. First, it transparently allows us to use all available information for the structural parameters via the use of priors. For example, if a parameter is bounded between 0 and 1, as will be the case for our key parameter that governs the degree of Ricardian behavior, then a natural assumption is a uniform prior between 0 and 1. Second, it allows for regularization, which is a natural requirement in macroeconomics, where sample sizes are often relatively small. Recent work by Giannone et al. (2026) emphasizes that prior choices in Bayesian IV models can be informative about instrument relevance, even when they appear diffuse on primitive coefficients. We therefore report the prior implied by our specification for the relevance of each shock instrument in Appendix I, Figure 11.

This prior puts little mass near the two degenerate cases of an irrelevant instrument and a perfectly measured shock, but otherwise leaves a wide range of relevance values. Third, Bayesian methods are already well-suited and supported for estimating state-space models. Finally, the likelihood principle (Berger and Wolpert, 1988) states that everything that can be learned about a given parameter from the data (including any instruments) is contained in the likelihood function. As such, the shape of the posterior distribution summarizes all evidence about the parameters of the model, obviating the need to make specific assumptions about whether parameters are identified to obtain valid statistical inference or needing weak-identification robust methods. Naturally, there is a trade-off: In contrast to semi-parametric frequentist methods, we need to commit to a fully parametric model. However, our model need not be a fully structural model - instead we use a VAR for X_t, a model many macroeconomists will be comfortable with as a reasonable description of macroeconomic dynamics.

Semiparametric quasi-Bayesian approaches, which do not require specifying a law of motion for X_t, do exist (Kwan, 1998; Kim, 2002; Chernozhukov and Hong, 2003; Gallant et al., 2017; Chib et al., 2018). We instead choose a fully parametric Bayesian framework to have access to the toolkit provided by linear Gaussian state space models in combination with the Bayesian approach, for example, for dealing with missing variables. Naturally, there are again tradeoffs when choosing a specific Bayesian approach, but it might be useful to point out that the B-HIVE estimator can easily be extended to accommodate features such as non-Gaussianity, stochastic volatility, or many of the other extensions of the linear Gaussian state space model that have been proposed in the literature.[15] Details on the construction of the likelihood function, the prior distributions we use, and the posterior sampling algorithm can be found in Appendices H, I, and L respectively.

4 Application

In this section we describe our time series data and macro shock instruments. Then we estimate the consumption function with **B-HIVE** and review the results, finding that consumers are non-Ricardian by a large margin.

4.1 Data

In order to estimate the behavioral consumption function (14), we require data on the directly observed variables (e.g. c_t), additional macro variables for a variety of robustness checks, and many macroeconomic shocks to use as instruments.

[15] A key assumption for those extensions is that one can still write them as conditionally Gaussian state space systems after introducing additional variables.

Name	Variable	Data Source	FRED code
Benchmark Time Series			
Household Net Worth	n_t	Fed Financial Accounts	TNWBSHNO
Market Value of Federal Debt	b_t	Dallas Fed	MVMTD027MNFRBDAL
Consumption	c_t	NIPA	PCEC
Personal Taxes	τ_t	NIPA	see note
Government Expenditures	g_t	NIPA	see note
Personal Income	y_t	NIPA	see note
Real 3-Month T-Bill Yield	q_t	Federal Reserve, SPF	TB3MS; see note
Additional Robustness Predictors			
3 Quarterly Macro Factors		FRED-QD (McCracken and Ng, 2020)	see note
Four-Quarter Income Forecast	f_t^y	Michigan Survey of Cons.	
Real-Rate Present-Value Proxy	f_t^q	Aruoba Term Structure	
Hours Growth		BLS	HOANBS
Average Income Tax Rate		Urban-Brookings TPC	
Hand-to-Mouth Income Share		Survey of Consumer Finances	

Table 1: Time Series Data

Notes: The benchmark has $X_t = (n_t, b_t, \tau_t, y_t, g_t, q_t)'$; consumption is the dependent variable, not a VAR predictor. In terms of FRED codes, net taxes are $\tau = W055RC1Q027SBEA - A063RC1Q027SBEA$, expenditures are $g = GEXPND - (GRECPT - \tau)$, and income excluding transfers is $y = PINCOME - A063RC1Q027SBEA$. Net worth and debt use the seasonally adjusted series TNWBSHNO_SA and MVMTD_SA. We divide annual-rate nominal flows by nominal potential GDP (NGDPPOT) and multiply stocks by four before dividing by NGDPPOT. The real yield subtracts the SPF CPI inflation forecast from the Treasury-bill yield. Core variables are not standardized. Additional predictors enter only the robustness specifications described in the text; the three macro factors are the first three principal components of FRED-QD after its recommended transformations.

4.1.1 Directly Observed Variables

Our main time series come from the national accounts. We consistently take the approach of selecting variables that most closely match objects with which households interact directly, e.g. y_t is mapped to *personal income* rather than GDP. Taxes τ_t are *personal taxes* net of transfers; they mainly include income taxes but exclude corporate and production taxes and tariffs. To be consistent with this tax definition, we take g_t as government expenditures net of receipts that are not included in net personal taxes. We subtract the same transfer series, A063RC1Q027SBEA, from personal income. As usual, c_t is total personal consumption expenditures. We take household net worth n_t from the financial accounts, which includes household ownership of government bonds but not their future tax liability. For government debt b_t, we use the market value of federal debt calculated by the Dallas Fed. Both stocks are seasonally adjusted. The main time series are quarterly, so we calculate real interest rates using three-month Treasury bills and subtracting the CPI inflation forecast from the Survey of Professional Forecasters. Table 1 summarizes our main time series, the economic

object they represent, and the source. Our baseline sample runs from 1951:Q4 to 2019:Q4. The state-space model infers missing observations, including real interest rates before the SPF inflation forecasts become available. We end the sample before the COVID pandemic, whose large effects may invalidate our linear approximation.

The nominal consumption time series is detrended and deflated by dividing by the CBO's estimate of nominal potential GDP. We do the same for the other nominal flows in X_t. Since these flows and nominal potential GDP are reported at annual rates, we multiply net worth and debt by four before dividing by nominal potential GDP. Equivalently, stocks and quarterly flows are expressed relative to quarterly potential GDP. We do not standardize consumption or the six benchmark predictors.

4.1.2 Rational Expectation Variables

The consumption function includes the rational present values of income v_t^y, government spending v_t^g, and the interest-rate variable v_t^q. We construct these from the estimated VAR.

The benchmark predictor vector is $X_t = (n_t, b_t, \tau_t, y_t, g_t, q_t)'$. In deviations from its unconditional mean, it follows a VAR(4):

$$X_t = \sum_{k=1}^{4} B_k X_{t-k} + \nu_t \tag{19}$$

where ν_t are i.i.d. reduced-form innovations. The h-period-ahead rational expectation is given recursively by

$$\mathbb{E}_t[X_{t+h}] = \sum_{k=1}^{4} B_k \mathbb{E}_t[X_{t-k+h}]$$

The desired rational expectation variables are given by

$$v_t^y = \sum_{j=0}^{\infty} \beta^j e_y \mathbb{E}_t[X_{t+j}]$$

$$v_t^g = \sum_{j=0}^{\infty} \beta^j e_g \mathbb{E}_t[X_{t+j}]$$

$$v_t^q = \sum_{j=0}^{\infty} \beta^j e_q \mathbb{E}_t[X_{t+j}]$$

where e_y, e_g, and e_q select the corresponding entries of X_t. We estimate the common discount factor β jointly with the VAR. Constants from the unconditional means are absorbed by the consumption intercept.

4.1.3 Behavioral Expectation Variables

The consumption function also includes households' perceived present values of income $\tilde{v}_t^y$ and future bond prices $\tilde{v}_t^q$. These expectations need not be rational and therefore cannot be recovered from subsequent realizations alone.

In our benchmark, we use rational expectations with unknown scaling coefficients as proxies for agents' behavioral expectations. This allows us to include the rational analogs v_t^y and v_t^q directly in our estimation equation, alongside current income and the interest rate.

In another method, we use data from surveys and asset markets to account for household expectations.[16] To construct an income forecast, we multiply current personal income before subtracting transfers by one plus the Michigan Survey of Consumers' expected income growth over the next four quarters and divide by nominal potential GDP. This four-quarter income forecast, f_t^y, replaces the rational income present value in the consumption equation; current income remains a separate regressor.[17] The state-space model infers missing observations before the survey begins in 1976:Q3.

Households' perceived bond-price present value depends on their forecasts of short-term real interest rates. Appendix C gives

$$\tilde{v}_t^q = -\overline{Q}^2 \left(r_t^{(1)} + \sum_{j=1}^{\infty} \beta^j \tilde{\mathbb{E}}_t \left[r_{t+j}^{(1)} \right] \right)$$

We use the Philadelphia Fed's ATSIX fitted real yields (Aruoba, 2020) to construct a proxy f_t^q for this perceived present value. The consumption function also contains the rational present value v_t^q, which enters through the government budget constraint in Proposition 2. We therefore retain v_t^q and replace the separate current-q_t regressor with f_t^q, estimating both coefficients freely. Appendix K describes the proxy's construction.

4.1.4 Instruments

To build our dataset of macro instruments, we collected structural shock series identified by the literature.[18] The benchmark uses 28 instruments from 19 sources, with an unbalanced

[16] This strategy is similar to that of Indarte et al. (2026) who also use surveys of forecasts over limited horizons, combined with rational term structures and modeling assumptions, in order to construct perceived present discounted values in a non-rational consumption function.

[17] The survey reports income forecasts at a single horizon. A perceived AR(1) income process with fixed nonzero persistence would allow this forecast to summarize the expected income path up to scale, but a more flexible process would require forecasts at additional horizons. We treat the four-quarter forecast as a proxy for perceived income wealth and estimate its coefficient freely. The survey also concerns household income rather than the net aggregate income concept in the consumption function.

[18] In related work, Adams and Barrett (2025) apply the monthly subset of these instruments, plus additional monetary shocks, to use an IV method to decompose how MPS depend on contemporaneous interest

sample and gaps. We assign them to six non-consumption shocks: eight monetary-policy instruments, seven government-spending instruments, eight tax instruments, two technology instruments, two oil instruments, and one uncertainty instrument. Table 2 also identifies the six-instrument robustness subset, which uses one instrument per shock.

Shock Source	Method	Number	Notes	Range	Six-IV subset
Monetary Policy Shocks		8			
Jarociński and Karadi (2020)	HFI	2	Target rate and Fed information	1990:M1-2016:M12	
Miranda-Agrippino and Ricco (2021)	HFI	1	Orthogonalized w.r.t. Greenbook	1991:M1-2009:M12	
Bauer and Swanson (2023)	HFI	1	Orthogonalized w.r.t. financial data	1988:M2-2023:M12	
Swanson (2024)	HFI	3	FFR, forward guidance, and LSAP	1988:M2-2023:M12	
Aruoba and Drechsel (2024)	Narrative	1	Natural language processing of Fed documents	1982:M10-2008:M10	✓
Government Spending Shocks		7			
Fisher and Peters (2010)	External	1	Excess returns of defense contractors	1947:Q1-2008:Q4	
Ramey (2016)	Narrative	1	Military news	1947:Q1-2013:Q4	
Romer and Romer (2016)	Narrative	2	Social Security expansions	1951:M1-1991:M12	
Fieldhouse et al. (2018)	Narrative	1	Government housing purchases	1952:M11-2014:M12	
Fieldhouse and Mertens (2023)	Narrative	2	Defense and non-defense R&D	1947:Q1-2021:Q4	✓
Tax/Borrowing Shocks		8			
Leeper et al. (2012)	External	1	Fiscal news from bond markets	1947:Q1-2007:Q4	
Phillot (2025)	HFI	4	Treasury news: 2-, 5-, 10-, and 30-year yields	1998:M10-2020:M01	
Mertens and Ravn (2012)	Narrative	2	Anticipated and unanticipated tax changes	1947:Q1-2007:Q4	
Lieb et al. (2024)	Narrative	1	News from Presidents' speeches	1951:Q4-2007:Q3	✓
Technology Shocks		2			
Fernald (2014)	External	1	Utilization-adjusted TFP	1947:Q2-2024:Q4	
Miranda-Agrippino et al. (2025)	External	1	Patent filing news	1982:M10-2014:M12	✓
Oil Shocks		2			
Kilian (2008)	External	1	OPEC conflict events	1971:Q1-2004:Q3	
Känzig (2021)	HFI	1	Oil supply news	1975:M1-2023:M6	✓
Uncertainty Shocks		1			
Piffer and Podstawski (2018)	HFI	1	Uncertainty shocks from intraday gold prices	1979:M1-2025:M4	✓

Table 2: Structural Shock Instruments

Notes: Number is the count of instruments from each source; italic rows give totals by structural-shock assignment. HFI denotes high-frequency identification. Source ranges can extend beyond the estimation sample, 1951:Q4–2019:Q4. Monthly observations are averaged within quarters, and the resulting instruments are standardized. A checkmark identifies the six-instrument subset; only the defense R&D series is used from Fieldhouse and Mertens (2023).

Four sources identify monetary policy shocks from high-frequency data around policy events: Miranda-Agrippino and Ricco (2021), Jarociński and Karadi (2020) who estimate both an interest rate shock and a Fed information shock, Bauer and Swanson (2023) who expand the set of Fed events used to estimate the shocks, and Swanson (2024) who decomposes the high-frequency asset price movements into a target rate shock, a forward guidance shock, and a large-scale asset purchase shock. A fifth source uses narrative methods: Aruoba and Drechsel (2024) build on ideas in Romer and Romer (2004), using natural language processing of internal documents to capture the Fed's information set.

rate surprises versus news about future policy across many horizons.

For government spending, we began with two series previously collected and harmonized by Ramey (2016). These series use military events in some way as a source of exogenous variation: Fisher and Peters (2010) estimate shocks from excess returns for defense contractor stocks, and Ramey (2011) uses narrative military shocks constructed from periodicals. To these, we add three sources using narrative methods. Romer and Romer (2016) identify permanent and temporary transfer shocks from Social Security expansions, Fieldhouse et al. (2018) use mortgage purchases from federal housing agencies, and Fieldhouse and Mertens (2023) use changes to defense and non-defense federal R&D appropriations.

Two sources of tax shocks also come from the Ramey (2016) collection: Mertens and Ravn (2012) use records of the delay between passage and implementation of tax legislation to isolate anticipated and unanticipated tax changes; Leeper et al. (2012) estimate expected tax changes using spreads between federal and municipal bonds. In addition, we use the Lieb et al. (2024) tax shocks constructed from presidential speeches and four federal borrowing instruments from Phillot (2025), measured at different Treasury maturities around auction announcements.

To measure productivity shocks, we take innovations to the utilization-adjusted TFP series from Fernald (2014), which is regularly updated. To account for news, we include the technology news shocks from Miranda-Agrippino et al. (2025), which uses patent applications orthogonalized with respect to macroeconomic conditions as an instrument for future productivity.

For oil shocks, Kilian (2008) identifies supply disruptions from conflicts in oil-producing countries, and Känzig (2021) uses high-frequency asset price data around OPEC announcements to identify news about future oil supply.

Finally, Piffer and Podstawski (2018) estimate macroeconomic uncertainty shocks using high-frequency movements in gold prices around international financial and political events.

4.2 Results

Our baseline includes current income and the interest-rate variable alongside their rational present values, with freely estimated coefficients (Section 4.1.3). We use all of the shocks in Table 2 as instruments, although we also conduct robustness checks using selected subsets of these shocks.

Figure 1 reports the posterior densities from our baseline specification. The first panel corresponds to the behavioral attenuation θ. Our prior on this parameter (the red line) is uniform on the unit interval, because while theory gives little guidance on the exact value of θ, the main mechanisms for Ricardian non-equivalence discussed in Sections 2.3 and D predict that it falls between zero and one, ranging from complete neglect of future taxes to Ricardian equivalence. The posterior density is overwhelmingly far from one: consumers

are not Ricardian.

The behavioral attenuation θ is implied by $\theta = 1+\phi_\tau/\phi_n$, a transformation of coefficients in equation (18). The second panel corresponds to ϕ_n, the quarterly marginal propensity to consume (MPC) out of wealth. Its prior is uniform on $(1-\beta, 1)$ conditional on the estimated discount factor, rather than imposing $\phi_n = 1-\beta$ (Table 6). The posterior is concentrated near 0.0058. The third panel reports $-\phi_\tau = \phi_n(1-\theta)$, the *MPC out of debt-financed transfers* (MPC-DFT). Its prior is implied by the joint prior on θ and ϕ_n. The estimated MPC-DFT is close to the quarterly wealth MPC because households internalize little of the future tax burden that finances those transfers.

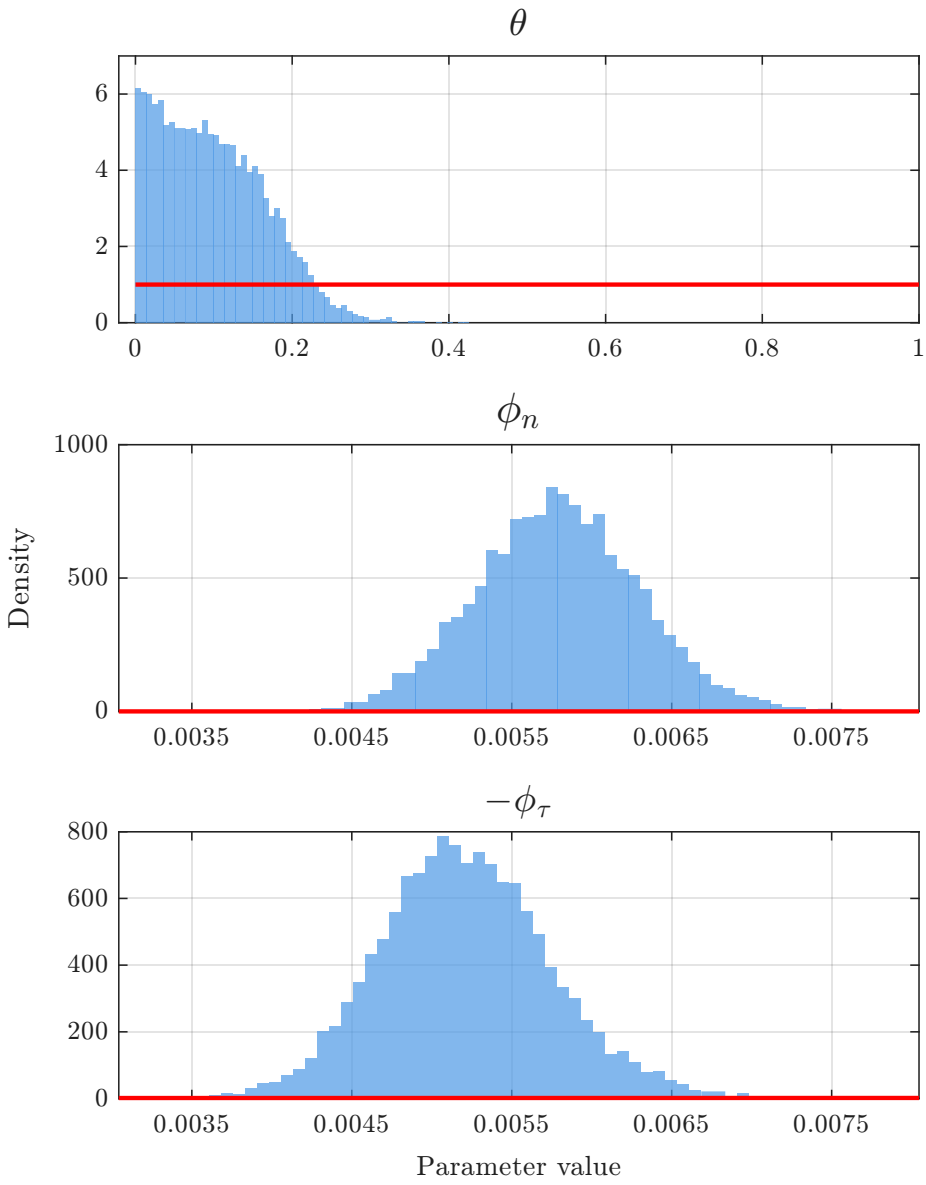


Figure 1: Marginal Posterior Densities for Three Consumption Function Parameters

We use a sequential Monte Carlo sampler (Herbst and Schorfheide, 2015) to approximate the posterior distribution. The panels plot θ, the quarterly wealth MPC ϕ_n, and MPC–DFT $-\phi_\tau$. The relation $\theta = 1 + \phi_\tau/\phi_n$ is applied to each draw; $\phi_n = 1 - \beta$ is not imposed. Priors are shown in red and posteriors in blue. Appendix L describes posterior sampling.

Table 3 reports posterior medians and 90% credible intervals. In the benchmark, the point estimate implies that consumers internalize 9.2% of future taxes, with credible interval $[0.8\%, 21.8\%]$. Most posterior mass lies far below the Ricardian value of one, conditional on the specification and priors. Since the maintained prior is continuous on $(0, 1)$, this is not a posterior-probability test of the point null $\theta = 1$.

The annual-rate wealth MPC $4\phi_n$ is 2.32%, somewhat below the micro evidence using illiquid wealth (Mian et al., 2013; Di Maggio et al., 2020; Chodorow-Reich et al., 2021). MPC–DFT is $-\phi_\tau = 0.519\%$: about half a cent of consumption per dollar transferred in the quarter. Unlike the stock-to-flow wealth coefficient, this flow-to-flow coefficient requires no conversion. It is smaller than the transfer MPCs in Parker et al. (2013, 2022); Baker et al. (2023); Johnson et al. (2006); Hausman (2016); Chetty et al. (2024). Cross-sectional transfer estimates cannot identify this aggregate MPC because they do not measure the consumption response to the shared future tax burden. This is the missing intercept problem discussed in the introduction.

The estimated consumption residual is highly persistent. Its autocorrelation has posterior median 0.948.

Specification	Attenuation (θ)	Annual-rate MPC ($4\phi_n$)	MPC-DFT ($-\phi_\tau$)
Benchmark	0.092 $[0.008, 0.218]$	0.0232 $[0.0198, 0.0265]$	0.00519 $[0.00436, 0.00614]$
Imperfect risk sharing	0.101 $[0.010, 0.368]$	0.0154 $[0.0115, 0.0202]$	0.00343 $[0.00205, 0.00463]$
Distortionary taxes	0.335 $[0.077, 0.674]$	0.0159 $[0.0123, 0.0198]$	0.00266 $[0.00112, 0.00408]$
Nonseparable utility	0.263 $[0.072, 0.447]$	0.0217 $[0.0184, 0.0256]$	0.00397 $[0.00320, 0.00481]$
External habits	0.189 $[0.040, 0.368]$	0.0559 $[0.0485, 0.0648]$	0.01144 $[0.00845, 0.01383]$
Six instruments	0.172 $[0.015, 0.587]$	0.0230 $[0.0185, 0.0272]$	0.00468 $[0.00221, 0.00612]$
Nondurable consumption	0.065 $[0.005, 0.246]$	0.0201 $[0.0173, 0.0231]$	0.00459 $[0.00363, 0.00553]$
Income-sum restriction	0.083 $[0.006, 0.339]$	0.0290 $[0.0241, 0.0350]$	0.00645 $[0.00481, 0.00804]$
Survey forecasts	0.562 $[0.057, 0.954]$	0.0154 $[0.0113, 0.0219]$	0.00169 $[0.00017, 0.00413]$
Larger X_t vector	0.076 $[0.006, 0.258]$	0.0230 $[0.0191, 0.0269]$	0.00520 $[0.00399, 0.00633]$

Table 3: Consumption Function Estimates

Notes: Entries are medians and 5th–95th percentiles under the normalized final SMC weights. We compute $\theta = 1 + \phi_\tau/\phi_n$ for each draw, although this does not hold exactly at the posterior medians. The wealth MPC is expressed at an annual rate as $4\phi_n$; MPC–DFT is the unscaled coefficient $-\phi_\tau$. Section 4.3 and Appendix K describe each alternative.

4.3 Robustness

Table 3 reports robustness checks that change the consumption function specification, forecasting information, or instruments. Appendix K gives additional implementation details for each case.

4.3.1 Alternative Consumption Function Specifications

Our behavioral consumption function depends on three components: two budget constraints and an Euler equation. The latter introduces misspecification risk, so we consider four alternatives that address possible distortions to the Euler equation.

First, we allow for generalized imperfect risk sharing among heterogeneous households. As discussed in Section 2.3, our benchmark consumption function is already consistent with occasionally binding liquidity constraints and a fixed set of hand-to-mouth households. However, it is possible to extend the benchmark to allow for more general forms of imperfect risk sharing. Appendix D.6 shows how the endogenous aggregate Euler wedge adds a perceived present-value term to the consumption function. Using the measured wedge of Berger et al. (2023), we assess whether our estimates depend on omitting distributional changes that affect aggregate intertemporal decisions. We do not find evidence that they do. The posterior median of θ is 0.101, close to the benchmark, with a 90% credible interval of $[0.010, 0.368]$.

Second, distortionary taxes may lead to non-Ricardian behavior if they enter the Euler equation; we account for this by including the present value of tax rates (Appendix D.4). Third, we allow the marginal utility of consumption to be affected by the labor-leisure choice, which adds the present value of hours growth (Appendix D.2); this is the "non-separable utility" line in Table 3. These specifications give median values of θ of 0.335 and 0.263, respectively, with intervals that overlap the benchmark's.

Finally, we consider external consumption habits (Appendix D.3), which add a weighted average of past consumption and retain the benchmark tax–wealth relationship. This specification implies larger MPCs and a median θ of 0.189, with a 90% interval of $[0.040, 0.368]$.

4.3.2 Alternative Implementation Choices

B-HIVE requires that our instruments satisfy the exclusion restriction, but there may be concerns that some instruments do not, especially when we use so many. Therefore, we repeat our baseline specification using only one carefully selected instrument per shock type, marked in Table 2.[19] This conservative specification gives a posterior median of

[19]The six instruments are from Fieldhouse and Mertens (2023); Aruoba and Drechsel (2024); Lieb et al. (2024); Känzig (2021); Miranda-Agrippino et al. (2025); Piffer and Podstawski (2018). We use the defense

$\theta = 0.172$, with a 90% credible interval of $[0.015, 0.587]$. The estimate is larger and less precise than in our baseline, but still implies that households internalize only about 17% of future taxes, which increases our confidence in the non-Ricardian conclusion we draw when using the larger instrument set. The annual-rate wealth MPC is virtually unchanged (2.30% versus 2.32%), while the MPC out of debt-financed transfers is slightly lower (0.468% versus 0.519%).

We also rerun our benchmark method using non-durable consumption instead of the standard aggregate. This definition implies a lower behavioral attenuation parameter θ, with a posterior median of 0.065 and a 90% credible interval of $[0.005, 0.246]$. Both MPCs are slightly smaller as one would expect: the annual-rate wealth MPC is 2.01% and the MPC out of debt-financed transfers is 0.459%. The credible interval for θ is slightly wider than in the benchmark, but the conclusion that households are not very Ricardian is unchanged.

The consumption function (14) implies additional restrictions between coefficients. We do not impose such restrictions in the benchmark, because not every cause of non-Ricardianism implies the same restrictions (see Appendix D.5). Still, we run estimate one alternative where we imposing an additional income-sum restriction that the perceived present-value of income and net worth have the same MPC coefficient. This leaves θ close to the benchmark while increasing both MPCs.

We next consider econometric alternatives. Our benchmark accounts for perceived income and bond-price present values using rational forecast sums and separate current-value terms. We check whether our conclusions depend on this treatment of household expectations by using survey income forecasts and a proxy constructed from the yield curve. The income forecast replaces v_t^y, and the term-structure proxy replaces current q_t; the rational present values of government spending and interest rates remain. This specification gives a higher estimate of θ, but with much greater uncertainty: the posterior median is 0.562, with a 90% interval of $[0.057, 0.954]$. Both MPCs are smaller than in the benchmark.

How sensitive are our results to the information set used to form expectations? Our benchmark vector X_t of predictor variables contains six series (see Section 4.1.2). Expanding the forecasting vector to fourteen variables leaves both MPCs nearly unchanged and gives a median θ of 0.076.

We report additional robustness checks in Appendix J. Across the board, they do not change our central conclusion. We do not use any SVAR-based shocks as instruments in our benchmark, as they may covary with other structural shocks if their identifying assumptions fail. Appendix J.1 assesses sensitivity to this choice and to the sources of identifying variation using alternative sets of instruments. We also test in Appendix J.2 whether our analysis is sensitive to our detrending method.

R&D series from Fieldhouse and Mertens (2023).

4.4 Non-Rational Expectations vs. Discounting

In Section 2, we argued that multiple different causes of non-Ricardian behavior lead to the same reduced-form consumption function, which we estimated without regard to the underlying cause. The specific cause changed only the interpretation of the behavioral attenuation parameter θ.

Moreover, our general representation of the consumption function can nest multiple causes of non-Ricardianism simultaneously. Our baseline method only identifies their effect through the composite wedge θ. However, by imposing additional structure, we can separately attribute non-Ricardian behavior to two causes: non-rational expectations and a discounting wedge ω. This is because non-rational expectations only affects consumption out of future taxes, income and so forth. But a discounting wedge also affects consumption out of current financial net worth.

To implement this decomposition, we maintain the light conditions from Proposition 2, but must make additional assumptions. First, we assume that households are either sophisticated or naive cognitive discounters.[20] Second, we assume that the present value of real taxes follows an AR(1), as in Section 2.3. Third, we must impose a value for β, the government's discount factor.

Proposition 10 in Appendix D.7 derives the consumption function in both of these cases: it matches the usual discounting wedge consumption function (Proposition 3) except the θ term becomes

$$\theta_{cd} = \begin{cases} \omega\lambda\frac{1-\beta\rho_\tau}{1-\beta\omega\lambda\rho_\tau} & \text{sophisticated cognitive discounting} \\ \omega\lambda\frac{1-\beta\rho_\tau}{1-\beta\omega\rho_\tau} & \text{naive cognitive discounting} \end{cases} \tag{20}$$

where λ is the cognitive discounting attenuation parameter. By taking a stance on the government discount factor β, and the autocorrelation of the present value of taxes ρ_τ, we can identify ω and λ from the estimated MPC $\phi_n = (1 - \beta\omega)$ and MPC-DFT $-\phi_\tau = (1 - \beta\omega)(1 - \theta_{cd})$.

Table 4 presents the results, calibrating the government's discount factor to $\beta = 0.9975$ (a 1% annual real return) and setting $\rho_\tau = 0.978$ for the persistence of the present value of taxes.[21] The calibrated government discount factor is distinct from the discount factor

[20] In both cases, cognitive discounting implies that future expectations are attenuated by a parameter λ. Sophisticated cognitive discounters recognize that they will form expectations in this way in the future, while naive households act as if they will have rational expectations in the future. See Appendix B for further details. Adams (2026) provides a more technical treatment of sophistication versus naivete in behavioral expectations.

[21] The calibration of ρ_τ uses a B-HIVE estimate of the autocorrelation of v_t^τ. In the risk-free-debt model, this is equivalent to the autocorrelation of debt. In the data, real debt is risky (Appendix D.1), so debt persistence need not equal the persistence of the present discounted value of taxes. The latter is the relevant object for this decomposition.

Mechanism	λ	ω
Naive Cognitive Discounting	0.0920	
Sophisticated Cognitive Discounting	0.8056	
Discounting Wedge + Rational		0.8056
Discounting Wedge + Naive C.D.	0.1045	0.9967
Discounting Wedge + Sophisticated C.D.	0.8083	0.9967

Table 4: Implied Parameter Values for Different Non-Ricardian Mechanisms

Notes: The table evaluates the mappings in Proposition 10 at the benchmark posterior medians $\theta = 0.0920$ and $\phi_n = 0.00580$, with parameters $\beta = 0.9975$ and $\rho_\tau = 0.978$. Rows (1)–(3) match only θ; rows (4)–(5) match both θ and the quarterly wealth MPC ϕ_n.

estimated to construct forecast sums in the benchmark. We evaluate the mappings at the benchmark posterior medians of θ and ϕ_n. First, naive cognitive discounting without a discounting wedge maps λ directly to θ. Second, sophisticated cognitive discounting requires a larger λ: attenuation compounds over the horizon, and future taxes are highly persistent. Third, a discounting wedge under rational expectations produces the same mapping as sophisticated cognitive discounting, but does not match the estimated MPC. Finally, we use equation (20) to match both θ and the quarterly wealth MPC under each behavioral assumption.

The results show that a discounting wedge alone cannot match both estimates. Matching θ with a discounting wedge alone requires $\omega = 0.8056$, which implies an annual-rate wealth MPC of 78.55%, compared with the estimated 2.32%. Matching the wealth MPC instead requires $\omega = 0.9967$, leaving most of the attenuation to non-rational expectations. The required cognitive discounting parameter is $\lambda = 0.1045$ for naive households and $\lambda = 0.8083$ for sophisticated households. This decomposition depends on the maintained calibrations and expectations assumptions. Its qualitative conclusion accords with Eichenbaum et al. (2025): intertemporal wedges alone are insufficient to explain the estimated degree of non-Ricardian behavior.

5 The Aggregate Consequences of Non-Ricardian Behavior

In this section, we study a two-country general equilibrium model with non-Ricardian households.[22] The purpose of this section is threefold.

First, what are the aggregate consequences of the non-Ricardian behavior that we have

[22]This approach has a long history in the twin deficits literature, following Frenkel and Razin (1986).

documented? To answer this question, we use our time series estimate of θ to quantify how much of the long-run decline in the US NFA position and current account can be explained by the increase in government debt.

Second, we show how long-run trends in external balances depend on the degree of non-Ricardianism. This provides a separate type of evidence to validate our estimates of θ identified using short-run variation.

Third, we use the model for a regression exercise, to demonstrate the econometric issues discussed previously. The exercise illustrates why OLS and IVs constructed from lagged aggregates fail to estimate the consumption function, while structural shocks can be used as instruments.

Appendix E presents a simplified small open economy version of the model that can be solved with pen and paper, and retains the main intuition and quantitative implications. Appendix F presents complete implementation details for the full general equilibrium model.

5.1 Model Assumptions

The economy contains representative domestic and foreign households, competitive domestic firms, and a government. The government finances exogenous spending with lump-sum taxes and one-period risk-free debt that is traded internationally.[23]

5.1.1 Households

The domestic household's preferences over current and future consumption are represented by

$$\tilde{\mathbb{E}}_t \left[\sum_{s=0}^{\infty} \beta^s u(C_{t+s}) \right] \tag{21}$$

where $\beta \in (0,1)$ is the household's discount factor, $u(\cdot)$ is increasing, strictly concave, and differentiable, and $\tilde{\mathbb{E}}_t$ is the household's subjective expectation.

The domestic household owns the domestic factors of production, so its income includes domestic output Y_t. It holds two assets: the capital stock K_t, which depreciates at rate $\delta \in (0,1)$, and risk-free bonds B_t, which trade at price Q_t. Its budget constraint is

$$Y_t + (1-\delta)K_{t-1} + B_{t-1} = C_t + T_t + K_t + Q_t B_t \tag{22}$$

The domestic household's savings decisions are characterized by two Euler equations. The

[23] We solve the linearized model using BEET (Adams, 2024), a toolkit for dynamic models with behavioral expectations.

Euler equation for bonds is

$$Q_t = \beta \tilde{\mathbb{E}}_t \left[\frac{u'(C_{t+1})}{u'(C_t)} \right] \tag{23}$$

while the Euler equation for capital is

$$1 = \beta \tilde{\mathbb{E}}_t \left[\frac{u'(C_{t+1})}{u'(C_t)} \left(R_{t+1} + 1 - \delta \right) \right] \tag{24}$$

where R_t is the rental rate.

Foreign households have the same preferences and expectations as domestic households. The foreign household consumes C_t^*, receives exogenous endowment income Y_t^* and holds bonds B_t^*. Its budget constraint is

$$Y_t^* + B_{t-1}^* = C_t^* + Q_t B_t^* \tag{25}$$

and its Euler equation is

$$Q_t = \beta \tilde{\mathbb{E}}_t \left[\frac{u'(C_{t+1}^*)}{u'(C_t^*)} \right] \tag{26}$$

Thus B_t denotes the domestic bond position, while B_t^* denotes the foreign bond position. Total government debt is $B_t + B_t^*$. Because B_t^* is a claim to one unit of consumption in period $t+1$, domestic NFA at the end of period t are

$$\text{NFA}_t = -Q_t B_t^* \tag{27}$$

In this model, the current account balance is the increase in NFA

$$\text{CA}_t = \text{NFA}_t - \text{NFA}_{t-1} = Q_{t-1} B_{t-1}^* - Q_t B_t^* = C_t^* - Y_t^* - (1 - Q_{t-1}) B_{t-1}^* \tag{28}$$

Thus the current account balance equals the trade balance less net interest paid to the foreign household. Because the foreign household does not pay domestic taxes, the behavioral forecasting mistakes do not directly affect the foreign household's perceived budget constraint, but do affect its forecasts of equilibrium prices.

5.1.2 Production

Domestic firms produce output using capital owned by domestic households. Production and the rental rate of capital are

$$Y_t = K_{t-1}^{\alpha} A_t^{1-\alpha} \tag{29}$$

$$R_t = \alpha \frac{Y_t}{K_{t-1}} \tag{30}$$

where A_t is productivity and $\alpha \in (0,1)$ is the capital share. The capital stock evolves according to

$$K_t = (1-\delta)K_{t-1} + I_t \tag{31}$$

Productivity grows at the constant rate g_A

$$A_{t+1} = (1+g_A)A_t \tag{32}$$

g_A is also the balanced growth rate of output, so for symmetry we assume that foreign endowment income grows at the same rate:

$$Y^*_{t+1} = (1+g_A)Y^*_t \tag{33}$$

5.1.3 Government

Government spending is exogenous and G_t/A_t is constant. The government finances its spending with taxes and debt held by domestic and foreign households. Its budget constraint is

$$G_t + B_{t-1} + B^*_{t-1} = T_t + Q_t(B_t + B^*_t) \tag{34}$$

Total government debt relative to productivity follows an AR(1) process:

$$\frac{B_t + B^*_t}{A_t} - \bar{b} = \rho_B \left(\frac{B_{t-1} + B^*_{t-1}}{A_{t-1}} - \bar{b} \right) + \epsilon^B_t \tag{35}$$

where $\rho_B \in (0,1)$, $\bar{b}$ is the long-run value of productivity-scaled debt, and ϵ^B_t is an unexpected iid shock. Total debt is exogenous, while taxes adjust to satisfy equation (34).

Let V^T_t and $\tilde{V}^T_t$ denote the rational and perceived present values of domestic taxes, which satisfy

$$V^T_t = T_t + Q_t \mathbb{E}_t[V^T_{t+1}] \qquad \tilde{V}^T_t = T_t + Q_t \tilde{\mathbb{E}}_t[\tilde{V}^T_{t+1}] \tag{36}$$

Current taxes are observed, but the domestic household need not fully account for the future taxes associated with government debt. Let v^τ_t and $\tilde{v}^\tau_t$ denote the first-order deviations of V^T_t/A_t and $\tilde{V}^T_t/A_t$ from their balanced growth paths (BGPs). As in the behavioral consumption function in equation (13), we impose

$$\tilde{\mathbb{E}}_t[\tilde{v}^\tau_{t+1}] = \theta \mathbb{E}_t[v^\tau_{t+1}] \tag{37}$$

where $\theta \in [0,1]$. When $\theta = 1$, the domestic household rationally forecasts its future tax burden. When $\theta < 1$, its forecast is attenuated. The foreign household pays no domestic taxes, so this restriction directly affects only the domestic household.

5.1.4 Equilibrium Definition

A *behavioral expectations equilibrium* is a sequence of prices and quantities satisfying:

1. The domestic and foreign households maximize expected utility: their budget constraints, Euler equations, and transversality conditions hold
2. Domestic firms produce by equation (29), satisfy equation (30), and capital evolves according to equation (31)
3. Fiscal variables satisfy the government budget constraint, debt process, and government solvency condition
4. The goods and government bond markets clear

The bond price Q_t is endogenous and ensures that the market for government bonds clears.

5.2 A Fifty-Year Increase in Government Debt

How much of the long-run decline in the US NFA position can be explained by the increase in government debt? To answer this question, we apply our model to a 50 year period where federal debt held by the public rose from 26% of GDP in 1976 to 98% in 2025. At the same time, NFA/GDP fell by 75.4 percentage points. We feed a path of fiscal shocks replicating the increase in debt into the linearized model and use our time series estimate of θ to calculate the implied average budget deficit, average current account balance, and change in NFA. Thus this exercise uses long-run movements in fiscal and external balances as a validation check of the θ that we estimated using short-run variation.

We parameterize the economy setting $\theta = 0.092$ per our baseline estimate, and otherwise choose a standard neoclassical calibration (Table 5). We set $\beta = 0.990$, corresponding to an annual discount factor of 0.961 and a calibrated quarterly MPC of 0.010.[24] We set $\alpha = 0.39$ using the average capital share and set $g_A = 0.0069$, which corresponds to annual real GDP growth of 2.78%. We set $\delta = 0.022$ using the average depreciation rate of private nonresidential fixed assets. To set the scale of the foreign economy, we denote the BGP ratio as $\mu \equiv \frac{Y^*}{Y}$, which we set to $\mu = 3.94$ using the average ratio of rest-of-world to US output-side real GDP in PWT 11.0 (Feenstra et al., 2015) over the 50 year period. We set the debt autocorrelation $\rho_B = 0.978$ to match our **B-HIVE** estimate. Finally, we set $\gamma = 1$ so that households have log utility.

The economy is initialized in 1976 on a BGP with government debt equal to 26% of annualized GDP and NFA equal to 4.3% of annualized GDP. We then choose debt shocks

[24] This MPC is slightly larger than our point estimate from the time series. We choose this value in order to imply an annualized MPC out of net worth that is closer to the 3% – 7% estimates from the literature discussed earlier. Appendix G tests parameter sensitivity.

to match a smooth increase in government debt/GDP of 72 percentage points over 50 years, with target path:

$$\frac{B_t + B_t^*}{4Y_t} = 0.26 + 0.72\frac{t}{200} \qquad t = 0, \ldots, 200 \tag{38}$$

The factor of 4 converts quarterly output to annualized GDP. We evaluate the ratio using endogenous output.

Government debt is AR(1), given by equation (35), so households do not expect the 50-year increase; instead they expect debt to revert to the long-run path. Accordingly, we implement the 50-year debt path with a sequence of unexpected debt shocks ϵ_t^B so that debt/GDP matches the path in equation (38). After year 50, the shocks stop and debt reverts towards its balanced-growth value. Table 5 reports the resulting equilibrium behavior for both the non-Ricardian model and a Ricardian calibration (where $\theta = 1$.)

Calibration

Parameter	Symbol	Value	Discipline
Behavioral attenuation	θ	0.09	Baseline estimate
Discount factor	β	0.99	Calibrated MPC
Capital share	α	0.39	Average U.S. capital share
Productivity growth	g_A	0.69%	Annual real GDP growth of 2.78%
Relative size of the foreign economy	μ	3.94	Average rest-of-world/U.S. GDP
Debt persistence	ρ_B	0.978	Autocorrelation of v_t^τ
Relative risk aversion	γ	1	Standard value
Depreciation rate	δ	2.2%	Annual rate of 8.6%

Fifty-Year Debt Exercise

Statistic	Data	Non-Ricardian Model	Ricardian Model
Initial government debt/GDP	26%	26%	26%
Government debt/GDP in year 50	98%	98%	98%
Average budget deficit/GDP	3.78%	3.08%	3.08%
Average current account/GDP	−2.40%	−1.43%	0.12%
50-year investment/GDP trend	0.23%	−0.11%	0.00%
Initial NFA/GDP	4.3%	4.3%	4.3%
NFA/GDP in year 50	−71.1%	−49.2%	4.3%

Table 5: Quarterly Calibration and Fifty-Year Outcomes

Notes: Data cover the United States from 1976–2025. The model is initialized at a BGP with an NFA position matching that of 1976. The budget deficit includes net interest. Investment is private nonresidential fixed investment, and NFA is measured by the U.S. net international investment position. Data averages weight years equally, while model averages weight quarters equally. The 50-year investment/GDP change is an estimated linear time trend. All model ratios use endogenous GDP; debt and NFA use annualized GDP. GDP ratios are reported in percent, and investment/GDP trends in percentage points, also denoted by %. Data are from the NIPA, BEA international and fixed assets accounts, OMB, and Penn World Table 11.0.

Given our estimated degree of non-Ricardian behavior, increased government borrowing explains a majority of the observed decline in NFA over the last 50 years. Table 5 shows that at $\theta = 0.092$, NFA fall from 4.3% to -49.2% of GDP. The resulting decline of 53.5 percentage points accounts for 71.0% of the observed 75.4 percentage point decline.[25] Similarly, the current account balance defined in equation (28) averages -1.43% of GDP, compared with -2.40% in the data.[26] Figure 2 plots these paths: as debt rises (first panel) the interest expense requires budget deficits to grow (second panel).[27] The government borrows from abroad to finance its deficits (third panel) and this borrowing accumulates as a shrinking NFA position (fourth panel). After 50 years, if the government runs primary surpluses to reduce its debt burden, the NFA and current account remain permanently low, while the government must run primary surpluses to pay interest abroad (dashed lines).

In contrast, if households are Ricardian, the 50-year increase in government debt has zero effect: the NFA and current account balance remain on their pre-1976 BGPs. Thus, we attribute the majority of the trends in these external balances to government borrowing and non-Ricardian households.

There is little crowding out of capital in this open economy; a fitted linear trend in investment/GDP implies a decline of only 0.11 percentage points over the 50-year borrowing episode. There is no sign of substantial crowding out in the macro time series (investment-to-GDP slightly rises in the data) so this is one reason it is crucial to consider the open economy element when modeling non-Ricardian behavior. Under a non-Ricardian closed economy, government borrowing must crowd out capital. Indeed, in the closed economy analog of this model, the decline in the investment-GDP ratio is about six times larger.[28] But in the open economy, foreign households absorb a majority of the increase in government debt, so there is only a small reduction in domestic capital accumulation.[29]

The matched debt path implies an average budget deficit of 3.08% of GDP, compared with 3.78% in the data. This discrepancy is mainly because the real value of debt is risky in the data, where shocks (such as unexpected inflation) can reduce the debt-to-GDP ratio. In contrast, real debt in the model is riskfree, so smaller deficits are consistent with the same

[25] At the bounds of the 90% credible interval, $\theta = 0.008$ and $\theta = 0.218$, the model accounts for between 61.1% and 77.5% of the observed decline in NFA. Our model abstracts from many elements that may explain the remaining gap. In particular, changes in observed NFA reflect valuation effects in addition to current account balances, and valuation effects have been especially large in recent years (Gourinchas and Rey, 2014; Atkeson et al., 2025).

[26] Quantitatively, this result is comparable to that of Kumhof and Laxton (2013), who study a two-country OLG model, and find that a permanent increase in the budget deficit produces a long-run increase in the current account deficit about 75% as large.

[27] The paths of budget deficits are similar but not exactly the same under non-Ricardian and Ricardian parameterizations. This is because of general equilibrium effects on interest rates and GDP, which are small.

[28] See Appendix G.2.

[29] This pattern is consistent with Feyrer and Shambaugh (2012) who find that roughly half of a U.S. tax increase is transmitted through the current account, with little effect on domestic investment.

debt burden.[30] Whether households are Ricardian or not has little effect on the deficits implied by the exogenous debt path, because general equilibrium interest rate effects are small.

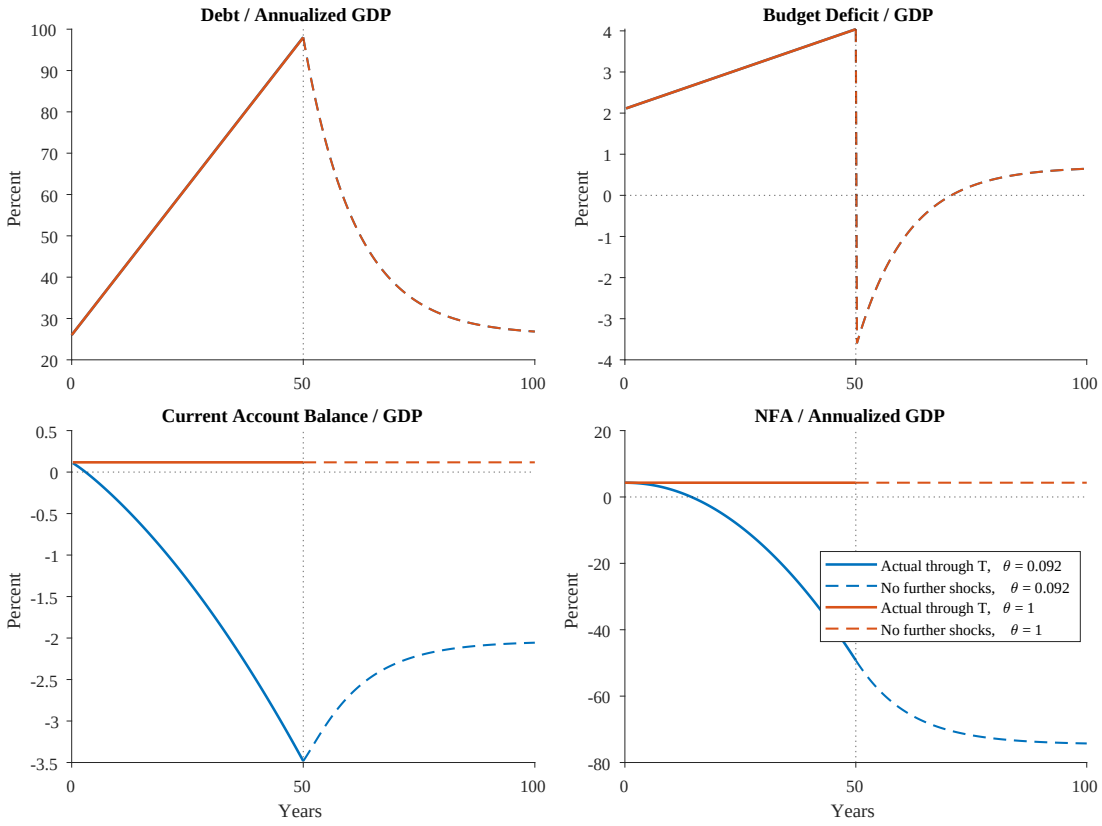


Figure 2: A Fifty-Year Increase in Government Debt

Government debt/GDP rises from 26% to 98% over the first 50 years. The blue lines set $\theta = 0.092$, while the orange lines impose Ricardian equivalence with $\theta = 1$. The vertical dotted line marks the final debt surprise. Solid lines show outcomes through year 50, while dashed lines show the subsequent outcomes with no further shocks. Debt and NFA ratios are reported with annualized GDP.

How do these large effects on the external accounts depend on the degree of non-Ricardianism? To answer this question, Figure 3 varies θ while holding the debt path and all other parameters fixed. As θ declines, domestic households save less to offset future taxes, shifting more of the increase in government debt abraod, reducing the current account balance and amplifying the NFA decline. Under Ricardian equivalence ($\theta = 1$), domestic households absorb the entire increase in debt, so NFA remain at 4.3% of GDP. When $\theta = 0$, NFA fall by 58.9 percentage points, to -54.6% of GDP. The average current account balance moves from 0.12% of GDP under Ricardian equivalence to -1.59% when

[30]Thus, matching debt rather than deficits is a conservative choice; if we instead matched average deficits, our model would produce larger current account deficits and NFA declines.

$\theta = 0$. In this otherwise neoclassical model, θ is the only parameter that can yield such large changes to the NFA.[31]

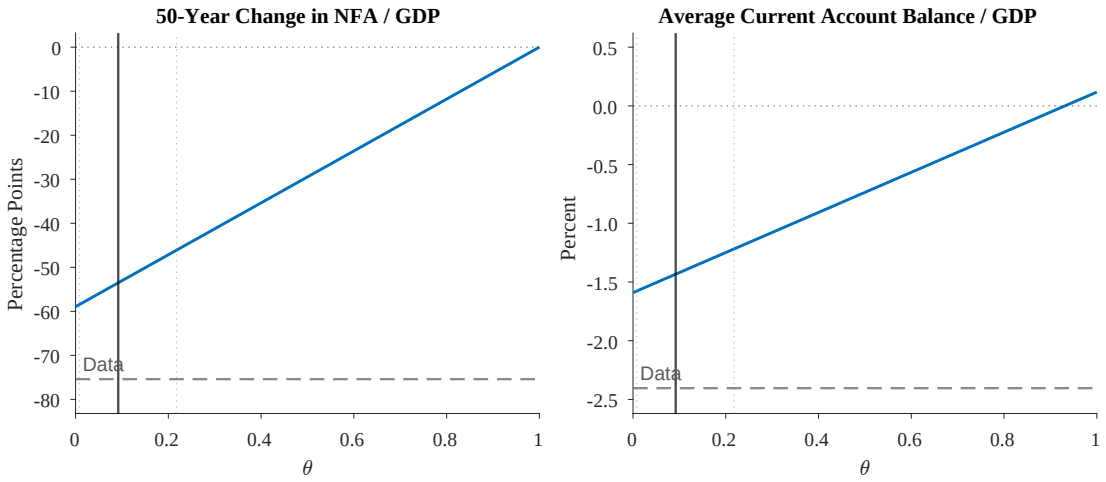


Figure 3: Ricardianism and External Balances

The panels report the change in NFA and the average current account balance over the first 50 years for different values of θ. The NFA change is reported in percentage points of GDP, and the average current account balance as a percent of GDP. The average current account balance gives equal weight to each quarter. The solid black vertical line denotes the posterior median $\theta = 0.092$, while the dotted gray vertical lines denote the endpoints of the 90% credible interval, $\theta = 0.008$ and $\theta = 0.218$. The dashed horizontal lines report the empirical values. All other parameters and the government debt path are held fixed.

While the NFA decline is similar in magnitude to the increase in government debt, it does not match it one-for-one, even if households are fully myopic ($\theta = 0$). Domestic households still absorb part of the increase in debt because non-Ricardian households are not hand-to-mouth. When $\theta = 0$, domestic households treat government bonds as net wealth, but their quarterly marginal propensity to consume out of wealth is only $1 - \beta = 0.010$. They receive the debt-financed transfers and initially save most of them. Over time, consumption out of accumulated bond wealth increases the need for foreign financing. Foreign households finance much of this additional consumption, while the reduction in capital accumulation remains small. The size of the foreign economy affects the bond price response required to induce foreign saving, but does not impose a portfolio share proportional to foreign income.

5.3 Population Regression Exercise

In this section, we use the linearized model to demonstrate the challenges involved in estimating Ricardianism, why OLS fails, and how IVs can resolve the problem.

[31] Appendix G reports how these outcomes vary with β, γ, ρ_B, and μ. None of these parameters affect the external accounts when households are Ricardian. When θ is small, only β has substantial effects: a smaller β value implies a larger MPC, which increases how much of a debt-financed transfer must be financed from abroad. Thus our small MPC is conservative, and larger MPCs increase the NFA decline.

To introduce these challenges, we modify the model by adding TFP, IST, risk, and demand shocks. The corresponding linearized equilibrium conditions and AR(1) shock processes are given in Appendix F. This section is only intended to be illustrative, and foreign borrowing is not needed to demonstrate the econometric problems, so we use the closed-economy limit of the model and linearize around zero government debt and NFA. We otherwise maintain the calibration in Table 5. In the linear equations, lowercase variables denote deviations from the BGP after quantities are scaled by productivity, except that τ_t denotes the deviation in taxes. Uppercase variables with bars denote balanced-growth values. In this model, the behavioral consumption function is

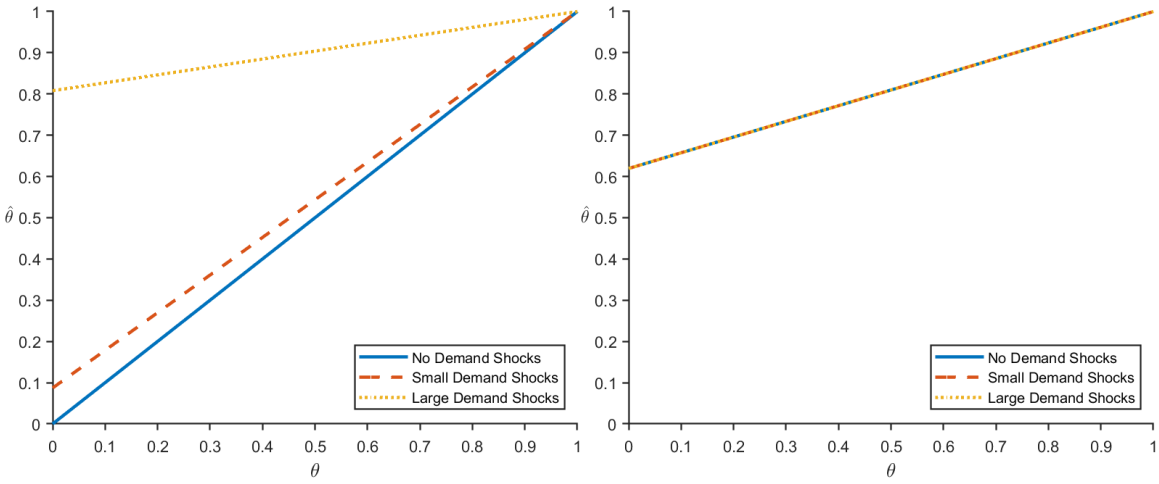
$$c_t = (1-\beta_A)\left(\frac{n_{t-1}}{1+g_A} - \frac{\theta b_{t-1}}{1+g_A} + \tilde{v}_t^y - (1-\theta)\tau_t\right) + \frac{(1+g_A)\overline{C}}{\gamma}\tilde{v}_t^q + \zeta_t^c \tag{39}$$

where $\beta_A = \beta(1+g_A)^{1-\gamma}$, n_{t-1} is domestic financial net worth, b_{t-1} is government debt, $\tilde{v}_t^y$ is perceived non-financial wealth, $\tilde{v}_t^q$ is the perceived consol value, and ζ_t^c is a demand shock.

Are government bonds net wealth? Behavioral households act as if they are. Equation (39) implies that if households do not have rational expectations, current bond holdings b_{t-1} affect consumption, *ceteris paribus*. Bonds appear in financial assets n_{t-1}, but households only net out the share θ of their bond holdings when making their consumption decision.

The residual ζ_t^c introduces an omitted variable problem. Even if the econometrician observes n_{t-1}, b_{t-1}, $\tilde{v}_t^y$, τ_t, and $\tilde{v}_t^q$, the demand shock affects consumption and the time-t endogenous variables, so OLS is biased. Even though taxes are exogenous, omitted variable bias still affects the coefficient on τ_t because we are estimating the effect of variation in taxes orthogonal to the other regressors.

Figure 4 reports the results. Panel 4a plots the population OLS coefficient on τ_t. The true coefficient is $-(1-\beta_A)(1-\theta)$, so we transform the coefficient using the true β_A to give an implied value of θ. When there are no demand shocks, the OLS coefficient implies the true θ. When demand shocks are present, the implied values are biased, and the bias increases with their variance.



(a) OLS

(b) OLS: Tax Shocks Only

(c) IV (Lagged Variables)

(d) IV (TFP, IST, and Risk Shocks)

Figure 4: Population Regression Coefficients for the Consumption Function

Each panel plots the population coefficient implied by a different regression of the consumption function in the general equilibrium model. The x-axis gives the true value of θ, which varies across model economies. Because the coefficients are calculated from the population moments of the model, there is no sampling variation. The solid blue lines ("No Demand Shocks") set $\sigma_{\zeta^c} = 0$, the dashed red lines ("Small Demand Shocks") set $\sigma_{\zeta^c} = 3\overline{C}$, and the dotted yellow lines ("Large Demand Shocks") set $\sigma_{\zeta^c} = 10\overline{C}$. We set the remaining shock standard deviations to one, except for $\sigma_\tau = 10$, and set $\rho_c = \rho_a = \rho_k = \rho_r = 0.5$. Each method directly estimates $-(1-\beta_A)(1-\theta)$, so each coefficient is transformed into an implied value of θ, plotted on the y-axis. The lagged-IV panel truncates values above $\hat{\theta} = 2$.

Applied macroeconomists have identified plausibly exogenous tax shocks with a number of methods (e.g. Mertens and Ravn, 2012 and Leeper et al., 2012). These shocks should be orthogonal to the demand shocks that create the omitted variable bias documented in Panel 4a. Can θ be identified by simply regressing consumption on exogenous tax shocks? No. Tax shocks also affect consumption through other channels, including interest rates. With this approach, those channels are omitted variables. Panel 4b shows that the resulting estimates are attenuated towards Ricardianism.

Using lagged macroeconomic variables as instruments does not work either. Early in the Ricardian equivalence literature it was clear that OLS might be biased; Hayashi (1982) and Feldstein (1982) proposed estimating the consumption function by IV with lagged variables. This solution is valid if demand shocks are uncorrelated. But in the model, demand shocks are autocorrelated. This persistence makes lagged observables correlated with the current consumption-equation error, so they do not satisfy the exclusion restriction. Panel 4c plots the population coefficients using lagged consumption, capital, and consol values as instruments. The method is consistent when there are no demand shocks, but its bias increases with the demand shock variance.

Macro shocks as instrumental variables can resolve the problem. We are interested in the coefficient on taxes, rather than separately identifying every coefficient in the consumption function. Therefore, we group the endogenous variables using their model-implied coefficients as $n_{t-1}/(1+g_A) + \tilde{v}_t^y + (1+g_A)\overline{C}\tilde{v}_t^q/(\gamma(1-\beta_A))$. Current taxes and lagged government debt enter as exogenous regressors. We use the TFP, IST, and risk innovations ϵ_t^a, ϵ_t^k, and ϵ_t^r as IVs for the endogenous composite. These IVs affect the composite, but not taxes, debt, or demand shocks. Panel 4d shows that regardless of the presence of demand shocks, the population IV coefficient recovers the correct effect of a tax change.

6 Conclusion

How Ricardian are we? Our time series estimates suggest: not very Ricardian. This is a challenging question to answer because it requires estimating the consumption function, for which microeconomic evidence is insufficient due to the missing intercept problem, while macroeconomic evidence suffers from omitted variables. To address these challenges, we derived a Bayesian limited information method that used structural macroeconomic shocks as IVs.

Our treatment of non-Ricardianism has broad implications. Because taxes distort consumption and investment, the degree of non-Ricardianism matters for public debt management, business cycle smoothing, optimal tax planning, monetary policy, and an endless variety of further topics. Moreover, our behavioral method of modeling non-equivalence

is easily embedded into other macroeconomic models that feature other frictions and are not necessarily behavioral. To demonstrate, we studied how non-Ricardianism affects a calibrated open economy, and found evidence for a strong twin deficits effect: with our estimated non-Ricardian parameter, the increase in government borrowing over the last 50 years explains a majority of the simultaneous NFA decline.

Our **B-HIVE** estimation method has broad applicability as well. The method is flexible out of the box, easily accommodating data with intermittent coverage and missing observations. It also handles expectations elegantly; rational expectations are internally consistent or can be proxied by external forecast data. And the Bayesian framework allows for informative priors based on theory – which we leveraged to estimate θ – plus all the usual benefits of Bayesian statistics. This method will be valuable for estimating structural macroeconomic equations in many contexts, either using the structural shocks that we have collected or by augmenting our dataset with further instruments.

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A Proofs

Proof of Proposition 1. Shift the budget constraint (3) forward one period and take expectations:

$$n_t + \tilde{\mathbb{E}}_t[y_{t+1}] = \tilde{\mathbb{E}}_t[c_{t+1} + \tau_{t+1} + \overline{B}q_{t+1} + \beta n_{t+1}]$$

$\tilde{\mathbf{E}}_{t,t+j}[x_{t+j+1}]$ denotes the iterated expectation (see Appendix B):

$$\tilde{\mathbf{E}}_{t,t+j}[x_{t+j+1}] = \tilde{\mathbb{E}}_t\tilde{\mathbb{E}}_{t+1}\tilde{\mathbb{E}}_{t+2}...\tilde{\mathbb{E}}_{t+j}\left[x_{t+j+1}\right]$$

iterate this equation forward and multiply by β:

$$\beta n_t + \sum_{j=1}^{\infty}\beta^j\tilde{\mathbf{E}}_{t,t+j-1}[y_{t+j}] = \sum_{j=1}^{\infty}\beta^j\tilde{\mathbf{E}}_{t,t+j-1}\left[c_{t+j} + \tau_{t+j} + \overline{B}q_{t+j}\right]$$

then replace βn_t in the time-t constraint (3) to get:

$$n_{t-1} + y_t + \sum_{j=1}^{\infty}\beta^j\tilde{\mathbf{E}}_{t,t+j-1}[y_{t+j}] = c_t + \tau_t + \overline{B}q_t + \sum_{j=1}^{\infty}\beta^j\tilde{\mathbf{E}}_{t,t+j-1}\left[c_{t+j} + \tau_{t+j} + \overline{B}q_{t+j}\right]$$

Substituting with $\tilde{v}_t^y$ and $\tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^\tau]$ simplifies the budget constraint to

$$n_{t-1} + \tilde{v}_t^y = c_t + \tau_t + \beta\tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^\tau] + \overline{B}q_t + \sum_{j=1}^{\infty}\beta^j\tilde{\mathbf{E}}_{t,t+j-1}\left[c_{t+j} + \overline{B}q_{t+j}\right]$$

Then, use the Euler equation (4) to write expectations of future consumption in terms of current consumption and bond prices:

$$\tilde{\mathbb{E}}_t[c_{t+1}] = c_t - \frac{\overline{C}}{\beta\gamma}q_t + \frac{\overline{C}}{\beta\gamma}z_t^d$$

$$\implies [j > 1]: \qquad \tilde{\mathbf{E}}_{t,t+j-1}\left[c_{t+j}\right] = c_t + \frac{\overline{C}}{\beta\gamma}(z_t^d - q_t) + \frac{\overline{C}}{\beta\gamma}\sum_{i=1}^{j-1}\tilde{\mathbf{E}}_{t,t+i-1}(z_{t+i}^d - q_{t+i})$$

thus the discounted sum of future consumption can be written as

$$\sum_{j=1}^{\infty}\beta^j\tilde{\mathbf{E}}_{t,t+j-1}\left[c_{t+j}\right] = \frac{\beta}{1-\beta}c_t + \frac{\overline{C}}{\gamma}\frac{1}{1-\beta}(z_t^d - q_t) + \frac{\overline{C}}{\gamma}\frac{1}{1-\beta}\sum_{j=1}^{\infty}\beta^j\tilde{\mathbf{E}}_{t,t+j-1}\left[(z_{t+j}^d - q_{t+j})\right]$$

so the budget constraint becomes

$$n_{t-1}+\tilde{v}_t^y=\frac{1}{1-\beta}c_t+\tau_t+\beta\tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^\tau]$$
$$+\left(\overline{B}-\frac{\overline{C}}{\gamma}\frac{1}{1-\beta}\right)q_t+\sum_{j=1}^{\infty}\beta^j\tilde{\mathbf{E}}_{t,t+j-1}\left[\left(\overline{B}-\frac{\overline{C}}{\gamma}\frac{1}{1-\beta}\right)q_{t+j}\right]$$
$$+\frac{\overline{C}}{\gamma}\frac{1}{1-\beta}z_t^d+\sum_{j=1}^{\infty}\beta^j\tilde{\mathbf{E}}_{t,t+j-1}\left[\frac{\overline{C}}{\gamma}\frac{1}{1-\beta}z_{t+j}^d\right]$$

and substitute in with $\tilde{v}_t^q$ and ζ_t :

$$n_{t-1}+\tilde{v}_t^y=\frac{1}{1-\beta}c_t+\tau_t+\beta\tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^\tau]+\left(\overline{B}-\frac{\overline{C}}{\gamma}\frac{1}{1-\beta}\right)\tilde{v}_t^q-\frac{1}{1-\beta}\zeta_t$$

Rearrange to isolate consumption:

$$c_t=(1-\beta)\left(n_{t-1}+\tilde{v}_t^y-\tau_t-\beta\tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^\tau]\right)+\left(\frac{\overline{C}}{\gamma}-(1-\beta)\overline{B}\right)\tilde{v}_t^q+\zeta_t$$

■

Proof of Proposition 3. Proposition 1 implies that with rational expectations and discount factor $\beta\omega$, the consumption function is

$$c_t=(1-\beta\omega)\left(n_{t-1}+\tilde{v}_t^y-\tau_t-\beta\omega\mathbb{E}_t[\tilde{v}_{t+1}^\tau]\right)+\left(\frac{\overline{C}}{\gamma}-(1-\beta\omega)\overline{B}\right)\tilde{v}_t^q+\zeta_t$$

The government budget constraint implies

$$b_{t-1}=\tau_t-g_t+\overline{B}q_t+\sum_{j=1}^{\infty}\beta^j\mathbb{E}_t[\tau_{t+j}-g_{t+j}]$$

$$=\tau_t-g_t+\overline{B}q_t+\beta\mathbb{E}_t\left[v_{t+1}^\tau-v_{t+1}^g+\overline{B}v_{t+1}^q\right]$$
$$=\tau_t+\beta\mathbb{E}_t\left[v_{t+1}^\tau\right]-v_t^g+\overline{B}v_t^q$$

Rearrange to write neatly as

$$\beta\mathbb{E}_t[v_{t+1}^\tau]=b_{t-1}-\tau_t+v_t^g-\overline{B}v_t^q \tag{40}$$

By definition $\tau_t=v_t^\tau-\beta\mathbb{E}_t[v_{t+1}]=\tilde{v}_t^\tau-\beta\omega\mathbb{E}_t[\tilde{v}_{t+1}^\tau]$. This implies that v_t^τ and $\tilde{v}_t^\tau$ are related by

$$\tilde{v}_t^\tau=\sum_{j=0}^{\infty}(\beta\omega)^j\mathbb{E}_t\left[v_{t+j}^\tau-\beta v_{t+1+j}^\tau\right]$$

which, per equation (15), simplifies to

$$= \frac{1-\beta\rho_\tau}{1-\beta\omega\rho_\tau} v_t^\tau$$

Combined with equation (40), this implies that $\beta\omega\mathbb{E}_t[\tilde{v}_{t+1}^\tau]$ can be written

$$\beta\omega\mathbb{E}_t[\tilde{v}_{t+1}^\tau] = \omega\frac{1-\beta\rho_\tau}{1-\beta\omega\rho_\tau}\left(b_{t-1} - \tau_t + v_t^g - \overline{B}v_t^q\right)$$

then use this result to replace $\beta\omega\mathbb{E}_t[\tilde{v}_{t+1}^\tau]$ in the consumption function. ■

Proof of Proposition 8. With rational expectations and zero steady-state debt, Proposition 1 gives the unconstrained households' consumption function

$$c_t^U = (1-\beta)\left(n_{t-1}^U + v_t^{y,U} - v_t^\tau\right) + \frac{\overline{C}^U}{\gamma}v_t^q + \zeta_t$$

Iterating the government budget constraint forward gives government debt per capita $b_{t-1} = v_t^\tau - v_t^g$. Substituting for taxes yields

$$c_t^U = (1-\beta)\left(n_{t-1}^U - b_{t-1} + v_t^{y,U} - v_t^g\right) + \frac{\overline{C}^U}{\gamma}v_t^q + \zeta_t$$

Aggregate this expression with constrained consumption $c_t^C = y_t^C - \tau_t$, using $n_{t-1} = \lambda n_{t-1}^U$ and $v_t^y = \lambda v_t^{y,U} + (1-\lambda)v_t^{y,C}$

$$\begin{aligned} c_t = (1-\beta)\left(n_{t-1} - \lambda b_{t-1} + v_t^y - \lambda v_t^g\right) - (1-\lambda)\tau_t \\ + \lambda\frac{\overline{C}^U}{\gamma}v_t^q + (1-\lambda)\left(y_t^C - (1-\beta)v_t^{y,C}\right) + \lambda\zeta_t \end{aligned}$$

The result follows from $(1-\beta)(1-\theta) = 1-\lambda$ and the definition of ς_t^C. ■

B Naive and Sophisticated Expectations

The behavioral expectations literature treats recursive expectations in multiple ways. This section elaborates, and clarifies the distinction between "naive" and "sophisticated" expectations that we adopt in the model.

In general, we define recursive behavioral expectations as

$$\tilde{\mathbf{E}}_{t,t+j}[x_{t+j+1}] \equiv \tilde{\mathbb{E}}_t\tilde{\mathbb{E}}_{t+1}\tilde{\mathbb{E}}_{t+2}...\tilde{\mathbb{E}}_{t+j}\left[x_{t+j+1}\right] \tag{41}$$

for some random variable x_{t+j+1}. How the behavioral expectation operators $\tilde{\mathbb{E}}_t$, $\tilde{\mathbb{E}}_{t+1}$, etc. interact is not defined without further assumptions, unlike the rational case where the law of iterated expectations always holds. Two common assumptions are naive and sophisticated expectations.

When using cognitive discounting with parameter $\lambda < 1$, the representations are clear:

$$\tilde{\mathbf{E}}_{t,t+j}[x_{t+j+1}] = \begin{cases} \lambda \mathbb{E}_t[x_{t+j+1}] & \text{(naive expectations)} \\ \lambda^{j+1}\mathbb{E}_t[x_{t+j+1}] & \text{(sophisticated expectations)} \end{cases} \tag{42}$$

In the model in Section 5, we assume that households have naive expectations, so $\lambda = \theta$, the non-Ricardian parameter in the consumption function (14). The following result shows that with sophisticated expectations, there is also a mapping from λ to θ:

Lemma 1 *If households are sophisticated cognitive discounters with parameter λ, then the attenuation factor defined in equation* (13) *for the present value of taxes is*

$$\theta = \lambda \frac{1 - \beta\rho_\tau}{1 - \beta\lambda\rho_\tau}$$

Proof. The process for taxes is

$$\tau_t = \rho_\tau \tau_{t-1} - \beta\epsilon_t^\tau + \epsilon_{t-1}^\tau$$

The expected present value of future taxes is

$$\tilde{v}_t^\tau = \tau_t + \sum_{j=1}^{\infty} \beta^j \tilde{\mathbf{E}}_{t,t+j-1}[\tau_{t+j}]$$

With sophisticated cognitive discounting

$$\tilde{\mathbf{E}}_{t,t+j-1}[\tau_{t+j}] = \lambda^j \mathbb{E}_t[\tau_{t+j}]$$

and the rational expectation is

$$\mathbb{E}_t[\tau_{t+j}] = \rho_\tau^j \tau_t + \rho_\tau^{j-1}\epsilon_t^\tau$$

Therefore

$$\begin{aligned} \tilde{v}_t^\tau &= \tau_t + \sum_{j=1}^{\infty} \beta^j \lambda^j (\rho_\tau^j \tau_t + \rho_\tau^{j-1}\epsilon_t^\tau) \\ &= \tau_t + \frac{\beta\lambda\rho_\tau}{1 - \beta\lambda\rho_\tau}\tau_t + \frac{\beta\lambda}{1 - \beta\lambda\rho_\tau}\epsilon_t^\tau \end{aligned}$$

Therefore the perceived present value of future taxes is $\frac{\beta\lambda\rho_\tau}{1-\beta\lambda\rho_\tau}\tau_t + \frac{\beta\lambda}{1-\beta\lambda\rho_\tau}\epsilon_t^\tau$ and the rational expectation is the same albeit with $\lambda = 1$. Thus the ratio is

$$\frac{\tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^\tau]}{\mathbb{E}_t[v_{t+1}^\tau]} = \lambda \frac{1 - \beta\rho_\tau}{1 - \beta\lambda\rho_\tau}$$

■

Lemma 1 implies that, given values for β and ρ_τ, and an estimate θ, the implied param-

eter λ is

$$\lambda = \frac{\theta}{1 - \beta\rho_\tau(1-\theta)}$$

B.1 Actual and Perceived Law of Motion

We use an actual law of motion (ALM) and perceived law of motion (PLM) approach to implement behavioral expectations in the general equilibrium model while maintaining internal consistency. The model admits a state space solution of the form

$$x_t = Px_{t-1} + Qz_t$$

where x_t is a vector of endogenous state variables and z_t is a vector of exogenous state variables. z_t follows

$$[ALM]: \qquad z_{t+1} = Nz_t + \vec{\epsilon}_{t+1}$$

where $\vec{\epsilon}_{t+1}$ is a vector of shocks. A rational expectations solution satisfies the linear equilibrium conditions given this ALM, as in Uhlig (2001). However, under behavioral expectations, we instead impose that expectations in the equilibrium conditions are formed based on

$$[PLM]: \qquad z_{t+1} = \tilde{N}z_t + \vec{\epsilon}_{t+1}$$

with $N \neq \tilde{N}$. The solution under this PLM is the behavioral equilibrium. Adams (2024) gives further details.

B.2 Representing Naive and Sophisticated Expectations with a PLM

In the general equilibrium model of Section 5, households are naive cognitive discounters regarding taxes only. Equations (113) and (114) give the corresponding ALM and PLM. The two laws of motion are identical except that the PLM multiplies by θ the coefficient mapping inherited government debt into current taxes. Because debt itself follows the same process under both laws of motion, θ is applied once to the tax forecast at every horizon.

If instead we were to model sophisticated cognitive discounting over taxes, the PLM would apply λ recursively to the persistence of the tax present value. If we were to model sophisticated cognitive discounting over all time series, the usual case, we would rescale the entire matrix N by λ.

C Consol Values and Yields

The market value of a consol bond (one period before any coupon payment) is given by equation (8). The analogous rational value of a consol bond is thus

$$v_t^q = q_t + \beta\mathbb{E}_t[v_{t+1}^q]$$

Why does this describe the value of a consol bond? A consol bond pays a constant coupon in perpetuity. Equivalently, the consol pays a one-period bond every period in advance of the associated coupon payment.

We can also relate the value of the consol to the yields at different horizons. The price Q_t of a one-period bond is related to its yield $R_t^{(1)}$ by

$$Q_t = \frac{1}{1 + R_t^{(1)}}$$

which is linearized as

$$q_t = -\overline{Q}^2 r_t^{(1)}$$

Thus the consol value maps to yields by:

$$v_t^q = -\overline{Q}^2 r_t^{(1)} + \beta \mathbb{E}_t[v_{t+1}^q]$$

$$\propto \sum_{h=0}^{\infty} \beta^h r_{t+h}^{(1)}$$

D Additional Model Extensions

D.1 Risky Debt

In this section, we relax the assumption that governments issue risk-free real debt. For example, this is important if governments issue risk-free nominal debt but the economy features inflation risk. We show that this extension yields the same behavioral consumption function as in Proposition 2.

Now we write risky debt $B_t^\dagger$ to denote the notional end-of-period real debt which grows by a risky factor R_{t+1}^B in the next period. The household's budget constraint (1) becomes

$$R_t^B B_{t-1}^\dagger + R_t^K K_{t-1} + Y_t^N = C_t + T_t + B_t^\dagger + K_t \tag{43}$$

while the government's budget constraint (11) becomes

$$R_t^B B_{t-1}^\dagger + G_t = T_t + B_t^\dagger \tag{44}$$

The household's Euler equation (2) now prices the risky debt:

$$1 = \beta \tilde{\mathbb{E}}_t \left[\frac{R_{t+1}^B u'(C_{t+1})}{u'(C_t)} \right] + Z_t^d \tag{45}$$

When linearized, these equations become

$$n_t^\dagger + y_t = c_t + \tau_t + \beta \tilde{\mathbb{E}}_t[n_{t+1}^\dagger] - \beta \bar{B}^\dagger \tilde{\mathbb{E}}_t[r_{t+1}^B] \tag{46}$$

$$\overline{R^B} b_{t-1}^\dagger + r_t^B \overline{B}^\dagger = \tau_t - g_t + b_t^\dagger \tag{47}$$

$$0 = \beta \tilde{\mathbb{E}}_t[r_{t+1}^B + \gamma \frac{1}{\overline{C}}(c_t - c_{t+1})] + z_t^d \tag{48}$$

where risky financial net worth $n_t^\dagger$ is now defined $n_t^\dagger \equiv \overline{R^B} b_{t-1}^\dagger + \overline{B}^\dagger r_t^B + \overline{R^k} k_{t-1}$ with the

assumption that $\overline{R^B} = \overline{R^K} = \beta^{-1}$.

Finally, define the risky debt consol value as

$$\tilde{v}_t^q = \beta\tilde{\mathbb{E}}_t[r_{t+1}^B] + \beta\tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^q] \tag{49}$$

and similarly for the rational expectations analog v_t^q.

The following Proposition shows that the behavioral consumption function is largely unchanged when government debt is risky. The only difference is that $\overline{R^B}b_{t-1}^\dagger$ enters in place of b_{t-1}, as expected: they both represent the time t value of outstanding debt.

Proposition 4 *If government debt is risky with budget constraint* (47) *and expectations satisfy equation* (13)*, then the consumption function is given by equation* (14) *albeit with* $n_t^\dagger$ *and* $\left(\overline{R^B}b_{t-1}^\dagger + \overline{B}^\dagger r_t^B\right)$ *in place of* n_{t-1} *and* b_{t-1} *respectively.*

Proof. The household equations (46) and (48) are equivalent to equations (3) and (4) by mapping $\beta\tilde{\mathbb{E}}_t[r_{t+1}^B] \to -q_t$ and $\overline{B}^\dagger \to \overline{B}$. Therefore the generic consumption function in Proposition 1 still holds with risky debt.

Next, define $x_t \equiv \overline{R^B}b_{t-1}^\dagger + \overline{B}^\dagger r_t^B$. The government budget constraint implies $x_t = \tau_t - g_t + b_t^\dagger$. Using $\overline{R^B} = \beta^{-1}$, this can be written as

$$x_t = \tau_t - g_t - \beta\overline{B}^\dagger\mathbb{E}_t[r_{t+1}^B] + \beta\mathbb{E}_t[x_{t+1}]$$

and iterating forward gives

$$x_t = \tau_t - g_t + \beta\mathbb{E}_t\left[\sum_{j=0}^{\infty}\beta^j(\tau_{t+1+j} - g_{t+1+j})\right] - \overline{B}^\dagger\sum_{j=1}^{\infty}\beta^j\mathbb{E}_t[r_{t+j}^B]$$

Using the risky-debt definition of the consol value v_t^q, this becomes

$$x_t = \tau_t + \beta\mathbb{E}_t\left[v_{t+1}^\tau\right] - v_t^g - \overline{B}^\dagger v_t^q$$

so

$$\beta\tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^\tau] = \theta\left(x_t - \tau_t + v_t^g + \overline{B}^\dagger v_t^q\right)$$

then use this result to replace $\beta\tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^\tau]$ in the consumption function (10). ■

D.2 Non-Separable Utility

In the baseline model we assumed that the marginal utility of consumption only depended on consumption. Sometimes, the utility function is thought to be non-separable in consumption C_t and labor L_t. In that case, the behavioral consumption function introduces an additional term that depends on the growth rate of labor.

Proposition 5 *If the utility function $u(C_t, L_t)$ is non-separable in labor L_t, then the behavioral consumption function becomes*

$$c_t = (1-\beta)\left(n_{t-1} - \theta b_{t-1} + \tilde{v}_t^y - (1-\theta)\tau_t - \theta v_t^g + \theta\overline{B}v_t^q\right) + \left(\frac{\overline{C}}{\gamma} - (1-\beta)\overline{B}\right)\tilde{v}_t^q + \alpha^\ell\beta\tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^{d\ell}] + \zeta_t$$

where $\tilde{v}_t^{d\ell}$ denotes the perceived present discounted value of future labor growth:

$$\tilde{v}_t^{d\ell} = \ell_t - \ell_{t-1} + \beta\tilde{v}_{t+1}^{d\ell}$$

and $\alpha^\ell \equiv -\frac{\bar{C}}{\gamma}\left(\frac{u_{CL}(\overline{C},\overline{L})}{u_C(\overline{C},\overline{L})}\right)$.

Proof. Instead of (2), the Euler equation with non-separable utility is given by

$$Q_t = \beta\tilde{\mathbb{E}}_t\left[\frac{u'(C_{t+1}, L_{t+1})}{u'(C_t, L_t)}\right] + Z_t^d \tag{50}$$

which linearizes as

$$q_t = \beta\tilde{\mathbb{E}}_t\left[\gamma\frac{1}{\overline{C}}\left(c_t - c_{t+1} + \alpha^\ell(\ell_t - \ell_{t+1})\right)\right] + z_t^d \tag{51}$$

Then, the derivation of Proposition 1 still applies, except wherever z_t^d appears, there is now an additional $\beta\tilde{\mathbb{E}}_t\left[\gamma\frac{1}{\overline{C}}\alpha^\ell(\ell_t - \ell_{t+1})\right]$ term. Therefore the consumption function (10) still holds, except wherever $\zeta_t = -\frac{\overline{C}}{\gamma}z_t^d + \beta\tilde{\mathbb{E}}_t[\zeta_{t+1}]$ appears, there is now an additional $\alpha^\ell\beta\tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^{d\ell}]$ term. Thus the consumption function is given by

$$c_t = (1-\beta)\left(n_{t-1} + \tilde{v}_t^y - \tau_t - \beta\tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^\tau]\right) + \left(\frac{\overline{C}}{\gamma} - (1-\beta)\overline{B}\right)\tilde{v}_t^q + \alpha^\ell\beta\tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^{d\ell}] + \zeta_t$$

Then following the same steps as the proof of Proposition 2 gives the desired result. ■

D.3 Consumption Habits

In this extension, we introduce the slow-moving external consumption habits of Campbell and Cochrane (1999).[32] This time non-separability distorts the Euler equation in a way that more closely fits the empirical relationship between consumption and asset returns. Let $C_{i,t}$ denote household i's consumption and let X_t denote the habit, a slowly depreciating stock of external consumption. Period utility is

$$u(C_{i,t}, X_t) = \frac{(C_{i,t} - X_t)^{1-\gamma} - 1}{1-\gamma} \tag{52}$$

[32]The contemporaneous keeping-up-with-the-Joneses specification of Galí (1994) is isomorphic to our benchmark because its aggregate Euler equation differs only through a rescaling of γ.

Each household takes X_t as given, so that habits represent keeping-up-with-the-Joneses incentives. The consumption Euler equation is therefore

$$Q_t = \beta\tilde{\mathbb{E}}_t\left[\left(\frac{C_{i,t+1}-X_{t+1}}{C_{i,t}-X_t}\right)^{-\gamma}\right] + Z_t^d \tag{53}$$

The habit depends on lagged aggregate consumption. To a first order approximation, its law of motion is

$$x_t = \rho_x x_{t-1} + \chi_c c_{t-1} \tag{54}$$

where $\rho_x \in (0,1)$, $\chi_c \in (0, 1-\rho_x)$, and $\overline{X} < \overline{C}$.

Proposition 6 *If preferences are given by equation* (52) *and the external habit follows equation* (54)*, then the aggregate behavioral consumption function is*

$$\begin{aligned} c_t = \kappa^h\left(n_{t-1} - \theta b_{t-1} + \tilde{v}_t^y - (1-\theta)\tau_t - \theta v_t^g + \theta\overline{B}v_t^q\right) \\ + \left(\frac{\overline{C}-\overline{X}}{\gamma} - \kappa^h\overline{B}\right)\tilde{v}_t^q + \alpha^x x_t + \frac{\overline{C}-\overline{X}}{\overline{C}}\zeta_t \end{aligned} \tag{55}$$

where

$$\kappa^h \equiv (1-\beta)\left(1 - \frac{\beta\chi_c}{1-\beta\rho_x}\right) \qquad \alpha^x \equiv \frac{\beta(1-\rho_x)}{1-\beta\rho_x} \tag{56}$$

Proof. Linearizing equation (53) and averaging across households gives

$$q_t = \beta\tilde{\mathbb{E}}_t\left[\frac{\gamma}{\overline{C}-\overline{X}}\left((c_t - x_t) - (c_{t+1} - x_{t+1})\right)\right] + z_t^d \tag{57}$$

Define the perceived present discounted values of consumption and the external habit by

$$\tilde{v}_t^c = c_t + \beta\tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^c]$$

$$\tilde{v}_t^x = x_t + \beta\tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^x]$$

Combining equation (57) with the household budget constraint as in the proof of Proposition 1 gives the generic consumption function

$$\begin{aligned} c_t = (1-\beta)\left(n_{t-1} + \tilde{v}_t^y - \tau_t - \beta\tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^\tau]\right) \\ + \left(\frac{\overline{C}-\overline{X}}{\gamma} - (1-\beta)\overline{B}\right)\tilde{v}_t^q + x_t - (1-\beta)\tilde{v}_t^x + \frac{\overline{C}-\overline{X}}{\overline{C}}\zeta_t \end{aligned}$$

Because households' perceived forecasts respect the habit law (54), the perceived habit value satisfies

$$\tilde{v}_t^x = \frac{x_t + \beta\chi_c\tilde{v}_t^c}{1-\beta\rho_x}$$

while the perceived intertemporal budget constraint from the proof of Proposition 1 implies

$$\tilde{v}_t^c = n_{t-1} + \tilde{v}_t^y - \tau_t - \beta\tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^\tau] - \overline{B}\tilde{v}_t^q$$

Substituting these expressions into the generic consumption function gives

$$c_t = \kappa^h \left(n_{t-1} + \tilde{v}_t^y - \tau_t - \beta \tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^\tau] \right) + \left(\frac{\overline{C} - \overline{X}}{\gamma} - \kappa^h \overline{B} \right) \tilde{v}_t^q + \alpha^x x_t + \frac{\overline{C} - \overline{X}}{\overline{C}} \zeta_t$$

Following the same steps as the proof of Proposition 2 gives the desired result. ■

If $\chi_c = 0$, then $x_t = 0$ and Proposition 6 reduces to the baseline behavioral consumption function. Otherwise, habits reduce the MPC from $1 - \beta$ to κ^h, while the MPC out of debt-financed transfers becomes $\kappa^h(1 - \theta)$. Thus the coefficients on net worth and taxes satisfy

$$\phi_n = \kappa^h \qquad \phi_\tau = -\kappa^h(1 - \theta)$$

so the behavioral attenuation is still given by $\theta = 1 + \phi_\tau / \phi_n$.

D.4 Distortionary Taxation

In the baseline, taxes appear in the budget constraints, and may distort the economy, but do not distort the intertemporal decision. In this section, we extend the model to allow for such effects.

Specifically, we allow for taxes on interest income, which enters the Euler equation for government bonds (2). Define $1 + i_t^b = Q_t^{-1}$ so that i_t^b denotes the pretax real interest rate on the bond. Interest income is taxed at rate τ_t^i, so the after-tax gross return is $1 + (1 - \tau_t^i) i_t^b$.

With non-zero steady state taxes, there will be a discounting wedge ω relative to the government.[33] Thus the Euler equation becomes

$$\left(1 + (1 - \tau_t^i) i_t^b \right)^{-1} = \beta \omega \tilde{\mathbb{E}}_t \left[\frac{u'(C_{t+1})}{u'(C_t)} \right] + Z_t^d \tag{58}$$

or in terms of the bond price Q_t:

$$\frac{Q_t}{\tau_t^i Q_t + 1 - \tau_t^i} = \beta \omega \tilde{\mathbb{E}}_t \left[\frac{u'(C_{t+1})}{u'(C_t)} \right] + Z_t^d$$

Proposition 7 *If there are taxes on interest such that the Euler equation is given by* (58)*, then the behavioral consumption function is*

$$c_t = (1 - \beta\omega) \left(n_{t-1} - \theta b_{t-1} + \tilde{v}_t^y - (1 - \theta)\tau_t - \theta v_t^g + \theta \overline{B} v_t^q \right) + \left(\frac{\overline{C}}{\gamma} \alpha^q - (1 - \beta\omega)\overline{B} \right) \tilde{v}_t^q - \alpha^i \tilde{v}_t^i + \zeta_t$$

where $\tilde{v}_t^i$ denotes the household's present discounted value of future taxes on interest:

$$\tilde{v}_t^i = \tau_t^i + \beta\omega \mathbb{E}_t[\tilde{v}_{t+1}^i]$$

and $\alpha^i \equiv -\frac{\bar{C}}{\gamma} \beta\omega^2 (1 - \beta)$, $\alpha^q \equiv \omega \left(1 - \beta\omega\overline{\tau}^i \right)$, and the steady state household discounting

[33] Strictly speaking, the household's discount factor $\beta\omega$ is the primitive parameter, and the government's steady state discount factor $\beta = \overline{Q}$ is determined by steady state taxes, but we stick to the wedge notation in order to be consistent with the rest of the paper.

wedge is given in terms of the government's discount factor β by

$$\omega = \frac{1}{1 - \overline{\tau}^i(1-\beta)}$$

Proof. To linearize around the deterministic steady state, use $\frac{\overline{Q}}{\overline{\tau}^i\overline{Q}+1-\overline{\tau}^i} = \beta\omega$. The government's discount factor is $\beta = \overline{Q}$, implying that the household wedge is $\omega = \frac{1}{1-\overline{\tau}^i(1-\beta)}$. The left-hand side linearizes as

$$\frac{1}{\overline{\tau}^i\overline{Q}+1-\overline{\tau}^i}\left(q_t - \frac{\overline{Q}}{\overline{\tau}^i\overline{Q}+1-\overline{\tau}^i}\overline{\tau}^i q_t - \frac{\overline{Q}}{\overline{\tau}^i\overline{Q}+1-\overline{\tau}^i}(\overline{Q}-1)\tau_t^i\right)$$
$$= \omega\left(1-\beta\omega\overline{\tau}^i\right)q_t + \beta\omega^2(1-\beta)\tau_t^i$$

so the linearized Euler equation is

$$\omega\left(1-\beta\omega\overline{\tau}^i\right)q_t = \beta\omega\tilde{\mathbb{E}}_t[\gamma\frac{1}{\overline{C}}(c_t - c_{t+1})] - \beta\omega^2(1-\beta)\tau_t^i + z_t^d \tag{59}$$

Therefore, in the derivation of the consumption function (Proposition 3), the Euler-equation component of the q_t term is rescaled by $\alpha^q = \omega\left(1-\beta\omega\overline{\tau}^i\right)$, while the budget-constraint component is unchanged. And wherever z_t^d appears, it is joined by $-\beta\omega^2(1-\beta)\tau_t^i$.

Following the same logic as in the proof of Proposition 5, the behavioral consumption function becomes

$$c_t = (1-\beta\omega)\left(n_{t-1} - \theta b_{t-1} + \tilde{v}_t^y - (1-\theta)\tau_t - \theta v_t^g + \theta\overline{B}v_t^q\right) + \left(\frac{\overline{C}}{\gamma}\alpha^q - (1-\beta\omega)\overline{B}\right)\tilde{v}_t^q - \alpha^i\tilde{v}_t^i + \zeta_t$$

■

When taxes on interest are $\overline{\tau}^i = \tau_t^i = 0$ then this reduces to the standard consumption function.

D.5 Liquidity Constraints

Liquidity constraints can also break Ricardian equivalence. This is because the consumption function (Proposition 1) is derived by iterating over future Euler equations, but when an agent is constrained, the Euler equation does not hold. Angeletos et al. (2024) argue that the discounting wedge from Section 2.3 is a good approximation of occasionally-binding liquidity constraints. But what if some agents are perpetually constrained?

A classic, tractable method of capturing this mechanism for non-Ricardianism is to introduce hand-to-mouth consumers (Campbell and Mankiw, 1989).[34] In this section, we describe a simple two-agent model, and show that the consumption function test of Ricardianism applies to this setting as well.

There is a measure $\lambda \in (0,1]$ of unconstrained consumers, and a measure $1-\lambda$ of constrained consumers. For simplicity, we assume government spending is constant with

[34]This method is still used in modern two-agent New Keynesian (TANK) models (Galí et al., 2007), which approximate non-Ricardian behavior in richer heterogeneous agent models (Auclert et al., 2024).

zero steady-state debt, both types of consumers receive the same tax process, and have rational expectations.

The unconstrained households' income y_t^U and the constrained households' income y_t^C may follow different processes. They add up to aggregate income by

$$y_t = \lambda y_t^U + (1-\lambda) y_t^C \tag{60}$$

We write the present discounted values of these income streams as $v_t^{y,U}$ and $v_t^{y,C}$ respectively.

The unconstrained choose consumption c_t^U satisfying the consumption function in Proposition 1. The constrained consume

$$c_t^C = y_t^C - \tau_t \tag{61}$$

and aggregate consumption is given by

$$c_t = \lambda c_t^U + (1-\lambda) c_t^C \tag{62}$$

Similarly, the unconstrained hold assets n_t^U while the constrained hold zero assets. Thus household-level assets are related to total assets by

$$n_t^U = \frac{n_t}{\lambda}$$

and similarly for bonds.

With this structure, Proposition 8 gives the aggregate consumption function.

Proposition 8 *If households have rational expectations, then the aggregate consumption function in the two-agent economy can be expressed as*

$$c_t = (1-\beta)\left(n_{t-1} - \lambda b_{t-1} + v_t^y - (1-\theta)\tau_t - \lambda v_t^g\right) + \lambda \frac{\overline{C}^U}{\gamma} v_t^q + \varsigma_t^C + \lambda \zeta_t \tag{63}$$

where the non-Ricardian parameter is now given by

$$\theta = 1 - \frac{1-\lambda}{1-\beta}$$

and $\varsigma_t^C \equiv (1-\lambda)\left(y_t^C - (1-\beta) v_t^{y,C}\right)$ *is a term that depends on the constrained income process.*

Proof: Appendix A

Unlike in the case of a discounting wedge, Proposition 8 shows that the consumption function in the two agent economy is similar but not quite equivalent to the behavioral consumption function from Proposition 2. Different restrictions relate the coefficients of the right-hand-side variables. And there is now a ς_t^C term which depends on the income process of the constrained households. We can estimate including this term directly, but it may also be convenient to focus on one of several special cases where the ς_t^C term is irrelevant (Corollary 1).

Current taxes still enter with coefficient $-(1-\beta)(1-\theta)$, but here θ reflects the share of constrained households. When $\lambda = 1$, $\theta = 1$ and households are Ricardian. With constrained households, the coefficient on existing debt is $-(1-\beta)\lambda$, rather than $-(1-\beta)\theta$ as in Proposition 2. Thus the restriction linking the tax and debt coefficients depends on the source of non-Ricardian behavior, so we do not impose such restrictions in our benchmark specification. Unlike the behaviovioral attenuation and discounting wedge causes of non-Ricardianism, this model also permits $\theta < 0$ if the constrained share $1-\lambda$ is sufficiently large.

Corollary 1 *If constrained households receive constant income $y_t^C = \bar{y}^C$ then the aggregate consumption function in the two agent economy reduces to*

$$c_t = (1-\beta)\left(n_{t-1} - \lambda b_{t-1} + v_t^y - (1-\theta)\tau_t - \lambda v_t^g\right) + \lambda\frac{\overline{C}^U}{\gamma}v_t^q + \lambda\zeta_t$$

Proof. When constrained income is constant, then its present discounted value is given by

$$v_t^{y,C} = \bar{y}^C + \beta\bar{y}^C + \beta^2\bar{y}^C + ... = \frac{\bar{y}^C}{1-\beta}$$

thus ς_t^C is given by

$$\varsigma_t^C = (1-\lambda)\left(y_t^C - (1-\beta)v_t^{y,C}\right) = (1-\lambda)\left(\bar{y}^C - (1-\beta)\frac{\bar{y}^C}{1-\beta}\right) = 0$$

and the result follows from Proposition 8. ■

D.6 Imperfect Risk Sharing

Borrowing constraints and other frictions can break perfect risk-sharing. Section 2.3 approximated this with a constant discount wedge, and Appendix D.5 approximated this with a two-agent model.

This appendix allows for a more general distribution of heterogeneous households. To do so, we rely on recent work by Berger et al. (2023), who show that a broad class of heterogeneous-agent economies admits a representative-agent (RA) representation in which imperfect risk sharing appears as an endogenous time-varying wedge in the aggregate Euler equation.

Let $\varphi_{i,t} \equiv C_{i,t}/C_t$ denote household i's share of aggregate consumption. With power utility, the RA Euler equation is

$$Q_t = \beta\tilde{\mathbb{E}}_t\left[\mathcal{W}_{t+1}\left(\frac{C_{t+1}}{C_t}\right)^{-\gamma}\right] + Z_t^d \tag{64}$$

where $\mathcal{W}_{t+1}$ is the endogenous time-varying wedge. For a group g of households that price

the the bond, Berger et al. (2023) calculate the wedge by

$$\mathcal{W}_{g,t+1} = \frac{1}{N_{g,t}} \sum_{i \in g} \left(\frac{\varphi_{i,t+1}}{\varphi_{i,t}} \right)^{-\gamma} \tag{65}$$

where $\varphi_{i,t}$ is household i's share of aggregate consumption at time t, and $N_{g,t}$ is the number of households. Euler equation (64) follows directly from Berger et al's Proposition 1 if agents form expectations by cognitive discounting with parameter λ, as it is equivalent to the rational model with discount factor $\beta\lambda$.

We normalize the steady-state wedge to one and let $\widehat{\mathcal{W}}_t \equiv \mathcal{W}_t - 1$ denote its deviation. Define its perceived present value by

$$\tilde{v}_t^w = \widehat{\mathcal{W}}_t + \beta \tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^w] \tag{66}$$

Proposition 9 *If aggregate consumption satisfies the RA Euler equation* (64)*, then the behavioral consumption function is*

$$c_t = (1-\beta)\left(n_{t-1} - \theta b_{t-1} + \tilde{v}_t^y - (1-\theta)\tau_t - \theta v_t^g + \theta \overline{B} v_t^q\right) + \left(\frac{\overline{C}}{\gamma} - (1-\beta)\overline{B}\right)\tilde{v}_t^q + \zeta_t - \frac{\overline{C}}{\gamma}\beta \tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^w] \tag{67}$$

Proof. Linearizing equation (64) gives

$$q_t = \beta \tilde{\mathbb{E}}_t \left[\frac{\gamma}{\overline{C}} (c_t - c_{t+1}) \right] + \beta \tilde{\mathbb{E}}_t[\widehat{\mathcal{W}}_{t+1}] + z_t^d \tag{68}$$

Relative to the benchmark Euler equation (4), there is an additional intertemporal wedge $\beta \tilde{\mathbb{E}}_t[\widehat{\mathcal{W}}_{t+1}]$. Following the derivation of Proposition 1, its contribution to the generic consumption function is

$$-\frac{\overline{C}}{\gamma} \sum_{j=1}^{\infty} \beta^j \tilde{\mathbf{E}}_{t,t+j-1}[\widehat{\mathcal{W}}_{t+j}] = -\frac{\overline{C}}{\gamma}\beta \tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^w] \tag{69}$$

Therefore the generic consumption function becomes

$$c_t = (1-\beta)\left(n_{t-1} + \tilde{v}_t^y - \tau_t - \beta \tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^\tau]\right) + \left(\frac{\overline{C}}{\gamma} - (1-\beta)\overline{B}\right)\tilde{v}_t^q + \zeta_t - \frac{\overline{C}}{\gamma}\beta \tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^w] \tag{70}$$

Following the same steps as the proof of Proposition 2 gives the desired result. ■

When risk sharing is perfect, $\widehat{\mathcal{W}}_t = \tilde{v}_t^w = 0$ and Proposition 9 reduces to the baseline behavioral consumption function. More generally, the measured wedge provides observations of $\widehat{\mathcal{W}}_t$, while its estimated law of motion, scaled by an unknown behavioral factor, determines $\tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^w]$.

D.7 Discount Wedges without Rational Expectations

This section derives the consumption function when households have both a discount wedge relative to the government, and non-rational expectations. If the present value of taxes are

AR(1) and households are cognitive discounters, Proposition 10 derives explicit expressions for the MPC and taxes coefficient.

Proposition 10 *If the government discounts by β and the household discounts by $\beta\omega$, the present value of taxes is AR(1) given by*

$$v_t^\tau = \rho_\tau v_{t-1}^\tau + \varepsilon_t^\tau$$

and households are sophisticated or naive cognitive discounters with behavioral attenuation λ, then the consumption function can be expressed as

$$c_t = (1-\beta\omega)\left(n_{t-1} - \theta_{cd} b_{t-1} + \tilde{v}_t^y - (1-\theta_{cd})\tau_t - \theta_{cd} v_t^g + \theta_{cd}\overline{B} v_t^q\right) + \left(\frac{\overline{C}}{\gamma} - (1-\beta\omega)\overline{B}\right)\tilde{v}_t^q + \zeta_t \quad (71)$$

where

$$\theta_{cd} = \begin{cases} \omega\lambda\frac{1-\beta\rho_\tau}{1-\beta\omega\lambda\rho_\tau} & \textit{sophisticated cognitive discounting} \\ \omega\lambda\frac{1-\beta\rho_\tau}{1-\beta\omega\rho_\tau} & \textit{naive cognitive discounting} \end{cases} \quad (72)$$

and where for quantity x, v_t^x denotes the present value using the government's discount factor β and rational expectations $\mathbb{E}_t$, while $\tilde{v}_t^x$ denotes using the household's discount factor $\beta\omega$ and expectations $\tilde{\mathbb{E}}_t$:

$$v_t^x = x_t + \beta\mathbb{E}_t[v_{t+1}^x] \qquad \tilde{v}_t^x = x_t + \beta\omega\tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^x]$$

Proof of Proposition 10. Proposition 1 implies that with discount factor $\beta\omega$, the consumption function is

$$c_t = (1-\beta\omega)\left(n_{t-1} + \tilde{v}_t^y - \tau_t - \beta\omega\mathbb{E}_t[\tilde{v}_{t+1}^\tau]\right) + \left(\frac{\overline{C}}{\gamma} - (1-\beta\omega)\overline{B}\right)\tilde{v}_t^q + \zeta_t$$

The government budget constraint implies

$$b_{t-1} = \tau_t - g_t + \overline{B}q_t + \sum_{j=1}^{\infty}\beta^j\mathbb{E}_t[\tau_{t+j} - g_{t+j}]$$

$$= \tau_t - g_t + \overline{B}q_t + \beta\mathbb{E}_t\left[v_{t+1}^\tau - v_{t+1}^g + \overline{B}v_{t+1}^q\right]$$

$$= \tau_t + \beta\mathbb{E}_t\left[v_{t+1}^\tau\right] - v_t^g + \overline{B}v_t^q$$

Rearrange to write neatly as

$$\beta\mathbb{E}_t[v_{t+1}^\tau] = b_{t-1} - \tau_t + v_t^g - \overline{B}v_t^q \quad (73)$$

By definition $\tau_t = v_t^\tau - \beta\mathbb{E}_t[v_{t+1}^\tau] = \tilde{v}_t^\tau - \beta\omega\tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^\tau]$. And the AR(1) assumption implies $\tau_t = (1-\beta\rho_\tau)v_t^\tau$.

If households are sophisticated cognitive discounters with attenuation λ, then $\tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^\tau] =$

$\lambda\mathbb{E}_t[\tilde{v}^\tau_{t+1}]$.[35] Therefore

$$\text{[sophisticated]} \qquad \tilde{v}^\tau_t = (1-\beta\rho_\tau)v^\tau_t + \beta\omega\lambda\mathbb{E}_t[\tilde{v}^\tau_{t+1}] = \frac{1-\beta\rho_\tau}{1-\beta\omega\lambda\rho_\tau}v^\tau_t$$

Combined with equation (73), this implies that $\beta\omega\tilde{\mathbb{E}}_t[\tilde{v}^\tau_{t+1}]$ can be written

$$\text{[sophisticated]} \qquad \beta\omega\tilde{\mathbb{E}}_t[\tilde{v}^\tau_{t+1}] = \omega\lambda\frac{1-\beta\rho_\tau}{1-\beta\omega\lambda\rho_\tau}\left(b_{t-1} - \tau_t + v^g_t - \overline{B}v^q_t\right)$$

then use this result to replace $\beta\omega\tilde{\mathbb{E}}_t[\tilde{v}^\tau_{t+1}]$ in the consumption function.

If households are naive cognitive discounters, then $\tilde{\mathbb{E}}_t[\tilde{v}^\tau_{t+1}] = \lambda\mathbb{E}_t[v^\tau_{t+1}]$. Therefore,

$$\text{[naive]} \qquad \tilde{\mathbb{E}}_t[\tilde{v}^\tau_{t+1}] = \lambda\sum_{j=0}^{\infty}(\beta\omega)^j\mathbb{E}_t[\tau_{t+1+j}]$$

$$= \lambda\sum_{j=0}^{\infty}(\beta\omega)^j\mathbb{E}_t[(1-\beta\rho)v^\tau_{t+1+j}] = \lambda\frac{1-\beta\rho_\tau}{1-\beta\omega\rho_\tau}\mathbb{E}_t[v^\tau_{t+1}]$$

Combined with equation (40), this implies that $\beta\omega\tilde{\mathbb{E}}_t[\tilde{v}^\tau_{t+1}]$ can be written

$$\text{[naive]} \qquad \beta\omega\tilde{\mathbb{E}}_t[\tilde{v}^\tau_{t+1}] = \omega\lambda\frac{1-\beta\rho_\tau}{1-\beta\omega\rho_\tau}\left(b_{t-1} - \tau_t + v^g_t - \overline{B}v^q_t\right)$$

then use this result to replace $\beta\omega\tilde{\mathbb{E}}_t[\tilde{v}^\tau_{t+1}]$ in the consumption function. ■

E Non-Ricardian Behavior in a Simple Small Open Economy

Section 5 demonstrated that when households are non-Ricardian, government borrowing reduces the net foreign asset (NFA) position in a two-country general equilibrium setting. This appendix provides a simpler version of the model that retains the main mechanism and intuition, but can be solved with pen and paper. It also provides a simple calculation that demonstrates how much of the United States' 50-year NFA decline is due to government borrowing with non-Ricardian households.

E.1 Model Assumptions

The simple model is of a small open economy. The household receives constant labor income $\overline{Y}$, pays lump-sum taxes net of transfers T_t, and trades one-period risk-free bonds at a fixed price $\overline{Q}$ with the rest of the world.

The household discounts utility by $\beta \in (0,1)$. The Euler equation and its first-order approximation around a steady state with $\overline{Q} = \beta$ are

$$\overline{Q} = \beta\tilde{\mathbb{E}}_t\left[\frac{u'(C_{t+1})}{u'(C_t)}\right] \qquad \Longrightarrow \qquad \tilde{\mathbb{E}}_t[c_{t+1}] = c_t \tag{74}$$

[35] Appendix B reviews some properties of sophisticated versus naive expectations.

As in the two-country model, B_t denotes domestic bond holdings and B_t^* denotes foreign holdings, so total government debt is $B_t + B_t^*$. We linearize around $\overline{B} = \overline{B}^* = \overline{T} = 0$ and $\overline{C} = \overline{Y}$, with lowercase variables denoting deviations in levels. Income and bond prices are constant, so $y_t = q_t = 0$, and government spending is zero. Thus the household and government budget constraints become

$$\begin{aligned} b_{t-1} &= c_t + \tau_t + \beta b_t \\ b_{t-1} + b_{t-1}^* &= \tau_t + \beta(b_t + b_t^*) \end{aligned} \tag{75}$$

Government debt is the only exogenous process and follows

$$b_t + b_t^* = \rho_B(b_{t-1} + b_{t-1}^*) + \epsilon_t^B \tag{76}$$

where $\rho_B \in [0, 1)$ and ϵ_t^B is an iid, mean-zero shock.

Households observe current taxes but may underweight the taxes that will finance outstanding debt. Define the rational and perceived tax present values by

$$v_t^\tau = \tau_t + \beta \mathbb{E}_t[v_{t+1}^\tau] \qquad \tilde{v}_t^\tau = \tau_t + \beta \tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^\tau] \tag{77}$$

and we assume that households' expectations of future taxes are attenuated by θ:

$$\tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^\tau] = \theta \mathbb{E}_t[v_{t+1}^\tau] \qquad \theta \in [0, 1] \tag{78}$$

The government budget constraint must hold for actual outcomes, but households' forecasts need not satisfy it.

E.2 The Consumption Function

We derive the behavioral consumption function with the same approach as in Section 2's more general setting.

Proposition 11 *The linear consumption function is*

$$c_t = (1 - \beta)\left[b_{t-1} - \theta(b_{t-1} + b_{t-1}^*) - (1 - \theta)\tau_t\right] \tag{79}$$

Proof. The household's budget constraint (75) is recursive in debt; telescope forwards using behavioral expectations:

$$b_{t-1} = c_t + \tau_t + \beta \tilde{\mathbb{E}}_t[c_{t+1} + \tau_{t+1}] + \beta^2 \tilde{\mathbb{E}}_t \tilde{\mathbb{E}}_{t+1}[c_{t+2} + \tau_{t+2}] + \cdots$$

The Euler equation (74) and definition of the perceived present value of taxes (77) imply

$$b_{t-1} = \frac{c_t}{1 - \beta} + \tau_t + \beta \tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^\tau] \tag{80}$$

Total debt must equal the rational present value of taxes:

$$b_{t-1} + b_{t-1}^* = v_t^\tau = \tau_t + \beta \mathbb{E}_t[v_{t+1}^\tau]$$

so use the attenuation (78) and substitute for $\tilde{\mathbb{E}}_t[\tilde{v}^{\tau}_{t+1}]$ in (80) to get

$$b_{t-1} = \frac{c_t}{1-\beta} + \tau_t + \theta\left(b_{t-1} + b^*_{t-1} - \tau_t\right)$$

and rearrange to yield equation (79). ■

The wealth MPC is $1-\beta$, while the MPC out of a debt-financed transfer is $(1-\beta)(1-\theta)$. If households were Ricardian ($\theta = 1$) then their consumption would only depend on their economy's debt to the rest of the world b^*_{t-1}. But, if they are non-Ricardian, taxes and transfers affect consumption, and households act as if domestically held government bonds are net worth.

E.3 Borrowing and External Balances

The economy-wide budget constraint implies that any variation in consumption must be financed from abroad. We define the NFA as the end-of-period position at market value:

$$\text{NFA}_t = -\overline{Q}B^*_t$$

which implies that the current account balance is

$$\text{CA}_t = \overline{Y} - C_t + (\overline{Q}^{-1} - 1)\,\text{NFA}_{t-1} = \text{NFA}_t - \text{NFA}_{t-1}$$

With $\overline{Q} = \beta$, deviations from the zero-NFA steady state satisfy

$$\text{nfa}_t = -\beta b^*_t \qquad \text{ca}_t = \text{nfa}_t - \text{nfa}_{t-1}$$

Subtracting the household budget constraint from the government budget constraint gives

$$\text{nfa}_t = \beta^{-1}\,\text{nfa}_{t-1} - c_t$$

$$\text{ca}_t = (\beta^{-1} - 1)\,\text{nfa}_{t-1} - c_t \tag{81}$$

In the current account, the $(\beta^{-1}-1)\,\text{nfa}_{t-1}$ term corresponds to asset income paid on the NFA, while $-c_t$ is the trade balance.

Government borrowing reduces NFA when households are non-Ricardian. The following proposition demonstrates how government borrowing determines external balances:

Proposition 12 *For any realized debt path over periods $t = 1, \ldots, T$, external balances satisfy*

$$\text{nfa}_T - \text{nfa}_0 = -(1-\beta)(1-\theta)\sum_{t=1}^{T}\beta(b_t + b^*_t) \tag{82}$$

and

$$\text{ca}_t = -(1-\beta)(1-\theta)\beta(b_t + b^*_t) \tag{83}$$

Proof. Substitute the government budget constraint into (79) to get

$$c_t = (1-\beta)\left[\beta^{-1}\,\text{nfa}_{t-1} + (1-\theta)\beta(b_t + b^*_t)\right]$$

Substitute into (81) to obtain (83). Since the current account balance is the change in NFA, summing current account balances from $t = 1$ to T gives

$$\begin{aligned}\text{nfa}_T - \text{nfa}_0 &= \sum_{t=1}^{T} (\text{nfa}_t - \text{nfa}_{t-1}) = \sum_{t=1}^{T} \text{ca}_t \\ &= -(1-\beta)(1-\theta) \sum_{t=1}^{T} \beta(b_t + b_t^*)\end{aligned}$$

which is equation (82). ■

Under Ricardian equivalence, $\theta = 1$ and government borrowing leaves the current account and NFA unchanged. But when $\theta \neq 1$, the twin deficits are connected: government deficits cause current account deficits.

E.4 A Linear Increase in Debt

In the general equilibrium model, we studied the United States' 50-year increase in debt. This section provides an analogous exercise in the small open economy, which leads to a simple calculation for how much an increase in government debt reduces the net foreign asset position.

Suppose that the market value of debt begins at $b_0 + b_0^* = 0$ and increases each period by $\frac{\Delta}{T}$, for T periods.[36] Normalize $\overline{Y} = 1$ so that the debt level is relative to annual GDP. This implies the realized path

$$\beta(b_t + b_t^*) = \frac{t}{T}\Delta \qquad t = 1, \ldots, T$$

This debt path is generated by fiscal shocks that satisfy equation (76). Households continue to forecast debt using the AR(1) process; they do not fully anticipate the realized debt path.

The observed increase in debt implies a large NFA decline in this simple economy. The changes are related by

$$\underbrace{\text{nfa}_T - \text{nfa}_0}_{-67\%} = -\underbrace{(1-\beta)}_{0.04}\underbrace{(1-\theta)}_{0.91}\underbrace{\frac{T+1}{2}}_{51/2}\underbrace{\Delta}_{72\%}$$

where the quantities correspond to the rounded values we used in Section 5.2's exercise: a $\Delta = 0.72$ increase in debt/GDP from 1976 to 2025, annual MPC $1 - \beta = 0.04$, and non-Ricardian attenuation $\theta = 0.09$ imply that NFA decline by 67 percentage points, compared to 75.4 percentage points in the data. Similarly, the average current account balance over the T periods is

$$\underbrace{\frac{1}{T}\sum_{t=1}^{T} \text{ca}_t}_{-1.34\%} = -\underbrace{(1-\beta)}_{0.04}\underbrace{(1-\theta)}_{0.91}\underbrace{\frac{T+1}{2T}}_{51/100}\underbrace{\Delta}_{72\%}$$

[36] Beginning the exercise with outstanding debt adds additional terms but remains tractable, and would imply that the NFA declines by even more.

and the 50-year increase implies an average current account deficit of 1.34% of annual income, compared to 2.40% of GDP in the data.

As in the two-country model, our estimated non-Ricardianism implies that government borrowing explains a majority of the NFA decline and average current account deficits. However, there are quantitative differences: the NFA declines by even more in the small open economy, which is missing several ingredients of the richer model, including two clearly attenuating forces. First, the general equilibrium model allows the interest rate to rise with government borrowing, which dampens household dissaving and thus international borrowing. Second, the general equilibrium model has endogenous production, where government borrowing is able to have a small crowding-out effect on capital, reducing income and thus consumption. Thus the general equilibrium setting makes more conservative predictions compared to what would be implied for a small open economy. However, these general equilibrium mechanisms are not clearly observed in the data: investment-to-GDP did not decline, nor did the US real interest rate rise substantially. This suggests that the small open economy exercise is a reasonable pen-and-paper approximation.

F Linearized Open Economy Model

The population regression exercise and the fifty-year debt experiment in Section 5 use the same linearized equilibrium conditions. The fifty-year experiment allows for foreign bond holdings and positive steady-state debt, while the population regressions use the nested special case with autarky and zero steady-state debt.

For the linearized model, we assume power utility, $u'(C) = C^{-\gamma}$ for both domestic and foreign households.

F.1 Deviation from Trend Variables

Lowercase quantities denote level deviations after scaling by productivity

$$x_t = \frac{X_t}{A_t} - \overline{X} \tag{84}$$

For prices and returns, $q_t = Q_t - \overline{Q}$ and $r_t = R_t - \overline{R}$. It is useful to define the growth-adjusted discount factor

$$\beta_A \equiv (1 + g_A)\overline{Q} = \beta(1 + g_A)^{1-\gamma} \tag{85}$$

We assume $\beta_A < 1$ so that the detrended present values converge. The present-value variables below discount deviations in income, taxes, and bond prices at β_A. The balanced-

growth values satisfy

$$\overline{Q} = \beta(1+g_A)^{-\gamma} \tag{86}$$

$$\overline{R} + 1 - \delta = \overline{Q}^{-1} \tag{87}$$

$$\overline{K} = (1+g_A)\left(\frac{\alpha}{\overline{R}}\right)^{\frac{1}{1-\alpha}} \tag{88}$$

$$\overline{Y} = \left(\frac{\alpha}{\overline{R}}\right)^{\frac{\alpha}{1-\alpha}} \tag{89}$$

$$\overline{I} = \left(1 - \frac{1-\delta}{1+g_A}\right)\overline{K} \tag{90}$$

To nest the closed economy, the parameter μ determines the size of the foreign economy. In this appendix, write each aggregate foreign quantity from Section 5 as $X_t^* = \mu X_t^{\mathrm{f}}$. For the fifty-year experiment, we normalize $\overline{Y}^{\mathrm{f}} = \overline{Y}$, so μ remains the foreign-to-domestic GDP ratio defined in Section 5.2. In particular, aggregate foreign bond holdings are $B_t^* = \mu B_t^{\mathrm{f}}$. The remaining balanced-growth values satisfy

$$\overline{T} = \overline{G} + \left((1+g_A)^{-1} - \overline{Q}\right)\left(\overline{B} + \mu\overline{B}^{\mathrm{f}}\right) \tag{91}$$

$$\overline{C} = \overline{Y} - \overline{I} - \overline{G} - \mu\left((1+g_A)^{-1} - \overline{Q}\right)\overline{B}^{\mathrm{f}} \tag{92}$$

$$\overline{C}^{\mathrm{f}} = \overline{Y}^{\mathrm{f}} + \left((1+g_A)^{-1} - \overline{Q}\right)\overline{B}^{\mathrm{f}} \tag{93}$$

Setting $\mu = 0$ removes foreign quantities from domestic market clearing without setting $\overline{C}^{\mathrm{f}} = 0$. For the population regression exercise, we set $\mu = 0$ and $\overline{B} = \overline{B}^{\mathrm{f}} = 0$, which gives $\overline{C} = \overline{Y} - \overline{I} - \overline{G}$ and $\overline{C}^{\mathrm{f}} = \overline{Y}^{\mathrm{f}}$. For the fifty-year debt experiment, we instead set $\mu = 3.94$ and choose steady-state positions satisfying

$$\frac{\overline{B} + \mu\overline{B}^{\mathrm{f}}}{4\overline{Y}} = 0.26 \qquad \frac{-\mu\overline{QB}^{\mathrm{f}}}{4\overline{Y}} = 0.043 \tag{94}$$

We initialize the economy at these balanced-growth positions. At $\mu = 0$, the equations for foreign quantities still contain the unit root, but it has no effect on domestic quantities or aggregate NFA.

F.2 Linearized Equilibrium Conditions

The domestic behavioral consumption function is

$$c_t = (1-\beta_A)\left(\frac{n_{t-1}}{1+g_A} + \tilde{v}_t^y - \tilde{v}_t^\tau\right) + \left(\frac{(1+g_A)\overline{C}}{\gamma} - (1-\beta_A)\overline{B}\right)\tilde{v}_t^q + \zeta_t^c \tag{95}$$

where ζ_t^c is the demand factor and the subjective consol value satisfies

$$\tilde{v}_t^q = q_t + \beta_A\tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^q] \tag{96}$$

The domestic resource constraint is

$$y_t = c_t + k_t - \frac{1-\delta}{1+g_A}k_{t-1} + \overline{I}z_t^k + \mu\left(\frac{b_{t-1}^{\mathrm{f}}}{1+g_A} - \overline{Q}b_t^{\mathrm{f}} - \overline{B}^{\mathrm{f}}q_t\right) \tag{97}$$

where z_t^k is the deviation of the investment price from one. Government bond holdings satisfy

$$b_t + \mu b_t^{\mathrm{f}} = \frac{B_t + \mu B_t^{\mathrm{f}}}{A_t} - \left(\overline{B} + \mu\overline{B}^{\mathrm{f}}\right) \tag{98}$$

For the population regression exercise, the stationary TFP factor $1 + a_t$ multiplies the production function in equation (29). Production and the rental rate become

$$y_t = \overline{Y}a_t + \frac{\overline{R}}{1+g_A}k_{t-1} \tag{99}$$

$$r_t = \overline{R}a_t + \overline{F}_{KK}k_{t-1} \tag{100}$$

where

$$\overline{F}_{KK} = (\alpha - 1)\frac{\overline{R}}{\overline{K}} \tag{101}$$

The perceived present value of domestic income, including changes in capital income, satisfies

$$\tilde{v}_t^y = (1-\alpha)y_t + \frac{\overline{K}}{1+g_A}r_t - \overline{I}z_t^k + \beta_A\tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^y] \tag{102}$$

The bond and capital returns satisfy

$$0 = \overline{Q}\tilde{\mathbb{E}}_t[r_{t+1}] + \overline{Q}(1-\delta)\tilde{\mathbb{E}}_t[z_{t+1}^k] + \overline{Q}^{-1}q_t - z_t^k + z_t^r \tag{103}$$

where z_t^r is the risk wedge between the two returns. Domestic financial net worth is

$$n_t = b_t + \overline{Q}^{-1}k_t \tag{104}$$

With government spending constant relative to productivity, the government budget constraint becomes

$$\tau_t = \frac{b_{t-1} + \mu b_{t-1}^{\mathrm{f}}}{1+g_A} - \overline{Q}(b_t + \mu b_t^{\mathrm{f}}) - \left(\overline{B} + \mu\overline{B}^{\mathrm{f}}\right)q_t \tag{105}$$

The perceived present value of taxes follows

$$\tilde{v}_t^\tau = \tau_t + \beta_A\tilde{\mathbb{E}}_t[\tilde{v}_{t+1}^\tau] \tag{106}$$

The foreign budget constraint is

$$c_t^{\mathrm{f}} + \overline{Q}b_t^{\mathrm{f}} + \overline{B}^{\mathrm{f}}q_t = y_t^{\mathrm{f}} + \frac{b_{t-1}^{\mathrm{f}}}{1+g_A} \tag{107}$$

The foreign Euler equation is

$$0 = \frac{\overline{Q}\gamma}{\overline{C}^{\mathrm{f}}}\tilde{\mathbb{E}}_t[c^{\mathrm{f}}_{t+1}] - \frac{\overline{Q}\gamma}{\overline{C}^{\mathrm{f}}}c^{\mathrm{f}}_t + q_t \tag{108}$$

NFA and the current account balance are

$$\mathrm{nfa}_t = -\mu\left(\overline{Q}b^{\mathrm{f}}_t + \overline{B}^{\mathrm{f}}q_t\right) \qquad \mathrm{ca}_t = \mathrm{nfa}_t - \frac{\mathrm{nfa}_{t-1}}{1+g_A} \tag{109}$$

Their balanced-growth values satisfy $\overline{\mathrm{NFA}} = -\mu\overline{Q}\overline{B}^{\mathrm{f}}$ and $\overline{\mathrm{CA}} = \left(1-(1+g_A)^{-1}\right)\overline{\mathrm{NFA}}$.

The endogenous variables in the model file are ordered as

$$(c_t, b_t, \tilde{v}^q_t, q_t, y_t, \tilde{v}^y_t, k_t, r_t, n_t, \tau_t, \tilde{v}^\tau_t, c^{\mathrm{f}}_t, b^{\mathrm{f}}_t)' \tag{110}$$

The exogenous variables are ordered as

$$(a_t, h_t, b_t + \mu b^{\mathrm{f}}_t, b_{t-1} + \mu b^{\mathrm{f}}_{t-1}, \zeta^c_t, z^r_t, z^k_t, y^{\mathrm{f}}_t)' \tag{111}$$

The exogenous fiscal state h_t denotes the component of taxes implied by debt at fixed bond prices

$$h_t = \frac{b_{t-1} + \mu b^{\mathrm{f}}_{t-1}}{1+g_A} - \overline{Q}(b_t + \mu b^{\mathrm{f}}_t) \tag{112}$$

Thus equation (105) sets the endogenous tax variable equal to $h_t - \left(\overline{B} + \mu\overline{B}^{\mathrm{f}}\right)q_t$. At the zero-debt balanced-growth path used in the population regression exercise, $h_t = \tau_t$. Total government debt similarly appears as an exogenous state and as the endogenous sum $b_t + \mu b^{\mathrm{f}}_t$, with equation (98) imposing their equality.

F.3 Actual and Perceived Laws of Motion

Total government debt follows the AR(1) process in equation (35). We normalize the fiscal innovation to raise h_t by one unit, which is equivalent to a unit tax innovation at zero steady-state debt. The fiscal block of the actual law of motion is

$$\begin{pmatrix} h_t \\ b_t + \mu b^{\mathrm{f}}_t \\ b_{t-1} + \mu b^{\mathrm{f}}_{t-1} \end{pmatrix} = \begin{pmatrix} 0 & (1+g_A)^{-1} - \overline{Q}\rho_B & 0 \\ 0 & \rho_B & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} h_{t-1} \\ b_{t-1} + \mu b^{\mathrm{f}}_{t-1} \\ b_{t-2} + \mu b^{\mathrm{f}}_{t-2} \end{pmatrix} + \begin{pmatrix} 1 \\ -\overline{Q}^{-1} \\ 0 \end{pmatrix} \epsilon^\tau_t \tag{113}$$

Together with $\tau_t = h_t - (\overline{B} + \mu\overline{B}^{\mathrm{f}})q_t$, equation (113) satisfies the government budget constraint in every period. An innovation ϵ^τ_t raises h_t by ϵ^τ_t and reduces total debt by $\overline{Q}^{-1}\epsilon^\tau_t$. Equivalently, the debt innovation in equation (35) is $\epsilon^B_t = -\overline{Q}^{-1}\epsilon^\tau_t$.

Households use the perceived law of motion

$$\begin{pmatrix} h_t \\ b_t + \mu b^{\mathrm{f}}_t \\ b_{t-1} + \mu b^{\mathrm{f}}_{t-1} \end{pmatrix} = \begin{pmatrix} 0 & \theta\left((1+g_A)^{-1} - \overline{Q}\rho_B\right) & 0 \\ 0 & \rho_B & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} h_{t-1} \\ b_{t-1} + \mu b^{\mathrm{f}}_{t-1} \\ b_{t-2} + \mu b^{\mathrm{f}}_{t-2} \end{pmatrix} + \begin{pmatrix} 1 \\ -\overline{Q}^{-1} \\ 0 \end{pmatrix} \epsilon^\tau_t \tag{114}$$

The PLM attenuates forecasts of h_t while leaving forecasts of total debt unchanged. Because h_t does not feed back into the debt process, equation (114) implies

$$\tilde{\mathbb{E}}_t[h_{t+j}] = \theta \mathbb{E}_t[h_{t+j}] \qquad j \geq 1 \tag{115}$$

Thus θ is applied once at every horizon. The same PLM is used for the domestic and foreign equilibrium conditions. In the zero-debt, zero-spending population calibration, $h_t = \tau_t$ and this implements the tax-expectation restriction in equation (37). At positive steady-state debt, the implementation attenuates the fixed-price component h_t; tax forecasts also include $-(\overline{B}+\mu\overline{B}^{\mathrm{f}})\tilde{\mathbb{E}}_t[q_{t+j}]$. The behavioral expectation for future interest rates depends on how the rates are determined endogenously by exogenous states, and the PLM distorts expectations of those states.

The remaining actual and perceived laws of motion are identical

$$a_t = \rho_a a_{t-1} + \epsilon_t^a \tag{116}$$
$$\zeta_t^c = \rho_c \zeta_{t-1}^c + \epsilon_t^c \tag{117}$$
$$z_t^r = \rho_r z_{t-1}^r + \epsilon_t^r \tag{118}$$
$$z_t^k = \rho_k z_{t-1}^k + \epsilon_t^k \tag{119}$$

The innovations are independent and mean zero.

At $\overline{B} = \overline{B}^* = 0$, summing the tax forecasts implied by the PLM gives

$$\tilde{v}_t^\tau = (1-\theta)\tau_t + \frac{\theta}{1+g_A}(b_{t-1} + \mu b_{t-1}^{\mathrm{f}}) \tag{120}$$

Substituting this expression into equation (95) gives the regression equation used in the population regression exercise

$$c_t = (1-\beta_A)\left(\frac{n_{t-1}}{1+g_A} - \frac{\theta}{1+g_A}(b_{t-1} + \mu b_{t-1}^{\mathrm{f}}) + \tilde{v}_t^y - (1-\theta)\tau_t\right) + \frac{(1+g_A)\overline{C}}{\gamma}\tilde{v}_t^q + \zeta_t^c \tag{121}$$

If $\gamma = 1$, then $\beta_A = \beta$ and the coefficient on taxes remains $-(1-\beta)(1-\theta)$.

G General Equilibrium Model Parameter Sensitivity

This appendix documents how the parameter values affect the main outcomes in Section 5's general equilibrium model.

G.1 External Balances

Figures 5–8 report how the change in NFA/GDP and the average current account balance/GDP in the fifty-year debt exercise from Section 5 depends on four parameters. We vary β, γ, ρ_B, and μ one at a time, leaving the remaining parameters and debt path at the values reported in Table 5. When varying γ, the MPC is no longer solely determined by β, so we hold the MPC fixed rather than the discount factor.

The discount factor β is the only parameter that substantially affects the main outcomes. This is because β controls the MPC; when the discount factor falls, households consume more out of permanent income. As a result, domestic households save less out of a positive fiscal shock, so additional financing must come form abroad. This reduces the current account and accelerates the NFA decline. Moreover, β is only relevant because households are non-Ricardian; when $\theta = 1$ it has no effect.

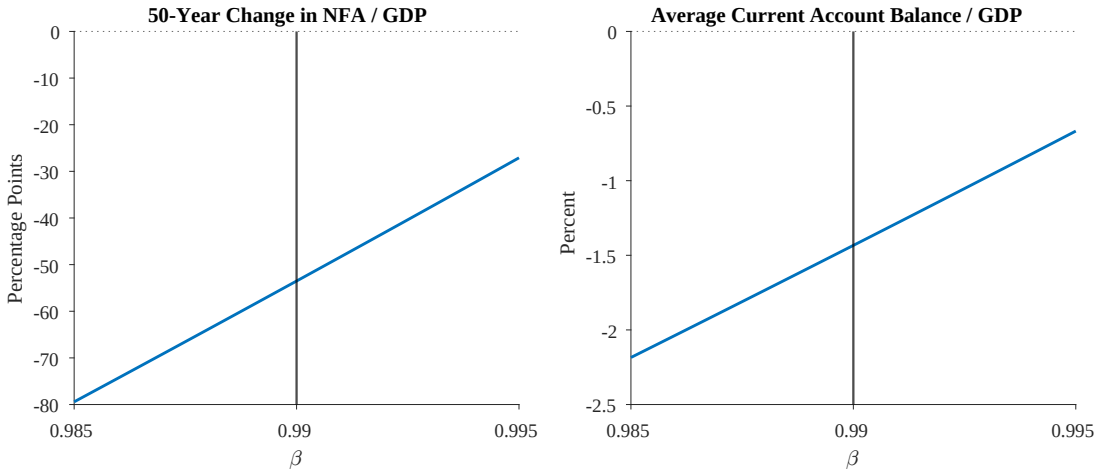


Figure 5: Sensitivity to the Discount Factor

The x-axis reports the parameter value of interest; remaining values are set as in Table 5. NFA changes are reported in percentage points of GDP, and average current account balances as a percent of GDP. The vertical lines mark the baseline calibration.

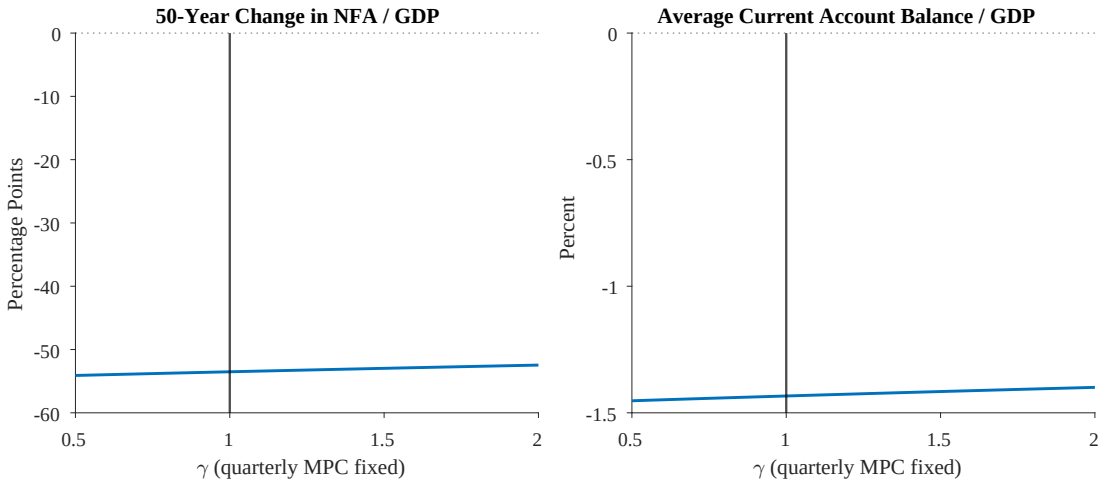


Figure 6: Sensitivity to Risk Aversion at a Fixed MPC

The x-axis reports γ. We adjust β to hold the quarterly MPC fixed at 0.010; remaining parameters are set as in Table 5. NFA changes are reported in percentage points of GDP, and average current account balances as a percent of GDP. The vertical lines mark the baseline calibration.

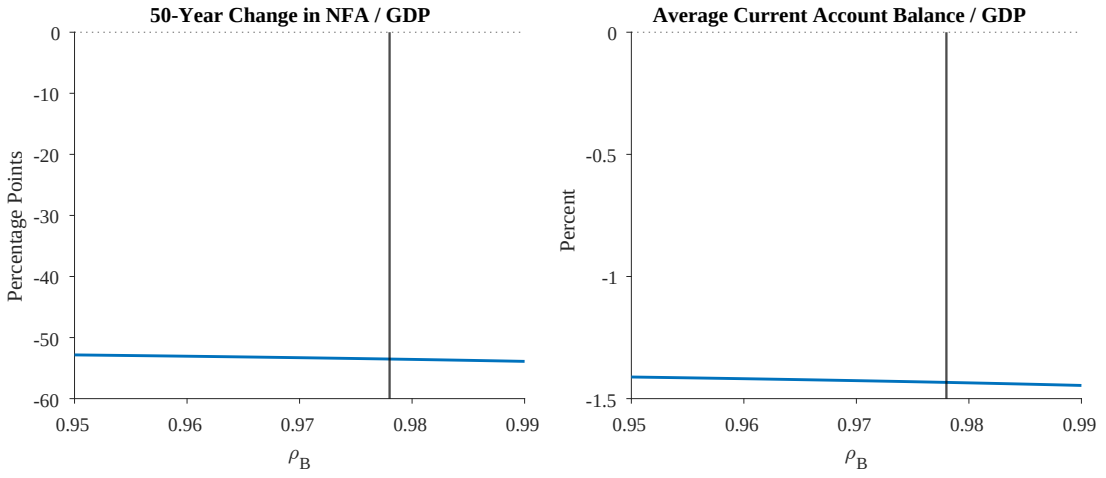


Figure 7: Sensitivity to Debt Persistence

The x-axis reports the parameter value of interest; remaining values are set as in Table 5. NFA changes are reported in percentage points of GDP, and average current account balances as a percent of GDP. The vertical lines mark the baseline calibration.

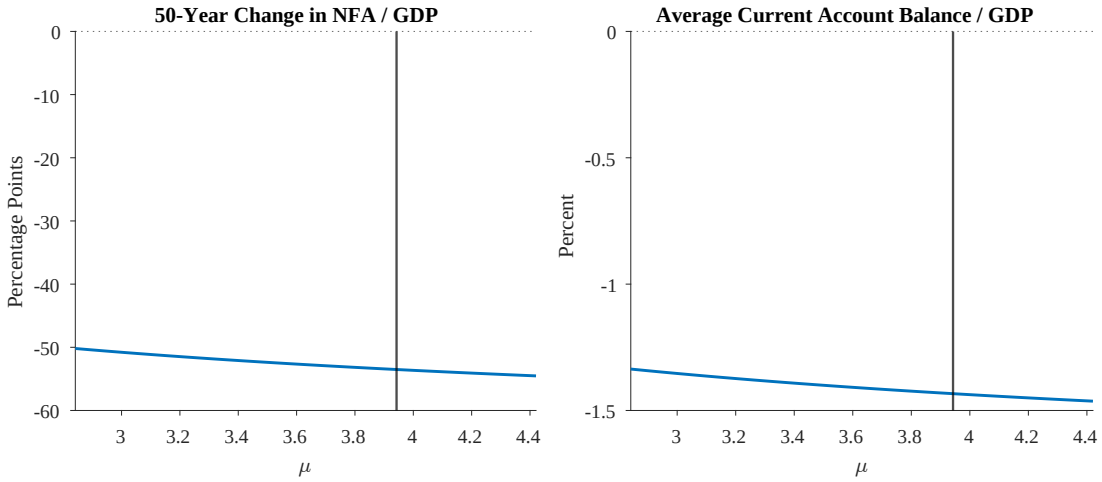


Figure 8: Sensitivity to the Relative Size of the Foreign Economy

The x-axis reports the parameter value of interest; remaining values are set as in Table 5. NFA changes are reported in percentage points of GDP, and average current account balances as a percent of GDP. The vertical lines mark the baseline calibration.

G.2 Investment

Figures 9 and 10 report how the fifty-year trend in I_t/Y_t varies with μ and θ in Section 5's debt exercise. We hold the government debt path fixed in both comparisons. When varying μ, we set initial NFA to zero throughout in order to include the closed economy case at $\mu = 0$. When varying θ, we maintain the baseline initial NFA and calibrated parameters.

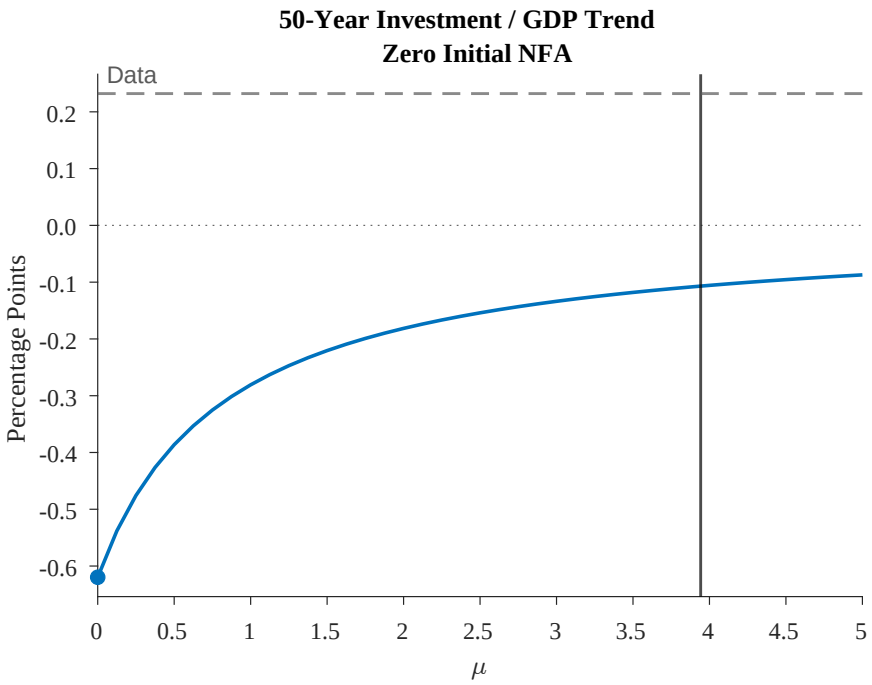


Figure 9: Investment Sensitivity to the Relative Size of the Foreign Economy

The 50-year trend is the coefficient from an OLS regression on a linear time trend multiplied by fifty years, expressed in percentage points of GDP. The vertical lines mark the baseline parameter values, and the dashed horizontal lines report the data counterpart using private nonresidential fixed investment.

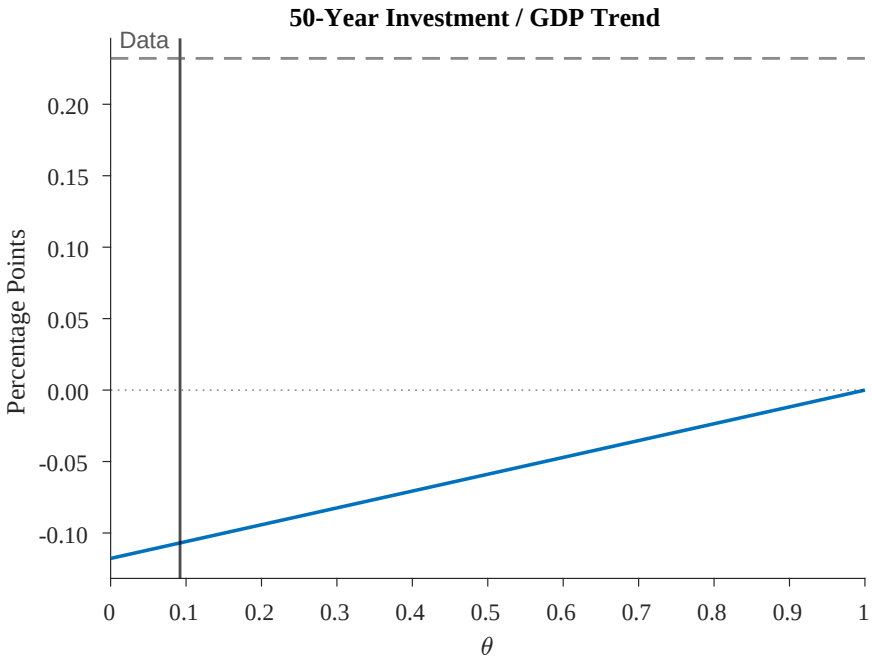


Figure 10: Investment Sensitivity to Non-Ricardian Behavior

The 50-year trend is the coefficient from an OLS regression on a linear time trend multiplied by fifty years, expressed in percentage points of GDP. The vertical lines mark the baseline parameter values, and the dashed horizontal lines report the data counterpart using private nonresidential fixed investment.

H The State Space Model

This appendix gives the state-space representation of the benchmark, with six forecasting variables, four VAR lags, 28 instruments, and an AR(1) consumption residual. We first present the system with complete observations and then describe how we handle missing data.

H.1 Preliminary Definitions

- $X_t \in \mathbb{R}^m$ is the vector of predictors and forecasting variables. It does not include c_t, because consumption is governed by the structural consumption equation.
- We assume a VAR(p) model for X_t:

$$X_t = \mu_X + A_1 X_{t-1} + \cdots + A_p X_{t-p} + G\varepsilon_t, \qquad \varepsilon_t = (\varepsilon_{c,t}, \varepsilon'_{X,t})' \sim N(0, I_{m+1}).$$

  The matrix $G \in \mathbb{R}^{m\times(m+1)}$ maps the structural innovations into X_t. The first innovation, $\varepsilon_{c,t}$, also drives the consumption residual:

$$\zeta_t = \rho_\zeta \zeta_{t-1} + h_c \varepsilon_{c,t}.$$

  Define $s_t = \zeta_t / h_c$ and $u_t = (s_t, \varepsilon'_{X,t})'$. Then

$$u_t = R u_{t-1} + \varepsilon_t, \qquad R = \mathrm{diag}(\rho_\zeta, 0, \ldots, 0).$$

  The VAR is driven by the innovations ε_t, not by the persistent state s_t.
- We form the companion state $\overline{X}_t = \left(X'_t, X'_{t-1}, \ldots, X'_{t-p+1}\right)' \in \mathbb{R}^{mp}$. The resulting companion form is given by

$$\overline{X}_t \;=\; \overline{\mu} \;+\; F\overline{X}_{t-1} \;+\; \overline{G}\varepsilon_t.$$

$$\overline{\mu} \;=\; \begin{pmatrix} \mu_X \\ 0 \\ \vdots \\ 0 \end{pmatrix} \in \mathbb{R}^{mp}, \qquad F \;=\; \begin{pmatrix} A_1 & A_2 & \cdots & A_{p-1} & A_p \\ I_m & 0 & \cdots & 0 & 0 \\ 0 & I_m & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & I_m & 0 \end{pmatrix} \in \mathbb{R}^{mp\times mp},$$

$$\overline{G} \;=\; \begin{pmatrix} G \\ 0_{m(p-1)\times(m+1)} \end{pmatrix}.$$

- We stack into the full latent state

$$Z_t \;=\; \begin{pmatrix} \overline{X}_t \\ 1 \\ u_t \end{pmatrix} \;\in\; \mathbb{R}^{(mp+1)+(m+1)}.$$

H.2 State Transition Equations

$$\widetilde{X}_t = \begin{pmatrix} \overline{X}_t \\ 1 \end{pmatrix}, \qquad Z_t = \begin{pmatrix} \widetilde{X}_t \\ u_t \end{pmatrix}, \qquad \varepsilon_t \sim \mathcal{N}\big(0, I_{m+1}\big).$$

$$Z_t = A\, Z_{t-1} + B\, \varepsilon_t,$$

$$A = \begin{pmatrix} \widetilde{F} & 0_{(mp+1)\times(m+1)} \\ 0_{(m+1)\times(mp+1)} & R \end{pmatrix}, \quad B = \begin{pmatrix} \widetilde{G} \\ I_{m+1} \end{pmatrix},$$

$$\widetilde{F} = \begin{pmatrix} F & \overline{\mu} \\ 0_{1\times mp} & 1 \end{pmatrix}, \qquad \widetilde{G} = \begin{pmatrix} \overline{G} \\ 0_{1\times(m+1)} \end{pmatrix}.$$

H.3 Measurement Equations

We observe

$$\mathcal{Y}_t = \begin{pmatrix} X_t^{\text{obs}} \\ c_t^{\text{obs}} \\ w_t \end{pmatrix} = \underbrace{\begin{pmatrix} H_m & 0_{\,m\times 1} & 0_{\,m\times(m+1)} \\ H_c & \phi_0 & \tilde{h}_c \\ M_X H_\ell & \mu_w & M \end{pmatrix}}_{C} \underbrace{\begin{pmatrix} \overline{X}_t \\ 1 \\ u_t \end{pmatrix}}_{Z_t} + \underbrace{\begin{pmatrix} v_{X,t} \\ v_{c,t} \\ \eta_t \end{pmatrix}}_{v_t},$$

where

$$H_c = \phi_n s_n' + \phi_b s_b' + \phi_\tau s_\tau' + \phi_y s_y' + \phi_q s_q' + \sum_{j\in\{y,g,q\}} \phi_{v^j} s_j'(I_{mp} - \beta F)^{-1},$$

and

$$\tilde{h}_c = \big[h_c, 0, \ldots, 0\big].$$

The row $\tilde{h}_c$ maps u_t into $h_c s_t = \zeta_t$. The selection vectors s_n, s_b, and s_τ pick n_{t-1}, b_{t-1}, and τ_t, respectively, from $\overline{X}_t$; s_y, s_g, s_q select current income, spending, and the bond price. H_ℓ selects X_{t-1}, and H_m selects X_t. The first column of M is zero, so instruments load on the non-consumption innovations in u_t, but not on s_t or its innovation.

The estimating equation uses the state-dependent part of each present value, $v_t^j = s_j'(I_{mp} - \beta F)^{-1}\overline{X}_t$. With a nonzero VAR intercept, the full discounted forecast sum is

$$\sum_{h=0}^{\infty} \beta^h \mathbb{E}_t[j_{t+h}] = v_t^j + \frac{\beta}{1-\beta} s_j'(I_{mp} - \beta F)^{-1}\overline{\mu}.$$

The second term is constant over time. The estimated ϕ_0 includes these constants multiplied by their consumption coefficients. Its prior in Table 6 applies to this effective intercept, not to an intercept defined before the constants are absorbed.

We assume the following distributions for the measurement errors:

$$v_{X,t} = 0, \qquad v_{c,t} = 0, \qquad \eta_t \sim \mathcal{N}(0, \Sigma_\eta).$$

The instrument measurement errors have diagonal covariance Σ_η, are serially independent, and are independent of ε_t. We fix the measurement-error variances of X_t and c_t to zero.

We initialize the stochastic states at their joint stationary mean and covariance for each parameter draw, retaining their cross-covariances. The constant state equals one with zero variance. This initialization requires $\rho(F) < 1$ and $|\rho_\zeta| < 1$, where $\rho(F)$ denotes the spectral radius of the VAR companion matrix.

H.4 Missing Observables

In each period, the Kalman filter uses only the observed entries of $\mathcal{Y}_t$, the corresponding rows of C, and the matching submatrix of the measurement-error covariance matrix (Durbin and Koopman, 2012). Missing observations contribute no measurement residual to the likelihood. Their underlying states are inferred from the transition equations and the observations that are available.

I Priors

The priors on the instrument loadings M and instrument measurement-error variances Σ_η imply a prior for instrument relevance. For an instrument assigned to one structural shock, define

$$\rho_i = \frac{M_i^2}{M_i^2 + \sigma_{\eta,i}^2},$$

the share of the instrument's residual variance attributable to the targeted structural shock after removing its predictable component. Figure 11 plots the implied prior, with median 0.40 and 90% interval [0.18, 0.69]. It puts little mass near zero or one. This measure concerns how well an instrument measures its assigned shock; its relevance for the consumption regressors also depends on the shock's effects through G.

Table 6 summarizes the priors. We use a hierarchical Minnesota prior for the free VAR coefficients in the first m rows of F, denoted F^{free} (Koop and Korobilis, 2010). Before the stability truncation, its mean conditional on the estimated persistence hyperparameter ρ is

$$\mathbb{E}[F^{free} \mid \rho] \;=\; \big[\underbrace{\rho \cdot I_m}_{\text{lag 1}} \quad \underbrace{0_{m \times m(p-1)}}_{\text{lags 2 to p}}\big].$$

Conditional on the tightness hyperparameters, the prior variance for the coefficient on variable j at lag ℓ in equation i is

$$\mathrm{Var}\big(F_{i,j}^{free,(\ell)} \mid \lambda_1, \lambda_2\big) \;=\; \begin{cases} \dfrac{\lambda_1^2}{\ell^2}, & i = j, \\ \dfrac{\lambda_2^2 \lambda_1^2}{\ell^2} \cdot \dfrac{\hat{\sigma}_i^2}{\hat{\sigma}_j^2}, & i \neq j, \end{cases}$$

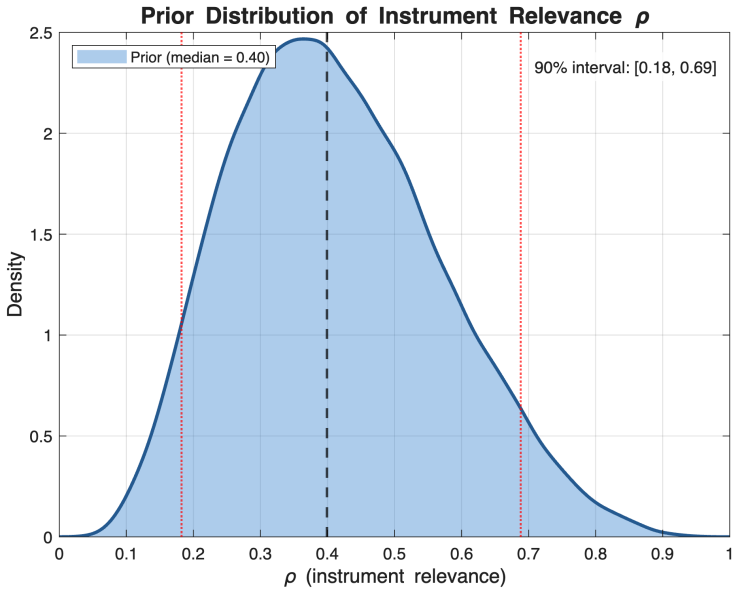


Figure 11: Prior distribution of instrument relevance. Instrument relevance is defined as $\rho_i = M_i^2/(M_i^2 + \sigma_{\eta,i}^2)$, where M_i is the loading of instrument i on its assigned structural shock and $\sigma_{\eta,i}^2$ is the measurement-error variance. The figure is simulated from the priors $M_i \sim \text{Gamma}(20, 0.025)$ and $\sigma_{\eta,i}^2 \sim \text{Gamma}(4, 0.1)$.

where λ_1, λ_2 are hyperparameters, and $\hat{\sigma}_k^2$ is the residual variance from a univariate AR(p) of variable k (estimated equation-by-equation on the available data).

Group	Parameters	Count	Prior	Hyperparameters / Notes
VAR dynamics	F (free rows)	144	Normal	Hierarchical Minnesota prior described below
MN hyperparameters	λ_1, λ_2, ρ	3	Mixed	$\lambda_1 \sim \text{Gamma}(4, 0.05)$; $\lambda_2 \sim \text{Gamma}(4, 0.125)$; $\rho \sim \text{Beta}(2, 2)$
Shock impacts on X	G (6×7)	42	Mixed	$G_{ij} \sim \mathcal{N}(0, 1.5^2\hat{\sigma}_i^2/7)$, except $G_{\tau,\varepsilon_\tau}$, G_{g,ε_g}, and $G_{q,\varepsilon_q} \sim \text{Gamma}(4, 0.025\hat{\sigma}_i)$ for the corresponding row i
VAR intercepts	μ_X	6	Normal	$\mu_{X,k} \sim \mathcal{N}(0, \sigma^2_{X_k})$
Consumption intercept	ϕ_0	1	Normal	$\mathcal{N}(0, \sigma_c^2)$
Selection coeffs	ϕ_n, ϕ_b, ϕ_τ	3	Mixed	$\phi_b \sim \mathcal{N}(0, [0.5\sigma_c/\sigma_b]^2)$; $\phi_n \mid \beta \sim U(1-\beta, 1)$; $\phi_\tau \mid \phi_n \sim U(-\phi_n, 0)$
Current and PV coefficients	$\phi_y, \phi_q,$ $\phi_{v^y}, \phi_{v^g}, \phi_{v^q}$	5	Normal	$\mathcal{N}(0, [0.5\sigma_c/\sigma_j]^2)$ for the corresponding $j = y, g, q$
Consumption innovation loading	h_c	1	Gamma	$\text{Gamma}(25, 0.02\sigma_c)$
Residual persistence	ρ_ζ	1	Truncated Normal	$\mathcal{N}(0, 0.2^2)$ on $(-0.95, 0.95)$
Discount factor	β	1	Beta	$\text{Beta}(31.5, 3.5)$; shared across y, g, q
Instrument loadings on lags	M_X	168	Normal	Mean zero; SD 0.025 for the n, b columns and 0.1 otherwise
Instrument intercepts	μ_w	28	Normal	$\mathcal{N}(0, 1)$
Shock impacts on instruments	M (nonzeros)	28	Gamma	$\text{Gamma}(20, 0.025)$; one nonzero loading per instrument
Instrument meas. variances	Σ_η (diagonal)	28	Gamma	$\sigma^2_{\eta,i} \sim \text{Gamma}(4, 0.1)$
Total		459		

Notes: Gamma distributions use shape and scale parameters; Normal distributions use means and variances. σ_c and σ_j are sample standard deviations of the transformed, unstandardized data, not measurement-error parameters. $\hat{\sigma}_i^2$ is the residual variance from a univariate AR(4). Stocks are divided by quarterly nominal potential GDP. This multiplies their ratios to annual-rate potential GDP by four, so the M_X prior standard deviations for their columns are divided by four. The joint prior implies $\theta = 1 + \phi_\tau/\phi_n \sim U(0, 1)$ and does not impose the income-sum restriction. Before the stability truncation, the listed priors are independent except for the stated conditioning and hierarchical Minnesota prior. The joint support requires $\rho(F) < 1$ and $\beta\rho(F) < 1$. Measurement-error variances for X_t and c_t are fixed to zero.

Table 6: Benchmark parameters and priors: six forecasting variables, four VAR lags, and 28 instruments.

J Additional Robustness Checks

J.1 Alternative Instruments

In section 4.3 we compared our benchmark set of instruments with the more limited set of 6 high quality IVs. In this section, we describe and report additional alternative IV sets. In each case, we replace four of the benchmark IVs with 4 alternatives.

In all cases, our main conclusions about non-Ricardian consumers are unchanged.

Close Analogs We make four replacements with instruments for similar economic shocks. First, we replace Swanson's FFR instrument with the standard monetary-policy shock from Jarociński (2024), who separates different types of policy news using financial market movements around Fed announcements. Second, we replace Ramey's government-spending news with the defense-spending news of Ben Zeev and Pappa (2017). They use a SVAR with medium-run restrictions to identify shocks that explain future defense spending but are orthogonal to current defense spending.

Third, we replace innovations to Fernald's utilization-adjusted TFP series with the unanticipated TFP shock of Ben Zeev and Khan (2015). Fourth, we replace Kilian's oil instrument, based on conflicts in oil-producing countries, with the oil-supply shock of Baumeister and Hamilton (2019). They use a Bayesian VAR incorporating prior information about elasticities to separately identify oil supply and demand shocks; we use only their oil-supply series. The results are reported in Table 7. The θ esitmate is somewhat larger – but still far from Ricardian – with a wider credible interval than our baseline.

Substantial Alternatives In this specification, we make four IV swaps that are more radically different. First, we replace temporary Romer–Romer Social Security payment amounts with an event indicator following Guerreiro et al. (2026), who identify months with large veterans' payments and other transfer events. Our indicator equals one in a month with either a temporary Romer–Romer payment or an additional transfer event from that study. Permanent Social Security transfers remain a separate instrument.

Second, the political-pressure instrument of Drechsel (2024) replaces Piffer–Podstawski. Drechsel uses archival data on presidential interactions with Fed officials to estimate political pressure shocks in a narrative sign-restricted SVAR (Antolín-Díaz and Rubio-Ramírez, 2018). Its assigned shock replaces the benchmark uncertainty shock and is distinct from the monetary-policy shock, leaving six non-consumption shocks.

Third, we replace the Miranda-Agrippino et al. technology news instrument with a reconstructed series of noise shocks to expectations over future TFP. Chahrour and Jurado (2022) develop a method to estimate the effects of these shocks; we construct the implied series following the method in Adams (2023).

Finally, we replace Aruoba–Drechsel's instrument based on natural language processing of Fed documents with the classic narrative monetary-policy instrument of Romer and Romer (2004).

Table 7 reports the estimates. Estimated MPCs are similar to the benchmark. The estimate MPC-DFT is somewhat smaller, implying larger values of θ, but it remains that households internalize less than a quarter of their future tax burden.

Specification	Attenuation (θ)	Annual-rate MPC ($4\phi_n$)	MPC-DFT ($-\phi_\tau$)
Benchmark	0.092 [0.008, 0.218]	0.0232 [0.0198, 0.0265]	0.00519 [0.00436, 0.00614]
Close Analogs	0.191 [0.025, 0.419]	0.0199 [0.0182, 0.0239]	0.00399 [0.00303, 0.00511]
Substantial Alternatives	0.220 [0.021, 0.660]	0.0223 [0.0165, 0.0301]	0.00421 [0.00189, 0.00630]

Table 7: Consumption Function Estimates with Alternative Instruments

Notes: Entries are medians and 5th–95th percentiles under the normalized final SMC weights. We compute $\theta = 1 + \phi_\tau/\phi_n$ for each draw; this relationship need not hold at the marginal posterior medians. The wealth MPC is expressed at an annual rate as $4\phi_n$; MPC–DFT is the flow-to-flow coefficient $-\phi_\tau$. All rows retain the benchmark consumption specification, priors, six forecasting variables, 28 instruments, and AR(1) residuals. The second replacement set pools temporary transfers and replaces the uncertainty, technology, and monetary-policy instruments as described above.

J.2 Alternative Detrending

In our benchmark, we detrended by normalizing aggregate quantities by the CBO's potential GDP series. In this section, we replace nominal potential GDP with a common fourth-order polynomial trend fitted to nominal GDP levels over 1951:Q4–2019:Q4. We divide consumption and the five nominal predictors by this fitted trend rather than subtracting it. The real interest rate, instruments, and other non-normalized variables are unchanged.

Table 8 reports the results. The median θ rises to 0.218, but its interval is wider and overlaps the benchmark's. The annual-rate wealth MPC is close to the benchmark, while MPC–DFT is lower.

Specification	Attenuation (θ)	Annual-rate MPC ($4\phi_n$)	MPC-DFT ($-\phi_\tau$)
Benchmark	0.092 [0.008, 0.218]	0.0232 [0.0198, 0.0265]	0.00519 [0.00436, 0.00614]
Common polynomial GDP trend	0.218 [0.018, 0.691]	0.0219 [0.0177, 0.0257]	0.00422 [0.00159, 0.00575]

Table 8: Consumption Function Estimates with Alternative Detrending

Notes: Entries are medians and 5th–95th percentiles under the normalized final SMC weights. We compute $\theta = 1 + \phi_\tau/\phi_n$ for each draw. The wealth MPC is expressed at an annual rate as $4\phi_n$; MPC–DFT is the flow-to-flow coefficient $-\phi_\tau$.

K Robustness Specifications

K.1 Implementation Details

This section gives implementation details for the robustness checks described in Section 4.3.

The specifications in Tables 3 and 7 use total consumption (except in the nondurables case), the 1951:Q4–2019:Q4 sample, four VAR lags, and the benchmark data transformations. Unless stated otherwise below, they retain the 28 instruments, the AR(1) consumption residual, and the priors in Table 6. Constant components of the present values are absorbed into the consumption intercept. Except for the imperfect-risk-sharing extension, when the dimension of X_t changes, we apply the same prior formulas at the new dimension. In particular, the Normal priors on the shock impacts in G have standard deviations $1.5\hat{\sigma}_i/\sqrt{m+1}$, where $m = \dim(X_t)$ is the number of variables in the VAR ($m = 6$ in the benchmark). Each additional forecasting variable adds one non-consumption innovation without an assigned instrument.

Nondurable consumption. We replace total consumption with personal consumption expenditures on nondurable goods (FRED series `PCND`).

Larger forecasting vector. We expand X_t from six to fourteen variables by adding the average income tax rate, hand-to-mouth household income, hours growth, the Michigan income forecast, the real-rate present-value proxy, and three FRED-QD factors. These variables enter the forecasting system, but do not directly appear in the consumption equation. The consumption regressors, coefficient restrictions, and their priors are unchanged.

Six instruments. We retain one instrument for each of the six non-consumption shocks: defense R&D appropriations for government spending, tax news in presidential speeches for taxes, Federal Reserve document news for monetary policy, patent-application news for technology, OPEC news for oil, and gold-price surprises for uncertainty. Table 2 identifies the sources.

Survey forecasts. We add the Michigan Survey of Consumers' four-quarter-ahead income forecast and a real-rate present-value proxy to X_t, giving eight variables. In the consumption equation, the income forecast replaces v_t^y, and the real-rate proxy replaces q_t. Current income and the VAR-based present values v_t^g and v_t^q remain. To construct the real-rate proxy, we take the negative discounted sum of the ATSIX fitted real yields at maturities of 3, 6, ..., 120 months at each quarter-end. Beyond 120 months, we hold the yield constant at its last observed maturity. We use a fixed discount factor of 0.995 to construct the proxy; the retained VAR-based present values use the estimated β. The income forecast is normalized by nominal potential GDP, while the real-rate proxy is standardized. The coefficient on the Michigan income forecast has prior $N(0, 0.5^2)$, while the coefficient on the real-rate present-value proxy has prior $N(0, 0.03^2)$. Missing forecasts are inferred within the state-space system.

Income-sum restriction. We impose $\phi_y + \phi_{v^y} = \phi_n$: the coefficients on current income and the present value of current and future income sum to the wealth coefficient. This restriction follows from Proposition 2 when perceived discounted future income is proportional to its rational counterpart and current income is known, as described in Section 3.1. We estimate ϕ_{v^y} with its benchmark prior and set $\phi_y = \phi_n - \phi_{v^y}$, dropping the separate prior on ϕ_y.

Distortionary taxes. We use the standardized average income tax rate a_t as a proxy for the distortionary tax rate, adding it to X_t and giving seven variables. We add $\phi_{\text{tax}} v_t^a$ to the consumption equation, where $v_t^a = \sum_{j=0}^{\infty} \beta^j \mathbb{E}_t[a_{t+j}]$. The new coefficient has prior $\phi_{\text{tax}} \sim N(0, 0.5^2)$.

Nonseparable utility. We add hours growth $\Delta \log H_t$ to X_t, giving seven variables, and add $\phi_{\Delta h} v_t^{\Delta h}$ to the consumption equation, where $v_t^{\Delta h} = \sum_{j=0}^{\infty} \beta^j \mathbb{E}_t[\Delta \log H_{t+j}]$. The new coefficient has prior $\phi_{\Delta h} \sim N(0, 0.5^2)$; the other consumption coefficients remain unrestricted apart from the benchmark tax–wealth restriction. The same estimated β discounts all present values.

External habits with partial restrictions. We add a weighted average of past consumption, m_t, to the benchmark consumption equation:

$$\begin{aligned} c_t &= \phi_0 + \phi_n n_{t-1} + \phi_b b_{t-1} + \phi_\tau \tau_t + \phi_y y_t + \phi_q q_t \\ &\quad + \sum_{j \in \{y,g,q\}} \phi_{v^j} v_t^j + \delta_h m_t + h_c \varepsilon_{c,t}, \\ m_t &= \rho_x m_{t-1} + (1 - \rho_x) c_{t-1}. \end{aligned}$$

The six observed forecasting variables are unchanged; m_t is one additional latent state with no separate innovation. We retain $\phi_\tau \mid \phi_n \sim U(-\phi_n, 0)$, hence $\theta = 1 + \phi_\tau/\phi_n \sim U(0, 1)$, but impose neither the income-sum restriction nor the theoretical equalities for debt, spending, or interest-rate coefficients. We use independent beta priors for the habit parameters. Habit-stock persistence, ρ_x, has prior mean 0.90 and standard deviation 0.05. The consumption coefficient on the habit stock, δ_h, has prior mean 0.143 and standard deviation 0.093.

We replace the benchmark wealth prior with

$$\operatorname{logit}(\phi_n) \mid \beta, \delta_h \sim N\big(\operatorname{logit}\big[(1-\beta)(1-\delta_h)\big],\, 0.5^2\big), \qquad \operatorname{logit}(u) = \log \frac{u}{1-u}.$$

The theoretical wealth coefficient is the conditional prior median. The prior allows $0 < \phi_n < 1$. The prior on β remains Beta(31.5, 3.5), and the other free consumption coefficients retain their benchmark priors. Unlike the benchmark, the consumption-function residual is iid, $\zeta_t = h_c \varepsilon_{c,t}$.

K.2 Data Details for Model Extensions

In our robustness checks, we use several additional time series, whose construction we describe here.

The survey robustness check replaces the rational income present value with a forecast proxy, as described in Section 4.1.3. We multiply current personal income before subtracting transfers by one plus the Michigan Survey of Consumers' expected income growth over the next four quarters and normalize by nominal potential GDP.

When constructing the Hand-to-Mouth (HtM) income process, we follow established practices as closely as possible. Using the U.S. Survey of Consumer Finances (SCF) for the years 1983 through 2022, we calculate the ratio of the average annual income of households

classified as HtM to that of all households in the survey. First, we classify households as HtM and omit certain observations by closely following Kaplan and Violante (2014). We consolidate all varieties of hand-to-mouth agents in their research into one HtM classification, determined by a household's biweekly income, net liquid wealth, and monthly credit limit. These variables are carefully constructed using the U.S. SCF full public dataset and the specifications provided by Kaplan and Violante. To construct average income measures, we weight the income for each household using the weights provided in the U.S. SCF dataset and calculate the average income within the desired groups accordingly. We limit our analysis to the years 1992 onward due to the non-uniform survey methodology used in the 1983 to 1989 surveys.

When estimating the extension with imperfect risk sharing, we use the 26 annual discount factor wedge observations for 1992–2017 measured by Berger et al. (2023) from the Consumer Expenditure Survey. We add the quarterly wedge $\widehat{\mathcal{W}}_t$ to X_t. The annual reading a_Y^w for calendar year Y enters the fourth-quarter measurement equation as

$$a_Y^w = \frac{1}{4}\sum_{q=1}^{4} \widehat{\mathcal{W}}_{Y,q},$$

with zero measurement error. The wedge observation is missing in other quarters and outside 1992–2017; the state-space model infers the quarterly wedge over the full 1951:Q4–2019:Q4 sample.

The augmented VAR(4) forecasts the wedge jointly with the other variables. We add $\phi_w v_t^{w,+}$ to the consumption equation, where the state-dependent future sum is

$$v_t^{w,+} = \beta s_w' F(I - \beta F)^{-1}\overline{X}_t,$$

and s_w selects the current quarterly wedge from $\overline{X}_t$. This is the state-dependent component of $\sum_{j=1}^{\infty} \beta^j \mathbb{E}_t[\widehat{\mathcal{W}}_{t+j}]$; its constant component is absorbed into the intercept. We use the same estimated β as for the other present values.

The new consumption coefficient has prior $\phi_w \sim N(0, 0.5^2)$. The additional VAR coefficients follow the hierarchical Minnesota prior in Appendix I. We use $s_a^2/4$, where s_a^2 is the sample variance of the annual readings, as the wedge's variance scale in the Minnesota and wedge-equation shock-impact priors. All benchmark parameters retain their priors, and the 28 instruments retain their shock assignments.

L Posterior Sampling Algorithm

We use a sequential Monte Carlo sampler to approximate the posterior distribution of our model (Herbst and Schorfheide, 2015). This algorithm is well suited to irregular posterior distributions with likelihood ridges, which can arise because the VAR coefficients also enter the consumption function through the present-value terms. Starting from prior draws, the sampler increases the likelihood exponent adaptively to target a relative conditional effective sample size of 0.90. At each stage, five Metropolis–Hastings sweeps update the particles using block proposals with adaptive scaling. We resample when the effective sample size falls below half the particle count. Table 9 reports the benchmark settings.

The larger forecasting-vector comparison uses 10,000 particles; the benchmark and other reported specifications use 15,000. The posterior densities in Figure 1 use equally weighted posterior draws. The medians and credible intervals in Table 3 use the final SMC particles with their normalized weights.

Setting	**Value**
Particles	15,000
Tempering schedule	Adaptive
Target relative conditional ESS	0.90
Maximum tempering stages	2,000
MH steps per stage (N_{MH})	5
Proposal scale	Adaptive by block
Resample threshold (ESS / particle count)	0.5
Resample method	stratified

Table 9: Sequential Monte Carlo settings used in the actual-data estimation.

M Additional Tables and Figures

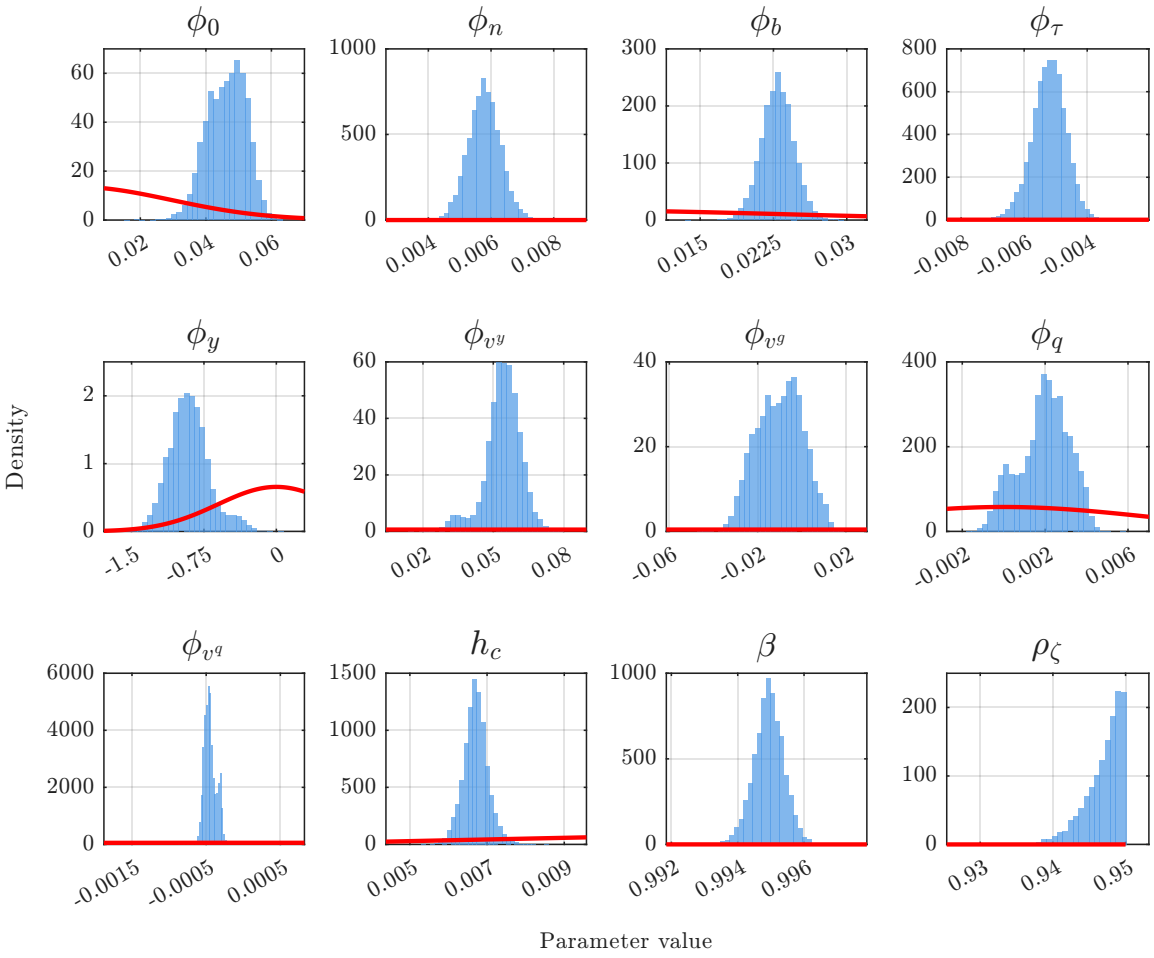


Figure 12: Priors and posteriors of the parameters of the consumption function